Recall Fluency: A Cognitive Source of Belief and Choice Heterogeneity

Pedro Bordalo* Nicola Gennaioli† Matthew O'Brien‡ Andrei Shleifer§

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Abstract

A growing literature shows that beliefs reflect cued recall from a database, subject to interference across experiences. We test this model in the HRS dataset which is unique in offering a respondent-level proxy for interference based on recall fluency in the word-list task, a standard paradigm in memory research. Examining beliefs across three domains – stock returns, home price growth, and life expectancy – we show expectations satisfy three key predictions. People with low interference (i.e. high recall fluency) exhibit: 1) more sensitivity to past experiences, both domain-specific and non domain-specific, 2) more sensitivity to current cues, and 3) less idiosyncratic heterogeneity. This implies that people with low interference are on average more optimistic about stocks, houses, and life expectancy, which correlates with higher stock and housing investment and a lower likelihood of retirement.

*University of Oxford

†Bocconi University

‡London School of Economics

§Harvard University

1 Introduction

A growing literature shows that regularities in human memory unify well documented features of measured beliefs, in particular distortions in the way beliefs react to news or current events (which reflects the role of cues), and the influence of remote DS and NDS personal experiences (which are due to the memory database and interference). Experimental and field evidence shows that exogenous changes in cues or the database of experiences affects recall, reasoning about forecasts, and the forecasts themselves, with qualitatively similar effects across abstract probability judgments, familiar economic domains, or assessment of novel risks.

So far, this work has explored variation in the cues that people are exposed to or in their past experiences, but not in their recall faculties. It is of course a fact of life that some people exhibit better memory than others. This variation is also robustly documented in abstract tasks in which people must recall familiar words from a list. Does variation in such recall fluency influence the way in which people form beliefs? Does a general-purpose feature of a person's mnemonic apparatus affect beliefs through the well-established mechanisms of cues and database across different domains? Addressing this question is important because variation in recall fluency could be an important source of belief and then choice heterogeneity across people, affecting economic outcomes. Just as people who are less patient save less and hence become less wealthy, so the beliefs of people with lower recall fluency may exhibit a systematically different reaction to news or experiences, shaping their choices and market outcomes.

In this paper we address this issue by suing the HRS database a bi-annual survey of a representative panel of US residents who are 50 years or older, with about 20 thousand people per wave. The HRS has measured beliefs in different domains, including future stock returns, home price growth and life expectancy for many years, which allows to study both cross sectional heterogeneity and reaction to events over time, including major shocks such as the 2008 financial crisis. It also allows to compute, for each respondent, a rich database of experiences, which may be domain-specific (DS) for some beliefs but non domain-specific (NDS) for others. Crucially, and uniquely among such surveys, it measures respondents' recall fluency in abstract word tasks, a standard paradigm of measurement and theory in the literature on memory, starting from Ebbinghaus (1885). We show that recall fluency declines somewhat with age but is a surprisingly stable trait that varies across people and exhibits large orthogonal variation with respect to other cognitive traits such as IQ.

Consistent with the memory literature, we take recall fluency to capture, at least in part, an inverse proxy for memory interference: people may be less able to recall familiar words from a recently studied list because many other words they know appear superficially similar and hence block retrieval of the target words in the list (just like confusion about a specific event often reflects the inability to separate it from many other related events). In this view, recall fluency captures a key force of mnemonic beliefs models, which allows us to use these models to make predictions on how individual level heterogeneity in recall fluency should shape beliefs. Specifically, for people with high recall fluency, beliefs should: 1) be more sensitivity to past experiences, both DS and NDS, 2) be more sensitivity to current events or cues, and 3) display less idiosyncratic heterogeneity.

Crucially, since recall fluency is a general-purpose technology, these predictions hold for a range of domains – stock returns, housing and life expectancy – that are unrelated to the word task where fluency is measured. Predictability from recall fluency emphasizes the role of memory in a robust way (compared to a link between beliefs and recollections, which may be due to reverse causality).

We find strong empirical support for these predictions. Beliefs reflect past experiences to an extent that increases in RF: experience effects are not mechanical, rather recalling them is key. This complements earlier results of similarity- or cue-dependent experiences effects, but identifies a persistent component. We also show that reaction to news stronger if RF higher and the news "matches" the personal experience database, that is, if they are particularly similar to a past experience. Thus, beliefs reflect not mechanical distortions in the use of statistics or experiences, but rather the ability to remember, and imagine, events the news are similar to. Variation in recall fluency generates strong heterogeneity in beliefs and instability in the reaction to news. Critically, variation in RF and in experiences, through their effects on beliefs, predict economically important choices such as investment in stocks, home purchases, and retirement decisions.

We break new ground by connecting beliefs and experiences to a measurement of general-purpose memory technology: recall fluency in standard laboratory memory tasks. Memory capacity emerges as a trait – like risk aversion, patience, or optimism – that can shape beliefs and choice.

2 Motivating Facts

2.1 The HRS survey

The Sample. The University of Michigan Health and Retirement Study (HRS) is a nationally representative biennial panel of U.S. adults over age 50. It introduces a new cohort roughly every six years and follows respondents (and their partners) until death. We consider ten waves, starting in 2002 because our main belief variables are not collected before. We have 34,677 unique respondents and 189,097 person-wave observations. Not all respondents appear in every wave due to staggered entry, mortality, attrition. Appendix Table X describes the survey and our sample. Our respondents have on average 67.7 years for age (with 11.4 st dev), 12.5 years of education (3,2 st dev), income of $57,574 ($105,371 st dev),[1] 53% of people are retired, 63% are married, 38% female.

Outcomes: Beliefs and Choices. The HRS elicits beliefs over various social and personal events. We study two macroeconomic domains, the stock and housing markets, and a personal domain, life expectancy. Beliefs about these domains play an important role in economic models, they are available in many waves, and are easily traceable to personal experiences.[2]

The stock market question asks respondents to assess the percent chance that mutual

[1] In this variable, we combine salaried income, self-employed income, bonuses, social security benefits, workers compensation, and pension. The combination of different income sources creates a high variance. For instance, just the 'salaried income' has a std of only 60k

[2] For instance, the HRS elicits also expectations about inflation and US economic performance, but only in a few waves (1996, 1998, 2000). It also elicits a range of personal outcomes which cannot however easily trace to experienced measured within the HRS survey (e.g. beliefs about inheritance or financial help from family members).

fund shares invested in blue chip stocks will be worth more a year from now. The housing question asks homeowners to assess the percent chance that their own home will be worth more a year from now. The life expectancy question asks respondents to assess the percent chance that they will live to a target age, where the threshold varies with respondents' age: it is 80 for people under 70, 90 for people between 75 and 80, 95 for people between 80 and 85, and 100 for people between 85 and 90.[3]

The HRS also measures a broad range of choices. To see if beliefs matter, we consider 6 choices tied to our domains of interest. For the stock market, we consider whether or not the respondent has made a new stock market investment in the past 2 years (regardless of amount), and the percentage of the respondent's IRA that is invested in stocks or mutual funds, which is restricted only to respondents who have an IRA. For the housing market, we consider whether the respondent has bought a new home in the past 2 years. For health choices, we consider whether the respondents expects to work past the age of 70 (restricted to respondents who are currently working), whether the respondent is currently retired or not, and the percentage of a respondents income invested in their pension (restricted to respondents on a defined contribution plan).

Demographics. The HRS report a vast array of indicators measuring respondents' cognitive faculties, health, financial and personal experiences, as well as standard demographics. We use age, gender, marital status, retired/not, income, and education as baseline controls in our regressions.

Experiences. Measured experiences comprise macroeconomic outcomes lived by respondents as well as personal economic and health outcomes. For macro experiences, we focus on experiences directly tied to beliefs.

Stock market and housing market experiences. We generate a respondent's database of annual real stock market returns over their lifetime (birth to 1-year ago), constructed from the Shiller S&P500 dataset (https://shillerdata.com/). In the housing market, we like-

[3]Exact survey wording for each dependent variable is reported in Table B.2.

wise measure housing returns from the Federal Housing and Finance Agency's (FHFA) home price index (https://www.fhfa.gov/data/hpi/datasets?tab=quarterly-data), which is estimated using sale and appraisal data. We match the respondents census region of birth – our most granular measure of location in the survey data – to the respective regional FHFA housing index. Unlike in the stock market, housing price data is only available from 1975 onwards. Hence, our measure of a respondent's housing market database consist of annual real housing market returns in the respondent's region of birth, spanning 1975 to present. These experiences are Domain Specific (DS) for stock and housing market beliefs, they are Non-Domain Specific (NDS) for life expectancy.

Health experiences. To generate analogous health Domain Specific experiences, we rely on the RHS dataset directly. In particular, we start by constructing measures of the following: i) childhood health and ii) current health. Childhood health is the first principal component of the following variables: missed time in school for health reasons (1,0), disabled as a child (1,0), number of childhood health conditions, number of severe childhood health conditions, and number of psychological childhood health conditions. Current health is the first principle components of the following variables: days spent in bed last month, number of health conditions, self-rated health (1-5), depressed (1,0), recent heart attack (1,0), new/worsening severe health condition (1,0), sleep PCA, number of daily tasks the respondents has difficulty doing on their own, whether the respondent has a functional helper (1,0), and number of tasks a functional helper helps with on a daily basis). We then generate a respondent's database of health experiences as the difference between health levels across surveys. In the respondent's first survey, we use the difference between health in the current survey, and childhood health.

Personal experiences. We use i) childhood health and ii) childhood socioeconomic status. Childhood health is as described above, and childhood socioeconomic status is the first principal component of the following variables: father's years of education, mother's years of education, father was unemployed as child (1,0), moved due to financial difficulty as a child (1,0), family received financial aid as a child (1,0), lived rural (1,0), spoke English at home (1,0), whether the respondent's mother worked as (1,0). These experiences, which are measured in the HRS, are NDS events for the stock and housing markets, but they can be DS for life expectancy (i.e. past personal health). We later discuss DS, NDS, and our model's predictions about them.

2.2 Recall Fluency and IQ

The HRS also measures key cognitive faculties, specifically memory and mathematical ability. Memory is our central focus, and in Section 2.3 we link both measures to beliefs.

Memory is measured using three tasks. In two tasks respondents first study a list of 10 words. Then, they are asked to list as many of these words as possible: i) right after the study period (Immediate Memory task), and ii) after completing a five minutes questionnaire on mental health (Delayed Memory task). Memory performance is then measured as successful recall of words in the list. In the third task, which is available in waves 2010-2020, respondents must name as many animals as possible in 60 seconds. Here memory performance is the number of animals recalled.

Starting from Ebbinghaus (1883), free recall of words from a list is a canonical tool for studying human memory in cognitive science. The advantage of this method is that it allows to study general purpose regularities in human recall in an almost idealized context. It abstracts from differences in education and world knowledge (the words are common and hence familiar to subjects), and does not entail symbol manipulation or logical rules, so it does not require reasoning. Based on these tasks, a vast body of work has unveiled four robust recall patterns.

The first is primacy: words studied earlier in the list are more likely to be recalled, due to repeated rehearsal. This phenomenon reflects a general force: recall of an event increases in its experienced frequency. The other patterns are: i) recency (words studied later in the list are more likely to be recalled compared to words in the middle), ii) contiguity (after recalling a word, a person is more likely to recall a contiguous word in the list), and iii) semantic proximity (after recalling a word, a person is more likely to recall a related word, e.g. car $\rightarrow$ brake).

The last three patterns point to the importance of similarity along different features.

Recency captures the similarity of mental context between the time in which the word was studied and the subsequent recall phase. Contiguity captures the similarity of mental context when the recalled and the contiguous words were studied. Semantic proximity captures similarity or relatedness of the words' meanings. Besides frequency and similarity, a key additional force is interference. It refers to the fact that cues prompting recall of specific word also inhibit, due to memory limits, the recall of words on the list that are less similar to the cue. Interference also arises when spontaneous thoughts cue words outside of the list, which may be erroneously recalled (the so called extra-list intrusions) prompting forgetting of words in the list.

Existing work shows that frequency similarity and interference do not operate uniformly. Performance in word recall tasks is shown to vary within a person across different tests and across people on average, with some subjects exhibiting higher performance than others. A variance decomposition exercise shows that cross-sectional variance in memory accounts for about two-thirds of the variation, with within person variation over time accounting for the rest. Inter-personal variation of memory performance is our key focus here.

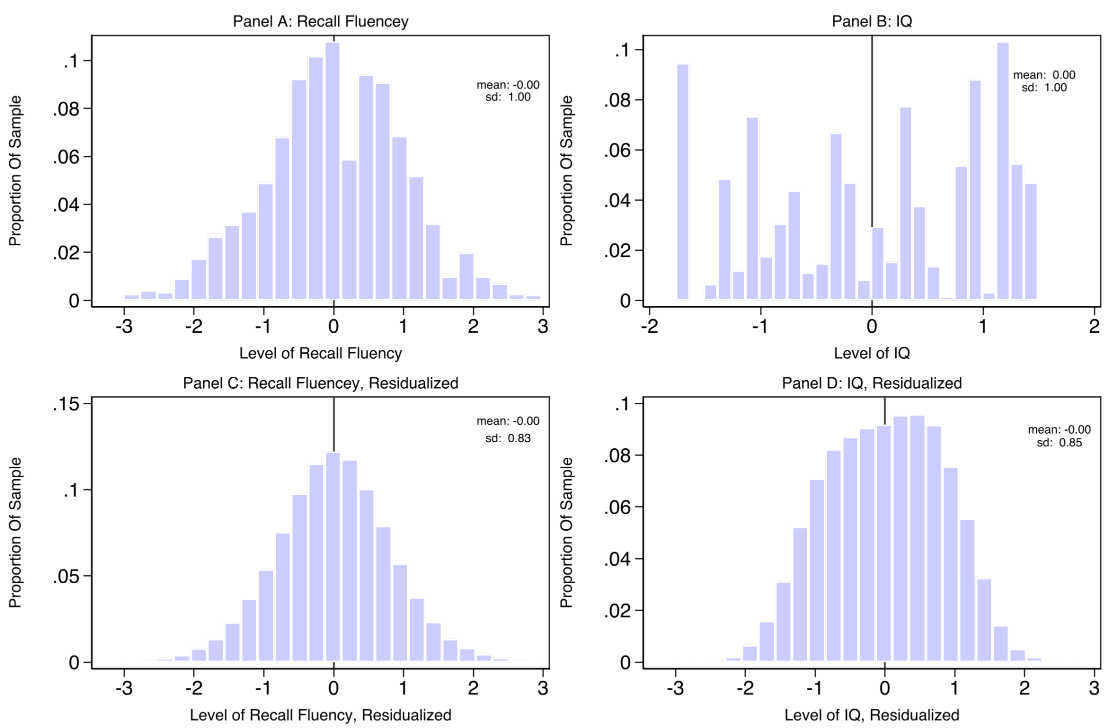
Mathematical ability is also measured in the HRS using three tasks. The first task, counting backwards, requires respondents to perform a sequence of subtractions starting from a given number. In the second task, number series, respondents must guess a rule by filling the last item of a series. In the third task, numeracy, respondents must convert percentages into absolute frequencies. Higher performance in each task indicates higher math ability. Counting backward is available in all years, number series is elicited in the 2010, 2012, 2016 and 2020 waves, and numeracy is only asked to first-time respondents. Mathematical ability affects the use of statistical information, a key driver of beliefs in standard models, also if someone recall every bit of information from memory. Controlling for mathematical ability therefore allows us to separate memory – our main object of interest – from this confound.[4]

[4] The HRS also elicits three additional cognitive measures that could, in principle, be incorporated into a broader cognitive index. General Knowledge consists of factual recall questions (e.g., identifying the current President or today's date). Word Meanings asks respondents to define common vocabulary items (e.g., "repair," "fabric," "plagiarize"). Verbal Analogies presents simple analogical reasoning tasks (e.g., "Mother is to Daughter as Father is to [blank]"). The first two measures align with the crystallized intelligence construct in the psychometric literature; the classification of Verbal Analogies is less clear. We do not incorporate these measures for two reasons. First, Verbal Analogies is available in only three survey waves (2012, 2014, and 2018), substantially limiting the estimation sample. Second, General Knowledge is administered to only half of respondents in each

To capture memory and mathematical ability at the respondent level, we construct two indices. The first one increases in memory performance and we call it called *Recall Fluency* (RF). The second index increases in mathematical performance. We use the shorthand "IQ" given the strong positive association between mathematical ability and this measure of intelligence. Each index is constructed by standardizing individual memory or math components across waves and by averaging, in each wave, a respondent's available standardized components.

Figure 1 reports the raw distribution of RF in Panel A, that of IQ in Panel B. It then reports in Panel C the residualized distribution of RF after regressing it on IQ and a set of demographics (age, income, education, marital status, retired/not, gender), reports in Panel D the that of IQ after regressing it on RF and the same demographics.

Figure 1: Distribution of Recall Fluency and IQ.



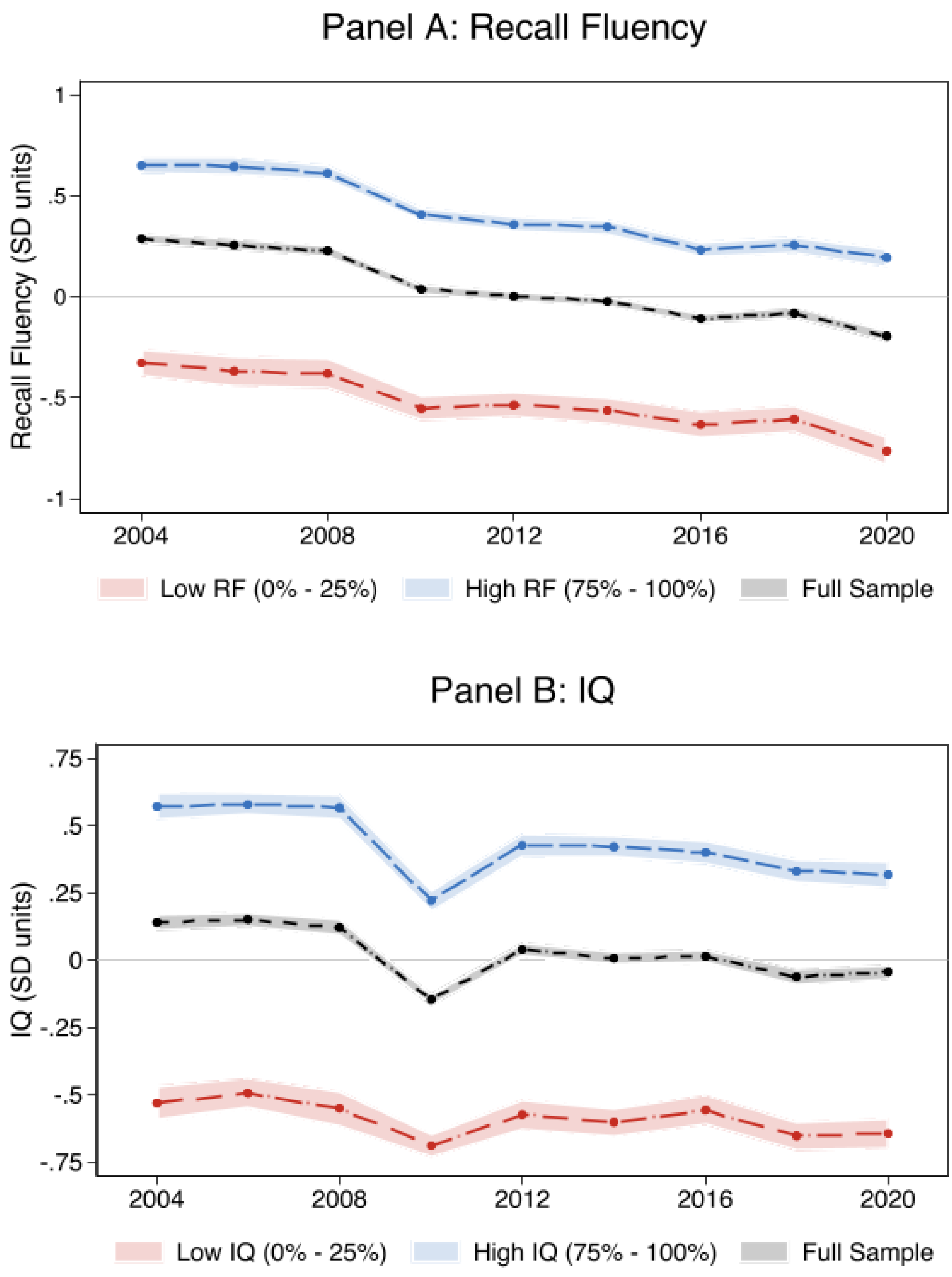
Notes: Panel A shows the raw distribution of Recall Fluency (RF) across all respondents and survey waves. Panel B shows the raw distribution of IQ. Panel C shows the distribution of RF after residualizing on IQ, age, income, education, gender, marital status, and retirement status. Panel D shows the distribution of IQ after residualizing on RF and the same demographics. Both indices are constructed as the standardized mean of their components and are expressed in standard deviation units. Sample: HRS respondents, waves 2002–2020.

wave, and Word Meanings is administered only to first-time respondents. Constructing an index that includes these measures would reduce the usable sample by more than half.

There is significant variation in RF and IQ across people and survey waves. The two indices are positively associated, their correlation is 0.4. Memory tasks do not require any mathematical ability, but some of the math tasks (e.g. counting backwards) benefit from strong memory. More broadly, there is a positive association between different dimensions of cognitive ability in our data, in line with previous findings (Spearman 1904, Greene 2024). Critically, Panels C and D show that even after controlling for demographics and the association between RF and IQ, substantial orthogonal variation in the proxies remains: the standard deviation of RF shrinks by only 0.17 and that of IQ by only 0.15. Moreover, regressing Memory (IQ) on demographics yields an R^2 of 0.24 (0.21), indicating that substantial variation persists after accounting for demographic factors. Adding IQ (Memory) to the regression raises the R^2 to 0.3 (0.28). The large independent variation allows us to study the role of RF when controlling for many confounds, including IQ.

Figure 1 reflects both stable interpersonal differences and within person variability. Is inter-personal variation significant? To what extent does it reflect age-dependent decay in cognitive faculties? Figure 2 show cross sectional and over time differences in RF and IQ. It focuses on the subset of 3,643 respondents present in the totality of our waves and reports the time evolution the index's average among all respondents (black line), among respondents who were in the top 25% of the index in 2002 (blue line) and among those who were in the bottom 25% in 2002 (red line). Panel A reports the results for RF, Panel B for IQ. Appendix Figure C.2 shows the same analysis using only sub-indices that are available throughout the entire sample period (2002–2020), confirming that the steady decline in both RF and IQ is not mechanically driven by the introduction of different sub-indices at different points in the sample.

Figure 2: Recall Fluency and IQ Over Time, By Quartile



Notes: The figure plots the average level of Recall Fluency (Panel A) and IQ (Panel B) in each survey wave from 2004 to 2020. Blue lines show respondents in the top 25% of the respective index in 2002; red lines show the bottom 25%; black dashed lines show the full sample. Shaded bands are 95% confidence intervals. The sample is restricted to COHORT 3 respondents present in all ten survey waves (2002–2020). 2002 is omitted as it is the base year for group assignment. Both indices are expressed in standard deviation units.

There is over time memory decay: average RF declines from 0.38 in 2002 to -0.20 in 2020, 0.58 standard deviations. Critically, the blue and red lines reveal large and stable interpersonal differences in RF. In 2004 the top quartile has RF of 1.03 and the bottom quartile of -0.65, a difference equal to 1.68 standard deviations.[5] In 2020 the difference between the two groups amounts to the comparable 1.4 standard deviation. Similar patterns appear, less pronounced, for IQ: the is a slight average decay from 0.18 in 2002 to -0.02 in 2020, amounting to 0.2 standard deviation, but stable cross sectional-gaps.

These patterns suggest that Recall Fluency is a stable personal trait. Table C.2 formalizes this observation by decomposing the total variance of RF—and of each of its subcomponents—into a cross-sectional component, capturing persistent differences across individuals, and an over-time component, capturing within-person fluctuations across waves. For the Memory Index, the majority of total variance reflects stable cross-sectional differences, with within-person variation accounting for the remainder. Figure C.1 provides complementary evidence by plotting the Spearman rank correlation between each pair of memory sub-components at lags of zero to five survey waves, corresponding to a horizon of up to ten years. Correlations remain high across all lags, confirming that individual rankings in memory performance are largely preserved over the full sample period.

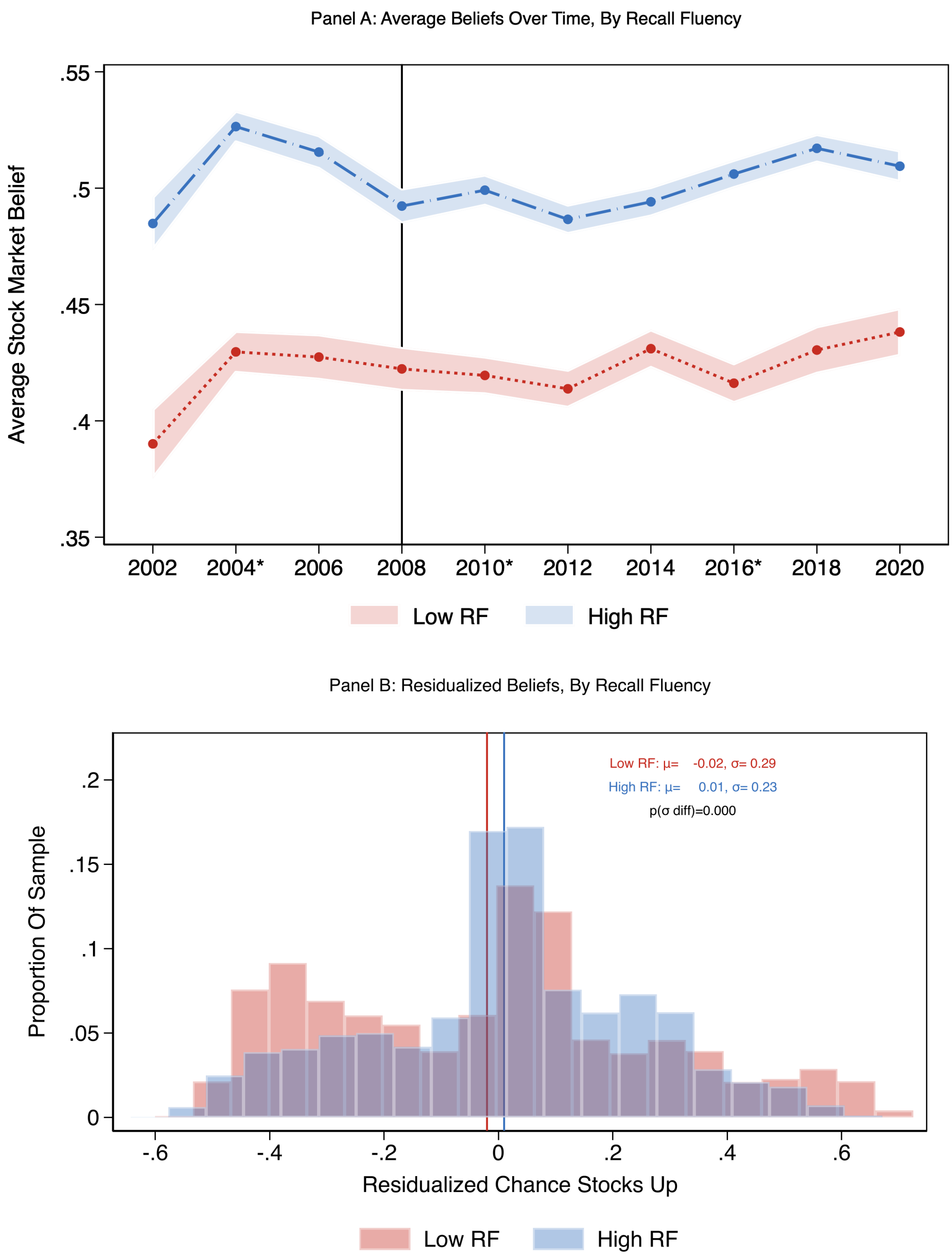
In sum, there is significant variation in how HRS respondents perform in standard word recall tasks, which have been historically used to unveil key regularities in human memory. RF is therefore a stable personal trait. We now show that such trait plays an important role in belief formation about aggregate and personal events, and – as we will see – in economic behavior.

[5] We cannot compare quartiles in 2002 because here variation in RF is also attributable to random shocks, which revert in 2004. A different way to quantify differences is to note that the "within" standard deviation due to age is 0.59 while the "between" standard deviation due to average differences is 0.68, implying that the latter accounts for 54% of the total variance in RF. If we residualize respondents' average RF by average IQ we see that the standard deviation drops by 0.13 standard deviations, so there is significant orthogonal variation in the two measures.

2.3 Recall Fluency and Beliefs

We report descriptive facts about the link between beliefs and the RF index. Figure 3 reports in Panel A the average estimate for the "percent chance stocks are worth more in 1year" across survey waves for the top (blue) and bottom (red) RF quartile. In Panel B it reports the overall distribution of estimates for the high and low memory groups, residualized by IQ, age, income, education, gender, retired and marital status.

Figure 3: Stock Market Beliefs by Recall Fluency.



Notes: Panel A shows the average raw stock market belief ("chance stocks will be worth more in one year") by survey wave, for the top (blue) and bottom (red) quartile of Recall Fluency. Panel B shows the distribution of stock market beliefs residualized on IQ, age, income, education, gender, marital status, and retirement status, for the high (blue) and low (red) RF groups. Shaded bands are 95% confidence intervals. Sample: HRS respondents who follow the stock market, waves 2002–2020.

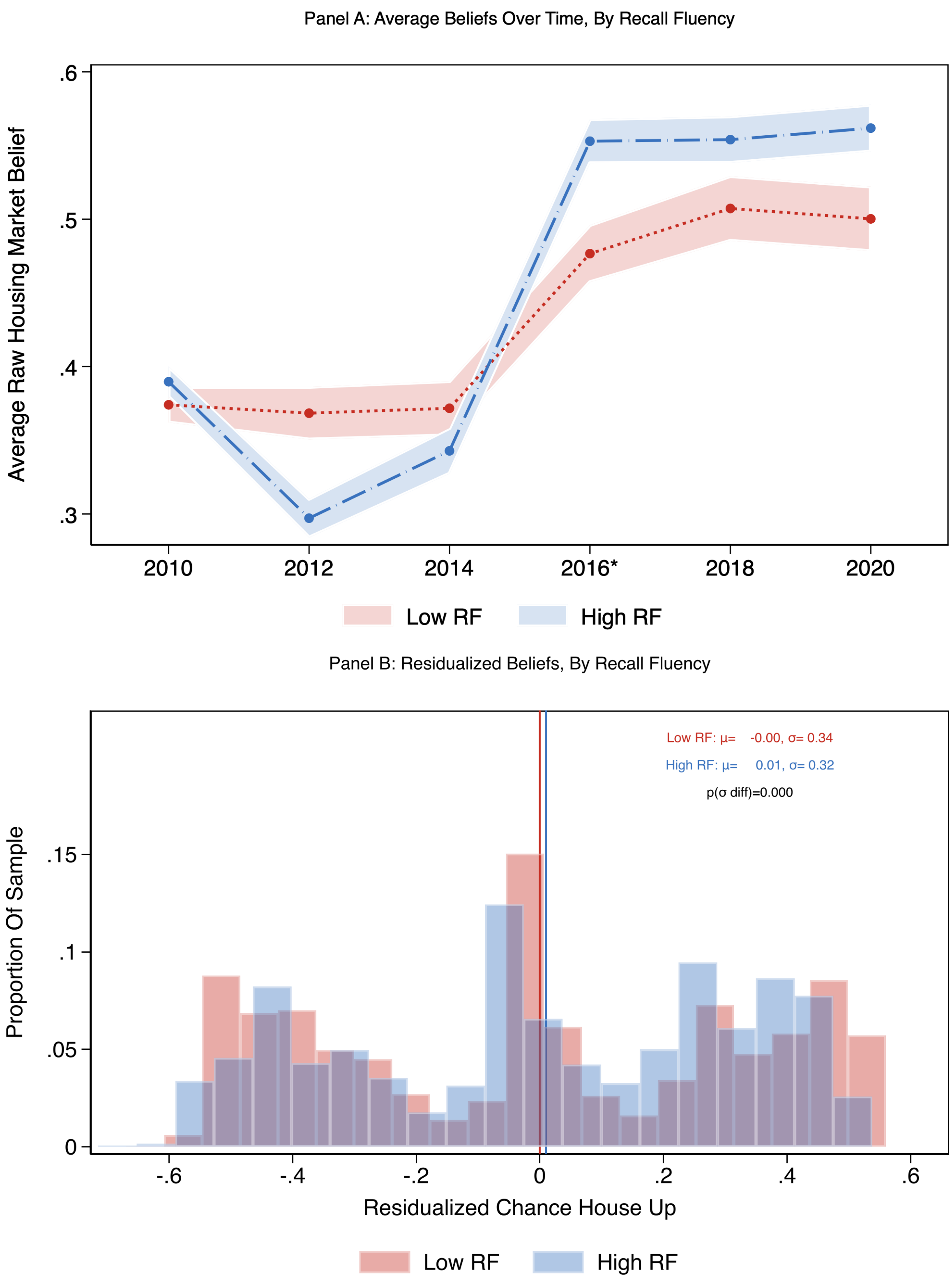
There is a sizable difference between high and low RF people: the former estimate an 8 percentage point higher probability of a stock market rise compared to the latter. The two groups also respond differently to the 2008 financial crisis (dashed vertical line). High RF individuals exhibit sharper boom-bust dynamics: their beliefs rise by 4.2 percentage points from 2002 to 2004, compared to 3.8 percentage points for low RF individuals—a comparable increase. The divergence is more pronounced during and after the crisis: high RF beliefs fall 2.3 percentage points from their peak, versus only 0.6 percentage points for the low RF group, and subsequently recover by 2.3 percentage points from trough to 2018, compared to 0.7 percentage points for low RF individuals.

Both groups are on average too pessimistic about stock returns: the average estimate of high RF individuals falls short of the actual frequency of stock market increases in the sample by 31 percentage points, while the low RF group falls short by 38 percentage points. The two groups exhibit similar distributional shapes, but high RF individuals display lower belief dispersion after controlling for covariates.

Figure 4 reports in Panel A the average estimated percent chance that the respondent's own home will be worth more in one year for the top (blue) and bottom (red) RF quartiles, and in Panel B the distribution of individual estimates for the two groups after residualizing on the same variables as in Figure 3. The question is asked to homeowners—those who fully own or are currently paying off their mortgage—corresponding to 44% of the sample, and is available from 2010 onwards.[6]

[6]The figure includes only respondents who faced the positively-framed version of the question, in which the probability of a home price increase is elicited directly, analogously to the stock market question. The regression results in subsequent sections include all homeowners regardless of question framing.

Figure 4: Housing Market Beliefs by Recall Fluency.



Notes: Panel A shows the average raw housing market belief ("chance home will be worth more in one year") by survey wave from 2010 onward, for the top (blue) and bottom (red) quartile of Recall Fluency. Restricted to positively-framed questions. Panel B shows the distribution of housing market beliefs residualized on IQ, age, income, education, gender, marital status, and retirement status, for the high (blue) and low (red) RF groups. Shaded bands are 95% confidence intervals. Sample: HRS respondents, waves 2010–2020.

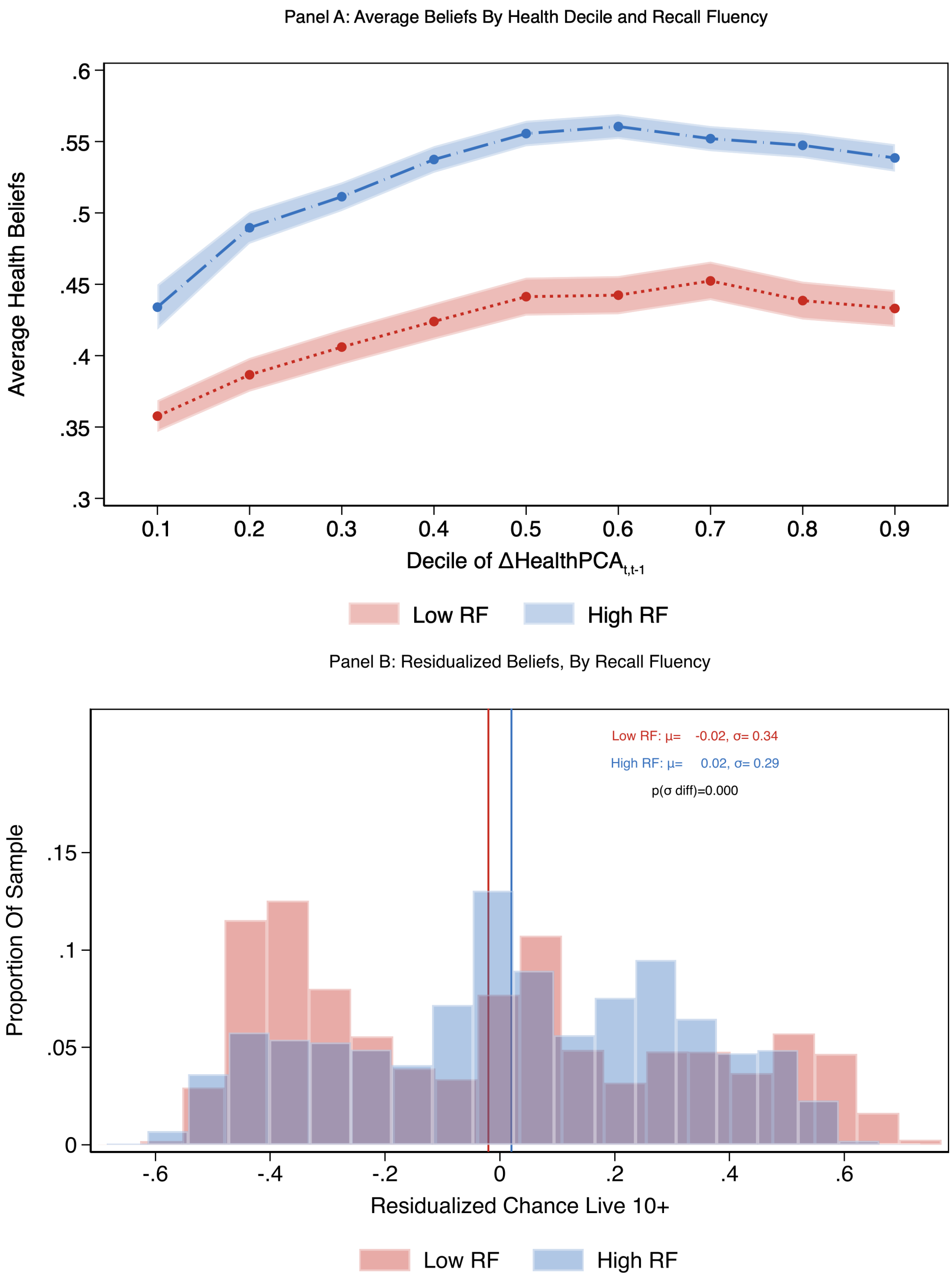
Again, significant differences tied to memory are apparent. While the average level of beliefs is comparable across groups, high RF individuals exhibit more volatile housing beliefs: their beliefs fall by 1.5 percentage points more than those of low RF individuals during the housing bust, but recover by 26.5 percentage points from trough to peak, compared to 13.2 percentage points for the low RF group.

Both groups underestimate the probability of house price increases relative to their actual frequency: by 28.2 percentage points for high RF and 29.7 percentage points for low RF individuals. The cross-sectional distribution of beliefs features fat tails in both groups, and low RF individuals exhibit greater belief dispersion than high RF ones—33 percentage points versus 31 percentage points, after controlling for covariates.

Finally, consider life expectancy, a personal event. Figure 5 reports in Panel A the average estimated probability of surviving at least ten more years for the top (blue) and bottom (red) RF quartile as a function of the respondent's current health status, proxied by the change in the first principal component of health indicators.[7] Panel B reports the cross-sectional distribution of beliefs, residualized by demographics.

[7] First principal component of the following variables: days spent in bed last month, # total health conditions, self-rated health, depressed, recent heart attack, New/Worsening severe health condition, sleep PCA, total number of task difficulties, has a functional helper, tasks helped with on a daily basis.

Figure 5: Life Expectancy Beliefs by Recall Fluency.



Notes: Panel A shows the average raw health belief ("chance live 10+ years") by decile of health change ($\Delta\text{HealthPCA}_{t,t-1}$), for the top (blue) and bottom (red) quartile of Recall Fluency. Panel B shows the distribution of health beliefs residualized on IQ, age, income, education, gender, marital status, and retirement status, for the high (blue) and low (red) RF groups. Shaded bands are 95% confidence intervals. Sample: HRS respondents, waves 2002–2020.

High RF individuals are significantly more optimistic about their life expectancy than low RF individuals at every level of current health status, and their optimism rises more steeply as health improves. The cumulative increase in beliefs along the high RF curve amounts to 10.5 percentage points, compared to 7.5 percentage points for the low RF group. As with stocks and housing, low RF individuals exhibit greater belief dispersion: the standard deviation of beliefs is 31 percentage points for low RF compared to 29 percentage points for high RF, after controlling for demographics.

These findings are surprising: individual performance in abstract laboratory word recall tasks is associated with sharp differences in beliefs about macroeconomic and personal events. RF can account for sizable baseline belief heterogeneity across people in terms of mean and standard deviation of opinions, but it also shapes their differential reaction to news. Why is this the case? A standard answer is omitted variables. High RF people have different information or preferences than low RF people. This possibility is not entirely plausible because, as we said, word recall tasks do not rely on knowledge or preferences and it is largely independent of demographics proxying for standard drivers of beliefs. Another possibility is that variation in RF captures variation in memory, which in turn causes differences in beliefs. But this possibility raises many difficult questions: why does RF affect beliefs in domains that are so different? Does it capture a parameter of a general purpose belief technology? And why does it affect beliefs and reaction to news so differently across domains? To seek to understand the role of RF and to try to address these questions, we now introduce a model of belief formation under selective memory.

In this model, memory is a general technology affecting beliefs in real world domains, and – in an analogy with production functions in economics – the parameters of this technology shape both performance in laboratory recall tasks, and hence RF, and ability to recall data from memory in the field, and hence beliefs. The model yields predictions that allow to: i) test memory against competing explanations, and ii) shed light on the differential role of RF in different domains.

3 The Model

We formalize recall based on memory research (Baddeley 199?, Kahana 2012), and link between memory to beliefs following the recall and simulation model of Bordalo et al (2023).

3.1 The Recall Technology

As in standard memory models, we rely on: i) a database of experiences, ii) the memory cue, and iii) a retrieval function mapping the cue to recollections from the database.

The database is a frequency distribution over categories of experiences. There is a mnemonic space E. A category is a trace/vector $e \in E$. Let f_e be the number of lived instances of $e \in E$; The database is $\mathbf{f} = \{f_e\}_{e \in E}$. Vector e could report the objective description/signal $s \subset \mathbb{R}$ of the experience, its subjective valence $v \subset \mathbb{R}$, and its semantic context $c \subset C$, where C is a context space, so that $e = (s, v, c)$. A stock return experience would then report a numerical return range, say $s = (2\%, 3\%)$, a marker for positive emotional valence, say $v = \mathbb{R}_{++}$, and context features such as the semantic "stock market" domain, the year, the country, etc. The database specifies the frequency of lived episodes falling in this and in all other categories.

Similarity and the Cue. The mnemonic space is endowed with a function $d : E \times E \rightarrow \mathbb{R}$ measuring the discrepancy between any two experiences $e, e' \in E$, where $d(e, e')$ increases in the difference among their features. If e' is a personal illness, it differs from a stock market experience along the objective, subjective, and context aspects, entailing a large $d(e, e')$. We shall not explicitly carry the features of experiences around, but it is useful to keep them in mind to follow the model's intuition for recall and belief formation.[8]

A cue is a subset of the mnemonic space $q \subseteq E$ that ignites similarity-based retrieval from it. The similarity $S(e, q)$ between the cue q and experience e is a bounded function that decays in their discrepancy $d(e, q)$. The probability of recalling e based on q from database $\mathbf{f}$ obeys:

[8] Rather than as a distribution over a metric space E (Kahana 2012), distributed memory models formalize the database it as a sum of experience vectors (Murdock). These approaches have similar implications for our analysis.

$$r_e(q, \mathbf{f}) = \frac{S(e, q) \cdot f_e}{\sum_{e' \in E} S(e', q) \cdot f_{e'}} \tag{1}$$

The numerator captures similarity and frequency: experiences more similar to the cue and more frequently lived are more likely to be recalled. The denominator captures interference: even if e is similar to q and frequent, its recall can be blocked by an even more similar to q or frequent experience $e' \neq e$.

3.2 The Recall Technology and RF

We now link Equation (1) to the RF index, which helps us identify sources of variation in RF across people. This in turn, will give us new predictions about the memory mechanism that that we test using the HRS expectations data. Our analysis here is heuristic but it could be precisely formalized using contextual drift models.

Study phase. Study of a word list shapes the database. A word in serial position $j = 1, \ldots, n$ entails an objective signal/phonetic w_j and a context tag c_j capturing the word's semantics and temporal features (we abstract from valence here but introduce when we study beliefs). Let $L \subset E$ be the set of traces (w_j, c_j) created in a word list task. Their encoding and rehearsal during study changes the database, which now attaches positive frequency f_j to each e_j.

Recall phase. The initial cue q_0 is the word list $(w_j)_j$ itself and the initial context c_0. As words are recalled, they shape the cue and subsequent retrieval. Primacy effects arise because the first word studied w_1 tends to be rehearsed more often than other words, increasing its encoded frequency f_1 and thus its recall in Equation (1). Recency effects occur because the last word studied w_n has a temporal context c_n similar to the initial context c_0, which increases its similarity-based recall in Equation (1).

As words are recalled, the evolution of the cue if often formalized using the AR(1) process $q_{t+1} = \rho q_t + (1-\rho)e_j$, where $(1-\rho) \in (0,1)$ is the impact of the last recollection e_j on a person's mental state (and hence on the cue). Contiguity effects arise because recall of e_j pollutes q_{t+1} with a context c_j that favors recall of the temporally contiguous words w_{j-1} or w_{j+1}, which share a similar context. Sematic proximity works analogously, via the semantic context in c_j.

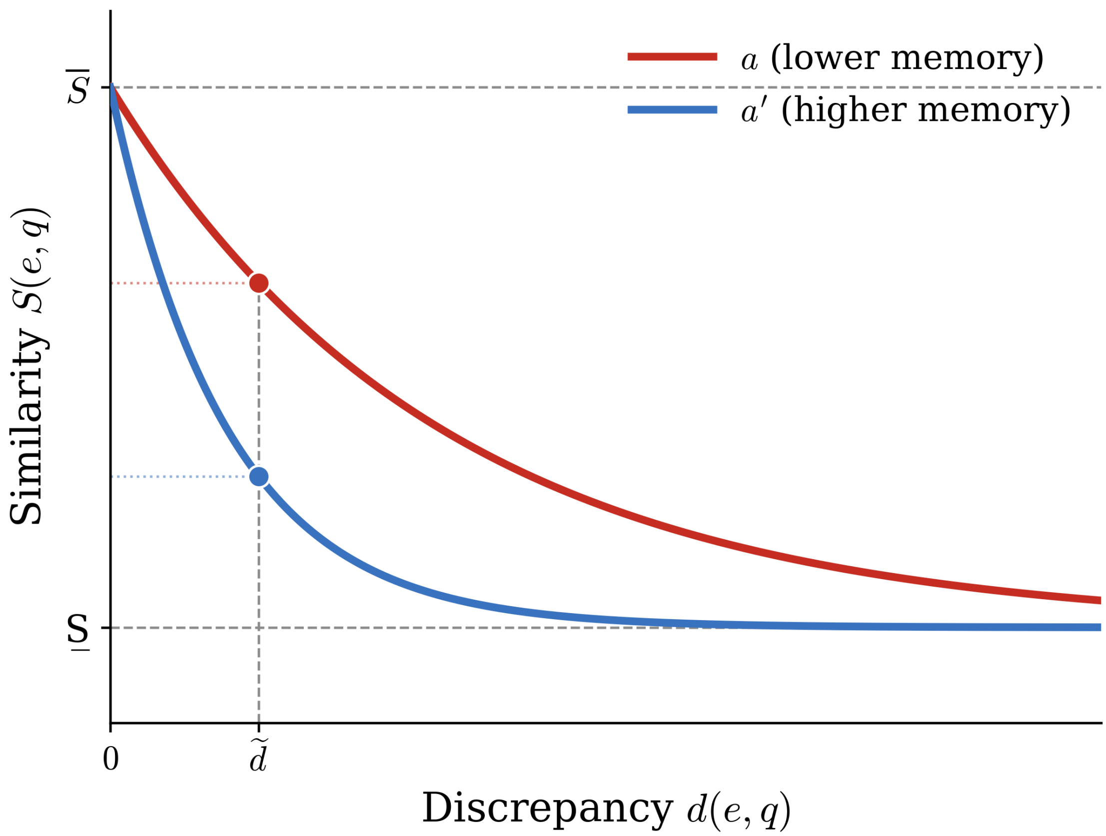
Forgetting is due to interference, the denominator in (1). In particular, recall of frequently rehearsed, or recent, or contiguous, or semantically similar words inhibits retrieval of words that do not have these properties, causing their forgetting. Critically, for the purpose of RF the most problematic kind of interference arises from irrelevant experiences: e_j can trigger recall of similar or frequent words that are *not* in the list (e.g. "apple" in the list may prompt recall of extra-list "pear") or simply distract the subject, causing forgetting of list members, also reducing RF.

Recall Fluency. By Equation (1), variation in RF across people can arise from differences in the encoded frequency f_j of vectors in L and their similarity $S(e_j, q_t)$ to the cue. Such differences cannot act uniformly across experiences: interference, the driver of forgetting, is shaped by the *relative* frequency and similarity of e_j to q compared to irrelevant experiences $e \notin L$. A lower frequency f_j of trace e_j relative to the universe of extra-list frequencies $\sum_{e \notin L} f_e$ evidently reduces recall of word w_j from the list in Equation (1). Such relative frequency may vary across people, for instance because some people encode recent events more weakly (their f_j is lower) or have more numerous irrelevant experiences (their $\sum_{e \notin L} f_e$ is higher). These forces may explain why RF declines with age: older people encode recent events less and have more irrelevant experiences than younger ones. (N: pls see evidence for this)

Differences in perceived similarity can also play a role. They can be due to differences in peoples' attention to the features of studied words, e.g. their temporal or semantic structure, during encoding and retrieval. Attention is indeed a key driver of similarity (Nosovsky) and crucially of learning, in word tasks (Murdock, others?) and in general. Widely used memorization techniques often rely on focusing attention on contextual associations between recall targets. Such attention encodes strong associations among targets, favoring their associative retrieval and reducing interference from non-targets.

To see this formally, consider for simplicity the similarity function $S(e,q) = \mathcal{S}\left[a \cdot d(e,q)\right]$ where $\mathcal{S}$ is decreasing and $a \in (0,1)$ increases in attention to features. People with higher a exhibit a sharper similarity gradient as shown by the blue compared to the red curve in Figure 6.

Figure 6: Similarity as a function of discrepancy, at high versus low attention.



Notes: Illustrative plot of the similarity function $S(e,q) = \mathcal{S}[a \cdot d(e,q)]$ with $\mathcal{S}$ decreasing and bounded between $\underline{S}$ and $\overline{S}$. The blue curve corresponds to higher attention a' (higher recall fluency); the red curve to lower attention a. At zero discrepancy both curves attain the upper bound $\overline{S}$; as $d(e,q)$ grows both decay toward the lower bound $\underline{S}$. At the intermediate discrepancy $\widetilde{d}$, a low-attention respondent still perceives the experience as similar to the cue, while a high-attention respondent perceives it as dissimilar. The figure is schematic and uses $\mathcal{S}(x) = \underline{S} + (\overline{S} - \underline{S})\exp(-x)$ for illustration.

For experiences whose features are identical to those of q, namely $d(e,q) = 0$, similarity is maximal at $\overline{S}$ regardless of attention a. Feature discrepancies instead cause attention to matter. An experience $\widetilde{e}$ with discrepancy $\widetilde{d}$ is perceived to be quite similar to q by a low attention person (red curve) but dissimilar from it by a high attention person (blue curve).[9] The probability of recalling any studied word in period t from list L obeys:

$$\Pr\left(e_j \in L | q_t\right) = \frac{\sum_{e_j \in L} \mathcal{S}\left[a \cdot d\left(e_j, q_t\right)\right] \cdot f_{e_j}}{\sum_{e' \in E} \mathcal{S}\left[a \cdot d\left(e_j, q_t\right)\right] \cdot f_{e'}} \tag{2}$$

By differentiating (2) around $a_0 = 0$ we can see that:

$$\frac{\partial \Pr\left(e_j \in L | q_t\right)}{\partial a} > 0 \quad \Longleftrightarrow \quad \frac{\sum_{e_j \in L} d\left(e_j, q_t\right) \cdot f_{e_j}}{\sum_{e_j \in L} f_{e_j}} < \frac{\sum_{e \notin L} d\left(e, q_t\right) \cdot f_e}{\sum_{e \notin L} f_e} \tag{3}$$

which links attention to RF: higher a increases the probability of recalling a target e_j if words in the list have on average more features in common with the task-cue q_t compared to other experiences. This is plausible, because items in the word list share many features: the time and space when they are studied, their phonetics, semantics, the goals of the task etc. In Figure 6, then, a target word w_j at short distance d_j from the task-cue q_t exhibits less interference from $\widetilde{e}$ for a person having higher attention to features, who then exhibits higher RF.

In sum, a selective recall model producing well established regularities can produce variation in RF either through variation in the encoded frequency of studied words or in attention to features. To understand the role of memory in the expectation facts of Section X, we adopt the attention channel, which we think is appropriate for two reasons. First, attention is a general-purpose factor shown to affect perception and behavior across many laboratory and field tasks, and varies systematically across people (Rosenberg Finn et al 2016, Unsworth Miller Strayer 2024, Hirmas Engelmann van der Weele 2024). Second, attention is parsimonious

[9] Figure 6 assumes bounded similarity, which implies that attention matters little for highly discrepant experiences, i.e. large $d(e, q)$. This property is not important for our results and can be relaxed with unbounded similarity.

because differences in a produce effects that are isomorphic to differences in encoded frequency (e.g., low attention to current context features reduces recall of recent experiences).[10] And considering only one factor simplifies the analysis.

3.3 Selective Recall and Beliefs

The respondent forms beliefs about event $T_X \subset \mathbb{R}$, where $X = SP, HP, LE$ is the domain: stock market (SP), housing (HP), life expectancy (LE). For stock market and housing T_X is a positive price increase in the next 1 year, for life expectancy it is being alive for at least the next ten years. Following Bordalo et al. (2024), people form beliefs by: i) selectively recalling $e \in E$ based on frequency, similarity, interference, and by ii) using the recalled experiences to imagine T_X. The cue for person i at time t is a vector in E that specifies: the current signal $s_{X,it}$ (recent stock return, house price growth or personal health), its semantic context $c = X$ (we abstract from temporal context here), and the respondent's affective state v_{it} (higher values stand for better mood), so $q_{X,it} = (s_{X,it}, v_{it}, X)$. These features ignite selective retrieval from the database. We first focus on the role of $s_{X,it}$ and $c_{X,t}$ but valence will also play a role.

Upon retrieving $e \in E$, the DM uses it to imagine an outcome in the target space $\mathbb{R}$ based on a simulation function $\sigma_X : E \rightarrow \mathbb{R}$ that obeys the following properties.

Assumption 1 *When forming beliefs on $X = SP, HP, LE$, simulation based on $e = (s, v, c)$ obeys:*

$$\sigma_X(e) = \begin{cases} h(s) & if \quad c = X \\ h_X(v) & if \quad c \neq X \end{cases} \tag{4}$$

where function h: $\mathbb{R} \rightarrow \mathbb{R}$ is increasing. Function h_X: $\mathbb{R} \rightarrow \mathbb{R}$ is increasing if and only if higher values in X are good for the DM. h_X takes a default value δ_X for a "neutral" experience

[10]We assume that attention to all features is a. In principle, for a total attention budget, people may vary in how they focus on one feature compared to others. This alternative formulation would not change our main results.

$(v = 0)$.

When the DM retrieves a Domain-Specific (DS) experience in the target domain, $c = X$, she imagines the future using an increasing function h of the recalled signal s. A DM is more optimistic about stock returns if she recalls a 10% rather than a 1% stock return. People recalling higher stock, housing or health experiences make higher forecasts in the respective domain.

When the DM retrieves a Non-Domain-Specific (NDS) experience from a non-target domain, $c \neq X$, she simulates based on affect, imagining a rosier future if the retrieved NDS experience has higher valence v. Recalling pleasurable personal financial experiences (i.e. earning a higher salary) helps imagine high stock returns and vice versa if bad experiences are recalled. If the retrieved experience is neutral, simulation is inhibited: NDS from routine activities such as having our usual breakfast, walking to the same office, etc., act as distractors not as simulators. In this case the agent picks an agnostic default δ_X in the response scale such as "I do not know", the midpoint, or a random element.

In sum, simulation is more powerful when the experience that is currently top of mind is more similar to the target. It is sharper for DS experiences, it carries to NDS experience that are congruent with the current affective state, but it eventually becomes insensitive to the features of a very different experience that differ both semantically and affectively from the cue. Realistically, simulation is also less effective if the NDS comes from a domain c that is very different from X, captured by a flatter h_X, as we formally and empirically show in Bordalo et al. (2026). Our predictions are strengthened if we allow for this property.

Based on recall and simulation, the future state in X imagined on average across many memory samples by an agent facing a cue $q_{X,it}$ and database $\mathbf{f}_i$ is equal to:

$$\mathbb{E}_{it}\left(s_X\right) = \sum_{e:\sigma_X(e)\in T_X} r_e\left(q_{X,it}, \mathbf{f}_i\right) \cdot \sigma_X(e) \tag{5}$$

The probability associated to the target event T_X then increases in such average imagination. Imagining higher future stock, prices, house prices, or a longer life leads to a higher assessed probability of positive stock or house price growth and of living for more than ten extra years.[11]

To study the implications of memory we linearize $S(e,q)= \mathcal{S}\left[a \cdot d(e,q)\right]$ at the zero attention (and hence constant similarity) benchmark $a = 0$, which gives the following result.

Proposition 1 *Let $|\mathbf{f}_{it}|$ be the total frequency in the database of agent i at time t and $d\left(q_{X,it}\right)$ the average discrepancy of $e \in E$ from the cue $q_{X,it}$ in the database. Average simulation of X obeys:*

$$\mathbb{E}\left(s_X|q_{X,it},\mathbf{f}_{it}\right) \approx \sum_e \frac{f_{e,it}}{|\mathbf{f}_{it}|} \cdot \sigma_X(e) + a_i \cdot \sum_e \frac{f_{e,it}}{|\mathbf{f}_{it}|} \cdot \left[d\left(q_{X,it}\right) - d\left(e,q_{X,it}\right)\right] \cdot \sigma_X(e) \tag{6}$$

The first term on the right-hand side says that the belief is anchored to average simulation in the database: a person is more optimistic about X (e.g. stock returns), if she had more positive DS experiences (high stock returns) or more NDS experiences that prompt stock market optimism (e.g. positive personal financial experiences) compared to an otherwise identical person.

The second term captures our key mechanism: higher RF a_i increases the sensitivity of beliefs to experiences that share more features with the cue than average. When thinking about a domain, say the stock market, high RF reduces interference from many neutral NDS experiences whose features are incongruent with the cue. As a result, it causes stock market beliefs to be more sensitive to context-congruent stock market experiences or valence-congruent NDS ones.

We now derive specific predictions by on the role of DS and NDS experiences by as-

[11] Results would change little if the assessed probability of the target event is modelled as the probability of recalling experiences that allow to simulate it, $\sum_{e:\sigma_X(e)\in T_X} r_e\left(q_{X,it},\mathbf{f}_i\right)$, but Equation (5) is more tractable.

suming the following tractable discrepancy function:

$$d\left(e, q_{X,it}\right) = \begin{cases} \delta \cdot \left(s_{it} - s\right)^2 & if \quad c = X \\ K & if \quad c \neq X,\ v \subseteq v_{it} \\ K + V & else \end{cases} \tag{7}$$

where $\delta, K, V > 0$ and δ is sufficiently low that $\delta \cdot \left(s_{it} - s\right)^2 < K$ for realistic signal values. If experience $e = (s, v, c)$ is DS, $c = X$, its discrepancy increases in the distance between its signal s and the signal in the cue s_{it}. When thinking about the stock market during a 10% return period, a 3% stock return experience is more discrepant than a 9% one. The strength of this effect increases in coefficient δ, that can capture the salience of s_{it}.

An NDS experience exhibits a discrepancy $K > 0$ due to context mismatch if it is valence congruent and an additional discrepancy $V > 0$ if it is also valence-incongruent. When thinking about stocks in a good mood ($v_{it} = \mathbb{R}_{++}$) positive valence NDS experiences exhibit lower discrepancy than negative valence ones. Both NDS experiences are however more discrepant than a DS experience due to context mismatch.

We now derive three key predictions linking RF to beliefs via selective memory. The first prediction concerns the link between RF and the prevalence of agnostic beliefs.

Prediction 1 (RF *and agnosticity*) *Let* $f_{n,it}$ *be the frequency of neutral experiences* ($v = 0$)*. The probability of an agnostic default is* $\frac{f_{n,it}}{|\mathbf{f}_{it}|}\left\{1 + a_i \cdot \left[d\left(q_{X,it}\right) - K - V\right]\right\}$*, which decreases in* a_i*.*

Due to higher attention to features a_i, high RF people are less likely to recall irrelevant (context-incongruent and valence neutral) experiences, reducing the use of the agnostic default. This is a key mechanism: higher RF fosters active simulation through less interference.

Prediction 2 (DS, RF *and beliefs*) *Let* $\overline{s}_{it}$ *be the average DS signal in the database.*

Upon a linear approximation of $h_1(s)$ in (4), there are positive constants $\eta_0, \eta_1, \eta_2, \eta_3$ such that beliefs obey:

$$\mathbb{E}\left(s_X | q_{X,it}, \mathbf{f}_{it}\right) \approx \eta_0 + \eta_1 \cdot \bar{s}_{it} + \eta_2 \cdot a_i \cdot \bar{s}_{it} - \eta_3 \cdot a_i \cdot cov\left[s, (s - s_{it})^2\right]. \tag{8}$$

The agent imagines a higher future state if she experienced a higher average state $\bar{s}_{it}$ in her DS experiences, $\eta_1 > 0$. This "experience effect" is entirely driven by frequency-based recall from the database. Due to similarity-based recall, though, there are two additional effects. First $\eta_2 > 0$ says that the marginal impact of $\bar{s}_{it}$ is stronger for a person with higher RF: when cued to think about, say, the stock market this person more sharply recalls DS stock market experiences, thus being more sensitive to their average value $\bar{s}_{it}$ in the database.

Second, among DS experiences, $\eta_3 > 0$ implies that higher RF fosters recall of signals s more congruent with the current signal s_{it}. This boosts beliefs if the current signal is closer to high past ones and more distant from low past ones, namely $cov\left[s, (s - s_{it})^2\right]$ is lower. Upon seeing a 10% stock return, a person is more likely to recall high experienced returns if these are close to 10%, in which case her beliefs are also more optimistic, the more so the higher is RF. Evidence for this form of cued recall has been found by Gennaioli et al. (2025) who show that the inflation expectations of US households are more sensitive to inflation levels that are close to past experiences, as captured by a negative covariance coefficient. Our key new prediction here is that this effect should be stronger for people with higher RF, who better recall the similar past.

In sum, by creating a sharper similarity gradient, higher attention and hence higher RF increases the sensitivity of beliefs to the DS database $\bar{s}_{it}$ compared to the NDS database, and increases – within the DS database – the sensitivity to experiences similar to the current signal. We now show a third role of RF: its effect across different NDS experiences.

Prediction 3 *Suppose that NDS experiences are frequent enough relative to DS ones. Then, there is an $\eta_4 > 0$ such that having lived a v-congruent NDS experience affects beliefs as* *follows:*

$$\left.\frac{\partial \mathbb{E}\left(s_X|q_{X,it},\mathbf{f}_{it}\right)}{\partial f_{e,it}}\right|_{v\subseteq v_{it}} \approx \frac{1}{|\mathbf{f}_i|}\cdot h_X(v)+\eta_4\cdot a_i\cdot h_X(v) \tag{9}$$

Having lived a valence congruent NDS fosters its recall and simulation, increasing the forecast if this NDS is "belief-boosting" $h_X(v)>0$ and decreasing them otherwise. Critically, if NDS experiences are many compared to DS ones, this effect increases with high RF. In this case, in fact, interference hinders recall of all experiences sharing any feature with the cue, including an affective feature. As a result, reducing interference via higher a_i increases the NDS effect. Thus, RF should amplify the baseline NDS effect (where the latter however can be small in a large database, when $1/|\mathbf{f}_i|$ is small).

In sum, a selective memory model of belief formation predicts a systematic relationship between RF and belief formation across domains that can be tested by studying how RF is connected to the sensitivity of beliefs to different DS and NDS experiences. These effects are not consistent with omitted variables explanations for the role of RF in Section 2.

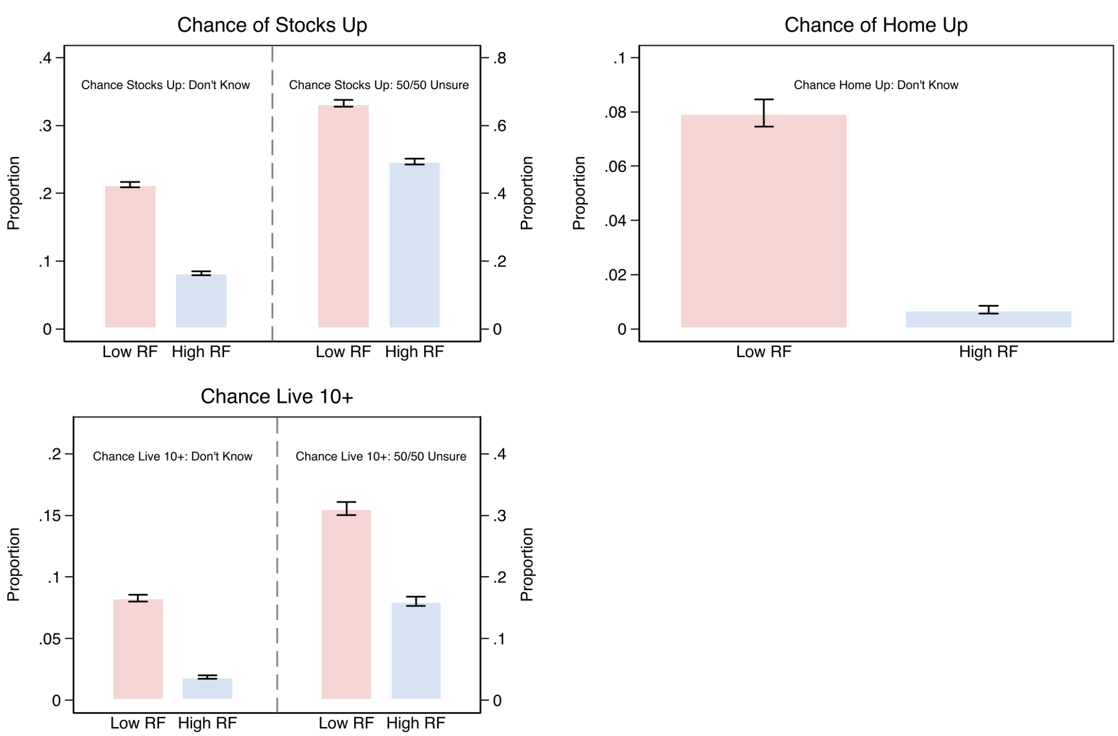
A mechanism whereby high RF proxies for better information cannot account for a heightened role of uninformative NDS experiences (but also for a stronger effect of personal DS experiences, because arguably a more informed person should also know more beyond her past experiences). Preference based explanations, according to which high RF proxies for motivated beliefs cannot explain the asymmetric role of RF in amplifying the reaction to positive and negative news for people having DS databases (the stronger impact of $cov\left[s,(s-s_{it})^2\right]\gtrless 0$). We now test the model's predictions across our three belief domains.

3.4 Agnosticity by Memory

We now test Prediction 1. Low RF people should be more likely to give agnostic answers than high RF people. This should be true in all belief domains, and arises because low RF people face more interference from neutral NDS and more generally from NDS that are not useful for simulation. Thus, they more often fail to imagine the future and hence report agnostic answers, consistent with Hassabis Shachter et al (cite) role of memory in imagination.

We measure agnostic answers in two ways: i) share of people answering by "I do not know", and ii) share of people answering with "50-50" *and* stating that they are highly uncertain of their answer. In both cases, respondents reveal to be unsure on how to think about the future. Figure 7 shows that prediction 1 strongly holds in the data: people in the bottom RF quartile are more likely to give agnostic answers than people in the top RF quartile.

Figure 7: Agnostic Beliefs By Recall Fluency Group.



Notes: Each panel reports, within one belief domain, the share of respondents giving an agnostic answer. The left cluster of bars reports the share of respondents who answer "Don't Know" and is read off the left-hand y-axis. The right cluster reports, among respondents who answer 50/50, the share who further state they are "unsure" (as opposed to "equally likely") and is read off the right-hand y-axis. The two clusters are separated by a dashed vertical line. Axis scales differ across panels. In the housing market there is no follow-up question to respondents who initially answer 50/50, so only the Don't Know cluster is shown for housing. The three panels correspond to stock beliefs (top-left), housing beliefs (top-right), and health beliefs (bottom-left). Red bars are the bottom recall-fluency quartile (low RF), blue bars are the top quartile (high RF). Error bars are 95% confidence intervals.

Consistent with the model, we see greater prevalence of agnostic answers for low RF people. Agnostic answers are typically explained with noise: people receive a signal, are aware of its limited informativeness, and thus shrink the answer toward the prior, exhibiting low confidence. Memory explains agnostic answer by nothing coming to mind. It can potentially imply that people do not bother to shrink, but rather report "I do not know" or anchor their estimate to a salient albeit arbitrary responses. This could also explain why answers of low memory people appear more dispersed and take extreme values in Figure 3, Figure 4 and Figure 5.

One potential concern is that low RF people may just be less willing exert effort, both in laboratory recall tasks, where they then recall fewer words, and in forecast elicitations, where they then give agnostic low effort answers. We now test the more specific Predictions 2 and 3, which rely on the precise memory mechanism, linking past experiences and beliefs. We will also see that memory and the report beliefs it entails shape actual economic choices, which shows that memory helps predict real world behavior, is not just a proxy for survey attention.

3.5 Domain Specific Experiences

To test Prediction 2 on the role of DS experiences we estimate Equation (8) in our three domains, stock market, housing, and life expectancy. The nature of domain specific experiences varies but the memory structure of Equation (8) and the role of recall fluency is common; The estimated probability $p_{X,it}$ in domain $X = SP, HP, LE$ by respondent i at time t should exhibit:

$$p_{X,it} \approx \gamma_0 + \gamma_1 \cdot \overline{s}_{it} + \gamma_2 \cdot a_i \cdot \overline{s}_{it} - \gamma_3 \cdot a_i \cdot cov\left[s, (s - s_{it})^2\right] \tag{10}$$

where all coefficients are positive. What varies across domains is the computation of DS regressors. In the stock market, $\overline{s}_{it}$ is the average yearly return experienced by the respondent in the past, the observable cue s_{it} is the recent stock return that triggers selective retrieval of similar past returns s encoded in the same respondent's DS stock market memories.

In housing, $\overline{s}_{it}$ is the average house price growth experienced by the respondent in the past, the observable cue s_{it} is the recent house price growth that triggers selective retrieval of similar past house price growth s encoded in the respondent's DS housing market memories.

In life expectancy, $\bar{s}_{it}$ is the average health state experienced by the respondent in the past, the observable cue s_{it} is the recent health state that triggers selective retrieval of similar past health states s encoded in the respondent's DS health memories.

To compute these proxies, for each respondent we construct a memory database of: 12 month stock returns and house price growth at the quarterly frequency since the date of birth, and the measured change in the personal health PCA in each HRS wave.[12] For the latter we have relatively few observations, equal to the number of waves a respondent participates to. We however measure health DS also by childhood health, which we include as a regressor. In stock market and house price regressions we can directly control for long term health outcomes using the stock or housing returns experiences throughout one's life.

Table 1 reports, for each domain, the estimation of the model Equation (10) in columns 1,3, 5. In the columns 2, 4 and 6 we add RF as a control to check that RF does not merely capture the level effect on beliefs but rather a sensitivity to past DS experiences. Table C.5 shows that the results are robust to adding survey-year fixed effects, ruling out common aggregate shocks as the driver. Table C.4 shows that the results hold when we additionally interact IQ with experiences, ruling out the possibility that effects are driven by respondents having different information sets.

The Table reveals the same general structure for beliefs in different domains, and the results strongly support the role of memory. In each domain: beliefs increase in "positive past DS", but especially so for high RF people. Cues also matter, as shown by the negative coefficient on the covariance term. The same news exhibits a stronger effect for people who have high RF and live through similar DS experiences. Similarity matters and works through RF. As a robustness check, Figure C.15–Figure C.17 report the same specification estimated separately within each RF quartile; the monotonically increasing interaction coefficients confirm that the amplification effect is not driven by the linear functional form assumption in Table 1.

[12] The results do not change if, following Gennaioli et al (2025) we use experiences starting at age 16. See Appendix Table AX.

Table 1: Database Regressions For $\overline{s_{it}}$ and $cov(s_k, (s_k - r_t)^2)$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance House ↑**		**Chance Live10+**	
Cognition						
RF	.	0.002	.	-0.013	.	-0.003
	.	[0.005]	.	[0.010]	.	[0.003]
IQ	0.025***	0.025***	0.016***	0.016***	-0.019***	-0.019***
	[0.002]	[0.002]	[0.003]	[0.003]	[0.002]	[0.002]
DS expiences						
$\overline{s_{it}}$ (γ_1)	0.000	0.001	0.027***	0.021***	0.029***	0.028***
	[0.001]	[0.003]	[0.002]	[0.006]	[0.001]	[0.002]
RF Interactions						
$RF * \overline{s_{it}}$ (γ_2)	0.005***	0.004***	0.004***	0.008***	0.007***	0.007***
	[0.000]	[0.001]	[0.001]	[0.003]	[0.000]	[0.001]
$RF * cov(s, (s - r_t)^2)$ (γ_3)	-0.004***	-0.004***	-0.030***	-0.030***	-0.001***	-0.001***
	[0.000]	[0.000]	[0.001]	[0.001]	[0.000]	[0.000]
N	139345	139345	49972	49972	116262	116262
AR2	0.04	0.04	0.05	0.05	0.11	0.11
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

Notes: Each column reports a panel regression of subjective beliefs on Recall Fluency (RF), IQ, direct-specific (DS) experience variables ($\overline{s_{it}}$, $(s_k, (s_k - r_t)^2)$), and their interactions with RF as shown in Equation 10. Demographic controls include age, gender, marital status, retirement status, income, and education. Columns 1–2 use stock market beliefs; columns 3–4 housing beliefs; columns 5–6 health beliefs. Columns 2, 4, 6 additionally include the RF base effect. Standard errors in brackets; * p<0.12, ** p<0.05, *** p<0.01.

To quantify economic magnitudes, Figure C.6, Figure C.7, and Figure C.8 report, for each domain, the effect of moving a single regressor from its 25th to its 75th percentile—holding the remaining two regressors at their respective sample means—expressed as a percentage of the dependent variable's sample mean.[13]

[13] Effects are computed as marginal effects from columns 2, 4, and 6 of Table 1, evaluated at the sample means of the conditioning regressors. All continuous regressors ($\overline{s}_{it}$, RF, IQ) are divided by their interquartile range prior to estimation, so that a one-unit shift in a standardised regressor corresponds exactly to a P25-to-P75 move in the original variable. The marginal effect of experiences at mean RF is $\hat{\gamma}_1 + \hat{\gamma}_2\,\overline{\mathrm{RF}}^*$, where $\hat{\gamma}_1$ and $\hat{\gamma}_2$ denote the estimated coefficients on $\overline{s}_{it}$ and RF $\times\ \overline{s}_{it}$ respectively, and $\overline{\mathrm{RF}}^*$ is the post-standardisation sample mean of recall fluency. Analogously, the marginal effect of RF at mean experiences is $\hat{\beta}_{\mathrm{RF}} + \hat{\gamma}_2\,\overline{s}^*_{it}$.

In stock market expectations the direct effect of that shift in past experiences, raising the subjective probability that stocks will go up by 1.9% of its mean (roughly 0.9 percentage points). An equivalent move along the recall-fluency distribution raises the same probability by 4.5% of its mean—more than twice as large. In line with the model's, the experiential record plays a larger role if it is more likely to be remembered. For housing market and health expectations, which concern the respondent's house price and health, more variation in beliefs reflects own experiences, but the role of memory is still large, corresponding to an 0.7pp and 1.8pp (approximately 1.3% and 3.7% of the respective means) increase in expectations in the two domains.[14]

Taken together, Figure C.6–Figure C.8 show that, despite different baseline magnitudes, recall fluency consistently amplifies the transmission from past experiences to forward-looking beliefs. The share of the combined Exp+RF effect attributable to the RF channel ranges from roughly 15% in housing to more than 70% in stocks, underscoring that heterogeneity in memory capacity is a quantitatively significant source of heterogeneity in beliefs across individuals. To verify that these magnitudes are not an artifact of the linear interaction specification, Figure C.15–Figure C.17 re-estimate Equation (8) non-parametrically within each RF quartile: the experience coefficients rise monotonically from the bottom to the top quartile in all three domains, confirming that even within memory quartiles, recall fluency amplifies the reaction to past experiences.

To benchmark these magnitudes against more familiar covariates, Figure C.9–Figure C.11 place the recall-fluency effect on the same percentage-point scale as the demographic controls in Table 1. In the stock-market regression, an interquartile shift in recall fluency raises the subjective probability that stocks will go up by approximately 2.1 percentage points—essentially identical to the effect of an interquartile shift in education ($\approx$ 2.1 percentage points over its 3-year IQR) and over five times the effect of an interquartile shift in household income ($\approx$ 0.4 percentage points over its $49K IQR). Gender—typically the largest demographic effect in these regressions—moves the same belief by roughly 4.9 percentage points (females relative

[14]To discuss: the covariance term is very strong for housing.

to males). The same ordering carries through to the housing and survival regressions: recall fluency is comparable in magnitude to education, dominates income, and is exceeded only by gender (and, in housing, retirement). Recall fluency thus operates on a magnitude scale familiar from the standard belief-formation covariates.

We conclude this Section with two exercises. First, we check whether the same role of memory arises in expectations over two other aggregate economic conditions: the probability of a major economic depression within the next ten years (available in four waves, 2002-2008), and the probability that inflation will exceed five percent over the next ten years (available in two waves only, 2006-2008). While our survey has much less data on these, the results can shed light on the role of memory on the many datasets on such expectations.

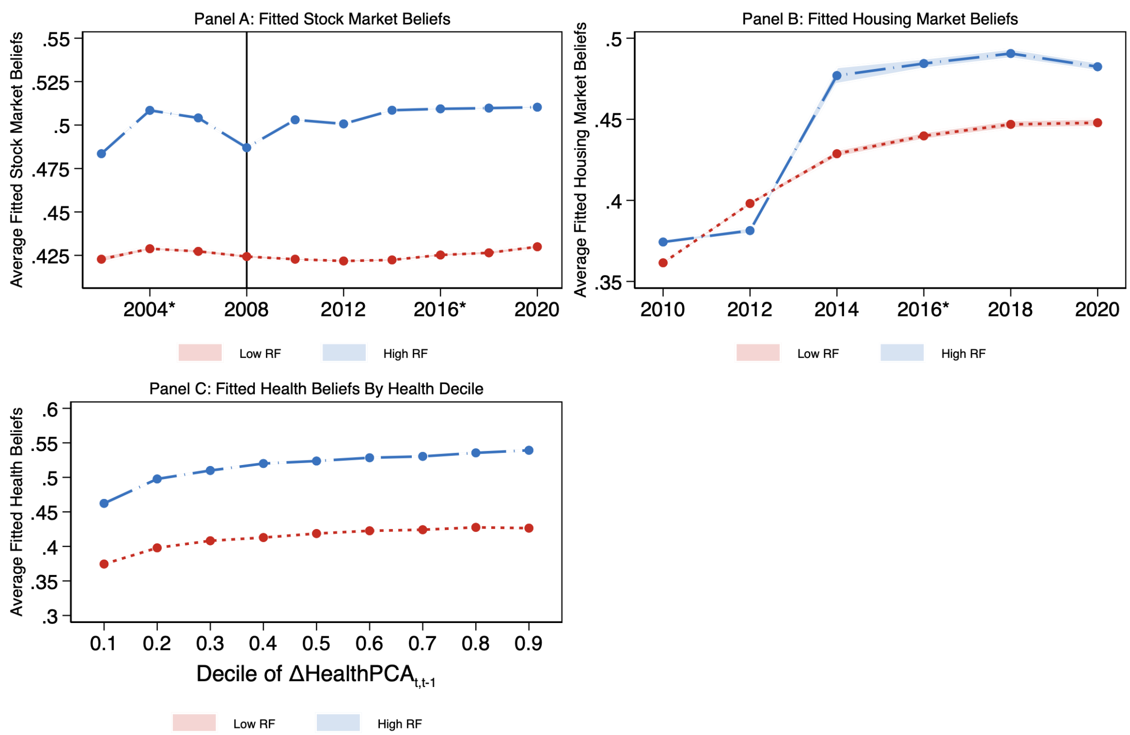
Table C.12 re-estimates Equation (8) pairing beliefs about a depression with lifetime NBER recession experiences, and beliefs about high inflation with lifetime inflation experience.[15] In both cases the interaction between recall fluency and past experiences enters with the predicted sign, and the cue-similarity term $RF \times cov(s,(s-r_t)^2)$ is sharply negative, replicating the core memory mechanism documented in Table 1.

In our second exercise, we assess the ability of the model in Equation 10 to capture the main qualitative patterns of expectations documented in the previous Section. Figure 8 reports model-implied beliefs for the top and bottom recall-fluency quartile across all three domains. In the stock market (Panel A), high-RF fitted beliefs exhibit a pronounced boom-bust cycle mirroring the raw evidence in Figure 3: they rise to a peak of 50.8 percent in 2004, fall by 2.1 percentage points to 48.7 percent in the wake of the 2008 financial crisis, and subsequently recover by 2.3 percentage points through 2020. Low-RF fitted beliefs, by contrast, decline by just 0.5 percentage points between 2004 and 2008 and recover by a comparable margin

[15] The depression belief is the HRS Section P item P034: respondents report the percent chance that the U.S. economy will experience a major depression sometime during the next ten years or so. The inflation belief is HRS item P116: the percent chance that the cost of living will rise by more than five percent per year on average over the next ten years; the verbatim HRS question wording for both items is reproduced in Table B.2. The paired DS experience $\overline{s}_{it}$ is, for the depression regression, the share of calendar years from birth through the survey quarter flagged as NBER recession years, and for the inflation regression, the respondent's average experienced quarterly CPI inflation rate over the same window; the accompanying covariance term $cov(s,(s-r_t)^2)$ is constructed analogously to the stock-return and house-price-growth databases of Table 1.

thereafter—consistent with the sharply attenuated boom-bust pattern in the raw data. In housing (Panel B), high-RF fitted beliefs for home price appreciation rise by 11.6 percentage points from 2010 to their 2018 peak (from 37.4 to 49.1 percent), while low-RF fitted beliefs rise by 8.5 percentage points over the same period (from 36.1 to 44.7 percent). The larger and faster post-crisis recovery for high-RF respondents mirrors the differential documented in Figure 4, and reflects the selective retrieval of an accumulating stock of positive housing-market memories as prices rebounded sharply between 2012 and 2018.

Figure 8: Fitted Beliefs Over Time for High vs. Low Recall Fluency.



Notes: Each panel shows fitted beliefs from database regressions for the bottom (red) and top (blue) quartile of Recall Fluency. Panel A shows fitted stock market beliefs over time from a regression of Equation 10 (column 1 of Table 1). Panel B shows fitted housing market beliefs over time from an analogous regression using HPI moments (Equation 10, column 3 of Table 1) . Panel C shows fitted health beliefs by decile of $\Delta\text{HealthPCA}_{t,t-1}$ from a regression using health experience moments (Equation 10, column 5 of Table 1). Shaded bands are 95% confidence intervals.

Panel C presents fitted health beliefs across deciles of $\Delta\text{HealthPCA}_{t,t-1}$. As in Figure 5, the fitted gradient is steeper for high-RF respondents: survival expectations rise by 9.2 percentage points from the bottom to the top health-improvement decile for high-RF individuals (from 45.0 to 54.1 percent), compared with 6.1 percentage points for their low-RF counterparts (from 36.5 to 42.6 percent). A level gap of approximately 8 to 11 percentage points in favour of high-RF respondents is preserved across the full health distribution. Taken together, the three panels confirm that the recall-fluency amplification mechanism is not an artifact of the pooled regressions: the model generates belief dynamics—a steeper boom-bust cycle in stocks, a sharper housing recovery, and a more responsive health belief gradient—that closely track the empirical patterns reported in Section 2.

3.6 Non Domain Specific Experiences

A distinct implication of memory is that beliefs may be shaped by experiences that are not domain specific, see Prediction 3. To test this prediction we add, for each domain, proxies for NDS and their interaction with RF. For stocks and housing we use childhood health and childhood socioeconomic status. These are remote and entirely personal experiences which have no relevance and contain no information for stock and housing markets. For life expectancy we use experienced stock and housing returns, which are also from a different domain and broadly not informative about health.[16] These proxies enter both in levels and interacted with RF, augmenting Equation 10 as:

$$p_{X,it} \approx \gamma_0 + \gamma_1 \cdot \overline{s}_{it} + \gamma_2 \cdot a_i \cdot \overline{s}_{it} - \gamma_3 \cdot a_i \cdot \text{cov}\Big[s, (s - s_{it})^2\Big] + \delta_1 \cdot nds_{it} + \delta_2 \cdot a_i \cdot nds_{it} + \varepsilon_{it} \tag{11}$$

Here δ_1 captures the direct effect of NDS on beliefs at the bottom percentile of RF (no memory), and δ_2 captures memory's amplification of those experiences. Since all regressors are IQR-normalised, each coefficient directly quantifies the belief change for a 25th-to-75th percentile shift in the corresponding variable.

[16] Discuss wealth effects.

Table 2: NDS Database Regressions For $\overline{s_{it}}$ and $cov(s_k, (s_k - r_t)^2)$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance House ↑**		**Chance Live 10+**	
IQ	0.025***	0.024***	0.016**	0.015***	-0.018***	-0.021***
	[0.002]	[0.002]	[0.006]	[0.003]	[0.002]	[0.002]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	0.001	0.001	0.027***	0.027***	0.028***	0.027***
	[0.001]	[0.001]	[0.007]	[0.002]	[0.002]	[0.002]
RF Interactions						
$RF * \overline{s_{it}}$ (γ_2)	0.004***	0.004***	0.004***	0.004***	0.007***	0.008***
	[0.000]	[0.000]	[0.001]	[0.001]	[0.001]	[0.001]
$RF * cov(s_k, (s_k - r_t)^2)$ (γ_3)	-0.003***	-0.004***	-0.030***	-0.030***	-0.001***	-0.001***
	[0.000]	[0.000]	[0.008]	[0.001]	[0.000]	[0.000]
NDS Experiences						
nds (δ_1)	-0.001	-0.003	-0.003	-0.007	0.002	0.001
	[0.001]	[0.002]	[0.002]	[0.005]	[0.003]	[0.001]
$RF * nds$ (δ_2)	0.001**	0.005***	0.002*	0.006**	0.008***	0.005***
	[0.001]	[0.001]	[0.001]	[0.003]	[0.001]	[0.000]
N	134136	139342	49593	49971	116262	100979
AR2	0.04	0.04	0.05	0.05	0.11	0.12
NDS Var	Chld.Health	Chld.SES	Chld.Health	Chld.SES	$\Delta_L sp500$	$\Delta_L hpi$
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

Notes: Each column reports a panel regression of subjective beliefs on Recall Fluency (RF), IQ, domain-specific (DS) experience variables ($\overline{s_{it}}$ and $\text{cov}(s_k, (s_k - r_t)^2)$), non-domain-specific (NDS) experiences, and their interactions with RF. NDS variables differ by domain: childhood health and SES index for stocks and housing; lifetime equity and housing returns for health. All continuous regressors (RF, IQ, $\overline{s_{it}}$, cov, NDS) are divided by their within-sample IQR prior to estimation, so each coefficient represents the change in beliefs associated with a move from the 25th to the 75th percentile of the regressor. Demographic controls include age, gender, marital status, retirement status, income, and education. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table 2 reports estimates of Equation 11. In all domains NDS matter—but almost exclusively for higher-RF individuals. Across all six columns the base NDS coefficient $\hat{\delta}_1$ is small and statistically indistinguishable from zero, while the memory-amplification term $\hat{\delta}_2$ is positive and significant throughout. NDS experiences do not shift beliefs for individuals near the bottom of the RF distribution; they become relevant only as recall fluency rises. The fact that NDS matters only through the RF interaction is consistent with the memory mechanism: these experiences are normatively irrelevant, so they can only affect beliefs by being recalled and entered into mental simulation. This finding also suggests that high-RF individuals are not simply better informed—they use memory more, including memory of events that are objectively uninformative about the domain in question. These results are robust to controlling for IQ-driven information differences (Table C.6) and to absorbing common time trends via survey-year fixed effects, as well as including the RF base effect (Table C.7).

Since all regressors are IQR-normalised, each coefficient directly quantifies the belief change for a 25th-to-75th percentile shift. Evaluated at the sample mean of RF, a one-IQR shift in NDS shifts stock-market beliefs by approximately 0.5 percentage points (childhood SES, column 2)[17] and health beliefs by 1.6 percentage points (lifetime equity returns, column 5). The corresponding DS experience effect, from the same regression evaluated at mean RF, is 0.9 percentage points for stocks and 4.1 percentage points for health, so the NDS channel represents roughly 40–60 percent of the DS channel in these two domains. In the housing market, by contrast, the NDS channel is substantially weaker: a one-IQR shift in childhood SES raises the subjective probability of house price appreciation by only 0.3 percentage points (column 4), and childhood health is negligible at 0.1 percentage points (column 3)—each less than one-tenth of the corresponding DS experience effect of 3.5 percentage points (we may want to try to elaborate on this). Figure C.12–Figure C.14 illustrate these magnitudes as a share of the dependent-variable mean and further decompose the heterogeneity generated by NDS databases relative to RF itself.

4 Recall Fluency, Expectations and Choices

Does memory affect economic choice by shaping, through recall fluency, the role of DS and NDS experiences? To address this question, we first correlate beliefs with choice. Next, we perform a two-stage exercise assessing the role of RF and experiences. In the first stage, we

[17] The marginal effect of NDS at mean RF is $\hat{\delta}_1 + \hat{\delta}_2\,\overline{\mathrm{RF}}^*$, where $\hat{\delta}_1$ and $\hat{\delta}_2$ denote the estimated coefficients on NDS_{it} and $\mathrm{RF} \times \mathrm{NDS}_{it}$ respectively, and $\overline{\mathrm{RF}}^* \approx 1.76$ is the post-standardisation sample mean of recall fluency, corresponding to an average respondent with raw recall-fluency index of zero. Analogously, the marginal effect of DS experiences at mean RF is $\hat{\gamma}_1 + \hat{\gamma}_2\,\overline{\mathrm{RF}}^*$.

regress a respondent's choice on a proxy for beliefs about a variable relevant to that choice:

$$C_{it} = \propto + \eta^{M} RF_{it} + \eta^{I} IQ_{it} + \theta B_{it} + \beta X_{it} + \varepsilon_{it} \tag{12}$$

where C_{it} is a choice recorded for respondent i at time t (such as new stock market investments), and B_{it} is the measured choice-relevant belief (such as belief about stock returns) of the same subject and the same time.

In the next step, we study whether, by shaping beliefs, baseline experiences and recall fluency matter for choice. To do so, we take the belief specifications of Table 2 (estimated from Equation 11) and we separate fitted beliefs into four components: $\widehat{B_{it}} = B(E_{it}) + B(RF_{it} \cdot E_{it}) + B(RF_{it} \cdot \text{Cov}_{it}) + B(Dems_{it})$. The first term $B(E_{it})$ is a pure "Experience" component, constructed by multiplying the respondent's experience of both DS and NDS by their respective estimated coefficients from Table Table 2.[18].

The second term, $B(RF_{it} \cdot E_{it})$, is an "Amplification" effect, constructed by multiplying the respondent's level $RF_{it} \cdot E_{it}$ by its estimated coefficient from Table 2. This generates the part of beliefs that are predicted by memory's amplification of experiences. Third is the "Reminder" component, $B(RF_{it} \cdot \text{Cov}_{it}) = -\widehat{\gamma}_3 \cdot a_i \cdot \text{cov}\left[s, (s - s_{it})^2\right]$, constructed by multiplying the respondent's RF-weighted cue-similarity term from Stage 1 by its estimated coefficient. This generates the part of beliefs predicted by memory's amplification of the within-period reminder, i.e. the degree to which recent realizations resemble the respondent's lifetime experience distribution. Finally, we take the remaining fitted belief, $B(Dems_{it}) = \widehat{\rho}' \cdot Dems_{it} + \widehat{\eta^{I}} \cdot IQ_{it}$, as the portion of beliefs predicted from demographic variables and IQ. Using all four components, we then estimate the following specification:

$$C_{it} = \propto + \eta^{M} RF_{it} + \eta^{I} IQ_{it} + \gamma B(E_{it}) + \mu B(RF_{it} \cdot E_{it}) + \nu B(RF_{it} \cdot \text{Cov}_{it}) + \rho B(Dems_{it}) + \beta X_{it} + \varepsilon_{it} \tag{13}$$

[18]This amounts to $B(E_{it}) = \widehat{\gamma}_1 \cdot \overline{s}_{it} + \widehat{\delta_{1,health}} \cdot ChildHealth + \widehat{\delta_{1,\ ses}} \cdot ChildSES$. Instead of estimating the impact of childhood health and childhood SES separately as we do in Table 2, we estimate $\widehat{\delta_{1,health}}$ and $\widehat{\delta_{1,ses}}$ jointly in regression that includes both regressors

where $\gamma > 0$ means that greater exposure to experiences fostering the choice relevant belief B_{it} fosters the choice itself; $\mu > 0$ means that this is especially true for people having high recall fluency; and $\nu > 0$ means that RF-driven amplification of the within-period reminder (the similarity between recent realizations and the respondent's lifetime experience distribution) also feeds through to behavior. The experiences in question are both DS and NDS. We measure choice and choice relevant beliefs as follows.

Stock Market. Choice variables are: *New Stock Market Investments 2y*, a dummy capturing whether the respondent has made any new stock market investments in the last 2 years; *% IRA in Stocks* measuring the proportion of the respondents IRA that is invested in stocks/mutual funds. This variable is restricted to people that own an IRA. The belief B_{it} relevant for these is the assessed probability of stocks going up, which should fosters both measures of stock investment.

Housing. The choice variable is *Bought New Home 2y*, a dummy taking value one if the respondent bought a new home in the past two years. The relevant belief measure is the probability of the respondent's own home increasing in value, which should foster this choice measure.

Life Expectancy. Choice variables include: *Chance Work Past 70*, a variable from 0 to 100 indicating the believed probability for the respondent to work after age 70 (higher value higher probability, the variable is restricted to people who are currently working and younger than 70). The second choice proxy referring to work decisions is *Retired*, a dummy taking value one if the respondent is currently retired. Importantly, in regressions where the choice variable and belief are health related (chance work past 70, retired, % income to pension), we include age fixed effects such as to only make within-age comparisons. The belief measure relevant to these choices is the probability attached to being alive in the next ten years, which should be associated with higher believed probability of working after 70 and to a lower probability of being currently retired. In these regressions we control for age, to avoid that effects are mechanically related to it. A third choice measure in this domain is *% Income to Pension*, namely the percentage of monthly pay the respondent contributes to their pension (restricted to people who are currently working and have a defined contribution pension scheme). Again, the relevant belief variable here is the probability of being alive in the next ten years, which should foster this choice.

In these regressions we control for a range of individual characteristics controlling for a person socioeconomic and family status, which may affect choices independently of beliefs. Specifically, we include age, education, gender, income, and married/not.

Table 3 reports, for each choice, the estimates of specification (3) in the first column and specification (4) in the second column. In this latter specification we also include a preference proxy, namely the self-reported degree of risk aversion.[19] In economics, risk aversion is considered a key driver of interpersonal differences in behavior, so including this proxy allows us to benchmark the explanatory role of memory against standard preference measures.

[19]To measure risk aversion, we take the standardized average of 3 proxies: (a) risk taking for employment, which asks the respondent if they would rather a job with higher expected pay but more variance vs. one with lower pay and more certainty; (b) risk taking for business inheritance, which asks whether they would put an inheritance into a safe asset or invest in a risky business; and (c) self-reported risk taking, measured on a scale of 1 to 5.

Table 3: Choice Regressions: Beliefs, Memory, and Economic Choices.

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Stock/Housing Market Choices					
	New Stock Market Inv.		**% IRA in Stocks**		**Bought a New Home**	
IQ (1 sd)	0.009***	-0.002	0.030***	0.008**	0.006***	-0.002
	[0.001]	[0.001]	[0.004]	[0.004]	[0.001]	[0.002]
Beliefs (ϕ)						
$\widehat{B}_{it}$ (θ)	0.052***	.	0.067***	.	0.008***	.
	[0.003]	.	[0.006]	.	[0.001]	.
$\widehat{B(E)}$ (γ)	.	-0.009***	.	-0.014***	.	0.002**
	.	[0.001]	.	[0.003]	.	[0.001]
$B(\widehat{RF \times E})$ (μ)	.	0.019***	.	0.020***	.	0.005***
	.	[0.001]	.	[0.003]	.	[0.001]
$B(\widehat{RF \times \text{Cov}})$ (ν)	.	-0.002**	.	-0.001	.	0.006***
	.	[0.001]	.	[0.002]	.	[0.001]
Risk Taking						
Risk Taking (1 sd)	.	0.004***	.	0.029***	.	0.004***
	.	[0.001]	.	[0.003]	.	[0.001]
Avg DV	0.10	0.10	0.56	0.56	0.04	0.04
N	129529	120357	36078	34263	45249	43865
Demographic Controls	Y	Y	Y	Y	Y	Y
	Panel B: Health Choices					
	Chance Alive Past 70		**Retired**		**% of Income to Pension**	
IQ (1 sd)	0.018***	0.014***	-0.006***	-0.003*	0.002	0.002
	[0.003]	[0.003]	[0.001]	[0.001]	[0.002]	[0.001]
Beliefs (ϕ)						
$\widehat{B}_{it}$ (θ)	0.035***	.	-0.010***	.	0.003	.
	[0.004]	.	[0.002]	.	[0.002]	.
$\widehat{B(E)}$ (γ)	.	0.013***	.	-0.020***	.	-0.002
	.	[0.004]	.	[0.002]	.	[0.002]
$B(\widehat{RF \times E})$ (μ)	.	0.018***	.	0.006***	.	0.004**
	.	[0.003]	.	[0.002]	.	[0.002]
$B(\widehat{RF \times \text{Cov}})$ (ν)	.	-0.001	.	0.002	.	-0.001
	.	[0.004]	.	[0.001]	.	[0.004]
Risk Taking						
Risk Taking (1 sd)	.	0.029***	.	0.002	.	0.001
	.	[0.003]	.	[0.001]	.	[0.002]
Avg DV	0.26	0.26	0.64	0.64	0.08	0.08
N	20280	20129	100979	91946	4352	4334
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort/Age FE	Y	Y	Y	Y	Y	Y

Notes: Two-stage regressions linking beliefs to economic choices, HRS 2002–2020. Stage 1 predicts beliefs from RF, IQ, domain-specific (DS) experiences and their interactions with RF, the RF × experience-covariance interaction (consistent with Table 1), and non-domain-specific (NDS) experiences (childhood health, childhood SES) and their interactions with RF, using the same specifications as Table 2. Stage 2 regresses each choice on components of Stage 1 fitted beliefs. Odd columns (1, 3, 5) use total predicted belief $\widehat{B}_{it}$ (θ). Even columns (2, 4, 6) decompose fitted beliefs into four additive components: $\widehat{B(E)}$ (experience-driven, γ); $B(\widehat{RF \times E})$ (memory amplification of past experiences, μ); $B(\widehat{RF \times \text{Cov}})$ (interaction of recall fluency with the domain-specific experience covariance, ν); and $B(\widehat{Dems})$ (controls-driven). A direct Risk Taking index is also included. Panel A outcomes: *New Stock Market Inv.*, an indicator equal to one if the respondent made a new stock market investment since the prior survey wave; *% IRA in Stocks*, the share of IRA assets held in stocks (0–1); *Bought a New Home*, an indicator for home purchase since the prior wave; person fixed effects. Panel B outcomes: *Chance Alive Past 70*, subjective probability of surviving past age 70 (0–100); *Retired*, an indicator for retirement status; *% of Income to Pension*, the share of income contributed to a pension plan (0–1); age and cohort fixed effects. Demographic controls (age, gender, marital status, income, education) in all columns. Coefficients scaled to 1 SD of each regressor. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table 3 reports the results. Across the three financial and health domains, a higher fitted belief raises the relevant choice probability in the odd columns (1, 3, and 5), with the exception of the pension-contribution outcome, where the total-belief coefficient is not statistically distinguishable from zero. Decomposing fitted beliefs into their four additive components (columns 2, 4, and 6), the memory-driven amplification of past experiences (μ) enters positively and significantly for every outcome, and for the stock-market and health outcomes it is the dominant belief-driven channel. The pure experience component (γ) and the reminder component (ν) have a heterogeneous role across domains.

Stock Market. A one-standard-deviation increase in fitted stock-price beliefs is associated with a 5.2 percentage-point (pp) rise in the probability of making a new stock market investment, a 52% increase relative to the 10% baseline investment rate. Decomposing these fitted beliefs, the memory-amplification of experiences contributes 1.9 pp (19% of the mean), while the pure experience component enters with the opposite sign at -0.9 pp (-9% of the mean) and the reminder-amplification component is marginally negative at -0.2 pp (-2%). A consistent pattern emerges for portfolio allocation: a one-SD increase in fitted beliefs raises the share of IRA assets held in stocks by 6.7 pp (12.0% relative to the 56% mean), with the amplification-of-experiences channel contributing 2.0 pp, the pure experience channel entering negatively at -1.4 pp, and the reminder channel statistically indistinguishable from zero. In both stock outcomes, the memory-amplification of past experiences is the channel through which fitted beliefs map into investment behavior; the pure experience channel, absent RF's amplification, does not.

Housing. In the housing domain, a one-SD increase in fitted house-price beliefs is associated with a 0.8 pp increase in the probability of buying a new home in the past two years, a 20% increase relative to the 4% baseline purchase rate. The pure experience component accounts for 0.2 pp (5.0% of the mean), the amplification-of-experiences component for 0.5 pp (12.5%), and the reminder-amplification component for 0.6 pp (15%)—the largest of the three channels, and the only outcome in the paper where RF's up-weighting of the within-period cue is both statistically and economically dominant. While the absolute magnitudes are modest, they are economically meaningful relative to the low base rate of home purchases among an older population.

Health and Retirement. Three choice outcomes are linked to survival beliefs. For the probability of expecting to work past age 70, a one-SD increase in fitted survival beliefs raises this outcome by 3.5 pp (13.5% of the 26% mean). Here the decomposition reverses the financial-domain ordering only partially: the pure experience channel accounts for 1.3 pp (5.0% of the mean), while the amplification-of-experiences component contributes 1.8 pp (6.9% of the mean) and is the larger of the two. The reminder-amplification channel is statistically indistinguishable from zero. Memory's amplification of past health exposure is thus the dominant driver of longer planning horizons, though direct experience also contributes meaningfully. For retirement status, higher survival beliefs predict a lower probability of being retired: a one-SD increase in fitted beliefs reduces the probability of being retired by 1.0 pp (1.6% of the 64% baseline), driven almost entirely by the pure experience channel at −2.0 pp, partially offset by a positive amplification-of-experiences coefficient of +0.6 pp. Finally, for the share of income contributed to a pension, the total-belief coefficient is not statistically distinguishable from zero, but the decomposition is informative: the amplification-of-experiences component enters positively and significantly at 0.4 pp (5% of the mean, $p < 0.05$), while neither the pure experience nor the reminder component is individually distinguishable from zero. The small sample for this outcome (approximately 4,300 observations) limits statistical power for the remaining components.

To address the possibility that RF affects choices directly—through channels such as general planning capacity or self-control, rather than through its role in shaping beliefs—Table C.10 in the appendix repeats the specification with RF included as a direct second-stage regressor. The main belief coefficients remain significant and of comparable magnitude, confirming that the belief-mediation channel documented in Table 3 is not an artifact of a direct effect of RF on choices.

References

A Proofs

B Survey Description and Variable Construction

Table B.1: Index Variable Construction.

Variable	Description	Min	Max	Mean
Panel 1: Recall Fluency Index				
Immediate Recall	Number of words recalled immediately from a 10-word list	0.00	10.00	5.40
Delayed Recall	Number of words recalled after a 5-minute delay	0.00	10.00	4.35
Retrieval Fluency	Number of unique animals named in 60 seconds, net of errors	0.00	40.00	16.09
Construction: Each component is standardized (demeaned, divided by s.d.) across the pooled sample. The index is the mean of the three standardized components. The index is then re-standardized to have mean zero and unit variance.				
Panel 2: IQ (Math) Index				
Counting Backwards	Number of correct serial-7 subtractions from 100	0.00	4.00	2.43
Number Series	Correct answers on number pattern completion (adaptive)	0.00	3.00	1.56
Numeracy	Correct answers on 3 arithmetic/probability questions	0.00	3.00	1.38
Construction: Each component is standardized (demeaned, divided by s.d.) across the pooled sample. The index is the mean of the three standardized components. The index is then re-standardized.				
Panel 3: Current Health PCA				
Self-Rated Health	Self-reported health on 5-point scale (poor to excellent)	1.00	5.00	2.91
Days in Bed	Days spent in bed due to illness in the past month *(inverted)*	-31.00	0.00	-0.77
Health Conditions (Full)	Count of all diagnosed health conditions *(inverted)*	-11.00	0.00	-2.55
Depressed	Indicator for depression diagnosis *(inverted)*	-1.00	0.00	-0.16
Health Change	Self-reported health change since last wave (3-point scale) *(inverted)*	-3.00	-1.00	-2.15
Health Conditions (Basic)	Count of 8 major diagnosed conditions *(inverted)*	-8.00	0.00	-2.15
New Heart Attack	New heart attack since last wave *(inverted)*	-1.00	0.00	-0.01
New/Worsening Severe	New or worsening severe health condition since last wave *(inverted)*	-1.00	0.00	-0.33
Sleep Quality	1st principal component of 4 sleep difficulty items *(inverted)*	-3.57	1.79	-0.00
Functional Limitations	Count of functional limitations (0–15 ADL/IADL scale) *(inverted)*	-15.00	0.00	-3.30
Has Helper	Indicator for receiving help with daily activities *(inverted)*	-1.00	0.00	-0.16
Tasks Helped With	Number of daily tasks receiving help (0–6) *(inverted)*	-6.00	0.00	-0.27
Construction: The first principal component is taken of the 12 components listed above. Components where a higher raw value indicates worse health are multiplied by -1 (marked *(inverted)*) so that higher values of every component, and of the resulting index, indicate better health. Missing values are imputed with the sample median.				
Panel 4: Childhood Health PCA				
Childhood Health Rating	Self-rated health before age 16 (5-point scale)	0.00	4.00	3.23
Missed School	Missed 1+ month of school due to health before age 16 *(inverted)*	-1.00	0.00	-0.11
Disabled as Child	Disabled for 6+ months before age 16 *(inverted)*	0.00	1.00	0.96
Childhood Conditions	Count of 16 childhood diseases/conditions *(inverted)*	-12.00	0.00	-0.63
Severe Conditions	Count of 5 severe childhood conditions *(inverted)*	-4.00	0.00	-0.21
Psychological Conditions	Count of 3 psychological conditions before 16 *(inverted)*	-3.00	0.00	-0.07

Continued on next page

Table B.1 – continued from previous page

Variable	Description	Min	Max	Mean
Construction: The first principal component is taken of the 6 subcomponents listed above. Childhood measures are asked once per respondent and carried forward across waves using the within-person mode. Components where a higher raw value indicates worse health are multiplied by -1 (marked *(inverted)*) so that higher values indicate better childhood health. Missing values are imputed with the sample median.				
Panel 5: Risk-Taking Index				
Subjective Risk Taking	Willingness to take risks, 0–10 self-report scale	-1.84	8.16	3.54
Risk Taking (Employment)	Prefers risky high-pay job over safe lower-pay job	0.00	1.00	0.15
Risk Taking (Business)	Prefers risky business gamble over safe cash-out	0.00	1.00	0.28
Construction: Each component is standardized (demeaned, divided by s.d.). The subjective risk-taking item is asked from 2014; pre-2014 values are imputed with the respondent's modal response. The index is the mean of the three standardized components.				
Panel 6: Current Socioeconomic Status PCA				
Education	Years of education completed	0.00	17.00	12.57
Income	Household income residualized on retirement status (millions USD)	-0.08	0.96	-0.00
Wealth	Total household wealth (millions USD)	0.00	35.50	0.18
Home Ownership	Indicator for owning a home	0.00	1.00	0.61
Not Unemployed	Indicator: not currently unemployed or laid off	0.00	1.00	0.97
Construction: The first principal component is taken of the 5 subcomponents listed above. Missing values are imputed with the sample median.				
Panel 7: Childhood Socioeconomic Status PCA				
Father's Education	Father's years of education	0.00	17.00	9.63
Mother's Education	Mother's years of education	0.00	17.00	9.88
Childhood Finances	Family financial status before 16 (3-point scale: 1=well-off, 3=poor) *(inverted)*	-3.00	-1.00	-1.77
Childhood Poor	Family was poor before age 16 *(inverted)*	0.00	1.00	0.69
Childhood Upper	Family was well-off before age 16	0.00	1.00	0.08
Father Unemployed	Father had extended unemployment before 16 *(inverted)*	0.00	1.00	0.79
Moved (Fin. Difficulty)	Family moved due to financial difficulty before 16 *(inverted)*	0.00	1.00	0.82
Family Received Aid	Family received financial aid before 16 *(inverted)*	0.00	1.00	0.85
Lived Rural	Lived in rural area before age 16 *(inverted)*	0.00	1.00	0.57
Spoke English	English spoken at home as a child	0.00	1.00	0.91
Mother Worked	Mother worked outside home during childhood *(inverted)*	0.00	1.00	0.44
Construction: The first principal component is taken of the 11 subcomponents listed above. Childhood measures are asked once per respondent and carried forward across waves using the within-person mode. Components where a higher raw value indicates lower childhood SES are multiplied by -1 (marked *(inverted)*) so that higher values indicate higher childhood SES. Missing values are imputed with the sample median.				

Notes: This table documents the construction of all index variables used in the paper. For each index, we list the component sub-variables, a brief description based on the underlying HRS survey item, and summary statistics (minimum, maximum, and mean) computed from the estimation sample. The *Construction* row at the foot of each panel describes how the index is aggregated from its components. Components marked *(inverted)* have been multiplied by -1 so that higher values indicate more favorable outcomes (better health, higher socioeconomic status), and the reported min, max, and mean reflect this inversion. Income and wealth are reported in millions of dollars.

Table B.2: Dependent Variable Survey Wording.

Variable	Exact HRS Survey Wording (by wave and HRS item code)
Chance Stock Market Up	**2002–2004 (P047):** "We are interested in how well you think the economy will do in the next year. By next year at this time, what is the percent chance that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today?" **2006–2014 (P047):** "We are interested in how well you think the economy will do in the future. By next year at this time, what is the percent chance that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today?" **2014 (P190, randomized branch):** "What do you think is the percent chance that the stock market will be higher in twelve months today?" **2016–2020 (P047):** "By next year at this time, what is the percent chance that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today?"
Chance Own Home Value Up	**2010 (P166):** "We are interested in how the value of your home will change in the future. On the same scale from 0 to 100, what do you think is the percent chance that by next year at this time your home will be worth more than it is today?" **2012–2020 (P166, randomized fill):** "We are interested in how the value of your home will change in the future. On the same scale from 0 to 100, what do you think is the percent chance that by next year at this time your home will be worth [more/less] than it is today?"
Chance Live 10+ Years	**2002–2004 (P029):** "(What is the percent chance) that you will live to be [X] or more?", where [X] is 80 if age 69 or younger, 85 if 70–74, 90 if 75–79, 95 if 80–84, and 100 if 85–89. **2006–2020 (P029):** "What is the percent chance that you will live to be [X] or more?", where [X] is 85 if under 65, 80 if 65–69, 85 if 70–74, 90 if 75–79, 95 if 80–84, and 100 if 85–89.
Chance Major Economic Depression	**2002–2008 (P034):** "What do you think are the chances that the U.S. economy will experience a major depression sometime during the next 10 years or so?"
Chance Cost of Living Rises >5%/yr	**2006–2008 (P116):** "Over the next 10 years what are the chances the cost of living will increase by 5 percent or more per year on average?"

Notes: This table reports the verbatim HRS Section P survey wording for each of the five dependent variables used in the paper, organized by estimation-sample wave and HRS item code. Wave scope for each DV is restricted to the waves in which the underlying item was fielded: *Chance Stock Market Up* covers 2002–2020; *Chance Own Home Value Up* covers 2010–2020 (P166 was introduced in 2010); *Chance Live 10+ Years* covers 2002–2020; *Chance Major Economic Depression* covers 2002–2008; and *Chance Cost of Living Rises >5%/yr* covers 2006–2008. In 2014 the stock-market question was administered under a randomized branch: half the sample (X504_4Random1_2 = 1) was asked the standard P047 wording, while the other half (X504_4Random1_2 = 2) was asked the alternative P190 wording referring to the stock market (rather than mutual funds in blue-chip stocks) and a twelve-month (rather than next-year-at-this-time) horizon. For *Chance Own Home Value Up*, 2012 onwards the fill phrase was randomized between "more" and "less" across respondents (variable X501 in 2012–2014; X083/P196 in 2016–2020); sign of the analysis variable is harmonized to the probability the home is worth more. For *Chance Live 10+ Years*, the target age fill depends on respondent age; in 2002–2004 the fill took five levels (80/85/90/95/100) while from 2006 onwards it took six levels (85/80/85/90/95/100), with the 85-year target added for respondents under age 65. The home-value and (2014-branch) stock-market questions are asked only to the relevant subsamples (homeowners; the X504_4Random1_2 = 2 branch). Source: HRS Section P questionnaires, hrsdata.isr.umich.edu.

C Additional Results

Table C.1: Variable Summary Statistics

	Mean	St.d	IQR	P10	P25	P50	P75	P90
Cognition								
Recall Fluency	0.01	0.90	1.14	-1.16	-0.55	0.04	0.58	1.10
– Immediate Memory	5.40	1.74	3.00	3.00	4.00	5.00	7.00	8.00
– Delayed Memory	4.35	2.08	3.00	1.00	3.00	4.00	6.00	7.00
– Retrieval Fluency	16.09	6.92	10.00	8.00	11.00	16.00	21.00	25.00
IQ	-0.00	0.88	1.61	-1.24	-0.81	0.07	0.80	1.18
– Counting Backwards	2.43	1.56	3.00	0.00	1.00	2.00	4.00	4.00
– Number Series	1.56	1.06	2.00	0.00	1.00	1.00	3.00	3.00
– Numeracy	1.38	1.03	2.00	0.00	0.00	1.00	2.00	3.00
Experiences (DS)								
$\overline{s_{it}}^{STOCKS}$	0.03	0.01	0.01	0.02	0.03	0.03	0.03	0.04
$\overline{s_{it}}^{HOUSE}$	0.03	0.01	0.01	0.02	0.02	0.03	0.03	0.04
$\overline{s_{it}}^{HEALTH}$	-0.04	0.28	0.22	-0.34	-0.15	-0.02	0.08	0.21
$cov(s,(s-r_t)^2)^{STOCKS}$	-0.00	0.01	0.01	-0.01	-0.01	-0.01	-0.00	0.01
$cov(s,(s-r_t)^2)^{HOUSE}$	0.00	0.00	0.00	-0.00	-0.00	-0.00	0.00	0.00
$cov(s,(s-r_t)^2)^{HEALTH}$	0.08	1.75	0.11	-0.29	-0.03	0.00	0.08	0.48
Experiences (NDS)								
Childhood Health PCA	-0.04	0.71	0.41	-0.95	-0.10	0.31	0.31	0.36
Childhood SES PCA	0.06	0.47	0.60	-0.63	-0.23	0.16	0.37	0.62
Δ_L S&P 500	1.71	0.54	0.90	0.90	1.24	1.85	2.13	2.31
Δ_L HPI	1.69	0.33	0.43	1.35	1.45	1.61	1.88	2.15
Beliefs								
Chance Stock Increase [0-100]	0.47	0.26	0.35	0.10	0.25	0.50	0.60	0.80
Chance Home Value Increase [0-100]	0.38	0.31	0.50	0.00	0.10	0.30	0.60	0.80
Chance Live 10+ Years [0-100]	0.48	0.32	0.55	0.00	0.20	0.50	0.75	0.93
Controls								
Age	67.61	11.42	17.00	54.00	59.00	67.00	76.00	83.00
Female [0,1]	0.59	0.49	1.00	0.00	0.00	1.00	1.00	1.00
Married [0,1]	0.63	0.48	1.00	0.00	0.00	1.00	1.00	1.00
Retired [0,1]	0.53	0.50	1.00	0.00	0.00	1.00	1.00	1.00
Income (1000s)	57.57	105.37	49.13	7.34	14.40	30.00	63.53	117.38
Years Education	12.57	3.28	3.00	8.00	12.00	12.00	15.00	17.00

Notes: HRS biennial waves 2002–2020. *Recall Fluency*: standardized mean of immediate recall (10-item list), delayed recall, and retrieval fluency (unique animals in 60s). *IQ*: standardized mean of Counting Backwards (serial subtraction of 7), Number Series, and Numeracy. DS experience variables $\bar{s}_{it}$: each respondent's lifetime average realized return (monthly S&P 500 for stocks, quarterly FHFA regional HPI for housing, wave-to-wave changes in a health PCA for health). The covariance term $\mathrm{cov}(s,(s-r_t)^2) \equiv \frac{1}{T_i}\sum_k (s_k-\bar{s}_i)(s_k-r_t)^2$ governs the direction of belief distortion under selective retrieval. NDS: *Childhood Health PCA* and *Childhood SES PCA* are first-principal-component summaries of retrospective childhood reports; Δ_L S&P 500 and Δ_L HPI are lifetime-to-date average realized returns on each index. Beliefs are elicited on a 0–100 scale and rescaled to $[0,1]$: chance stocks rise next year, home values rise locally, and respondent survives 10+ more years. Controls measured at interview; income in 1000s USD; female indicator $=1$ if female, 0 if male. IQR denotes the interquartile range (P75 − P25).

Table C.2: Variance Decomposition of Cognitive Indices and Sub-Components.

	Memory Index	Math Index	Immed. Mem.	Delayed Mem.	Named Animals	Count. Back.	Number Ser.	Numeracy
Variance								
$\mathrm{Var}(y_{it})$	0.806	0.776	1.000	1.000	1.000	1.000	1.000	1.000
$\mathrm{Var}_i(\overline{y}_i)$ (cross-section)	0.501	0.522	0.556	0.573	0.749	0.612	0.694	0.690
$\mathrm{E}_i[\mathrm{Var}_t(y_{it})]$ (over time)	0.305	0.254	0.444	0.427	0.251	0.388	0.306	0.310
Proportion								
$\mathrm{Var}(y_{it})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\mathrm{Var}_i(\overline{y}_i)$ (cross-section)	0.622	0.673	0.556	0.573	0.749	0.612	0.694	0.690
$\mathrm{E}_i[\mathrm{Var}_t(y_{it})]$ (over time)	0.378	0.327	0.444	0.427	0.251	0.388	0.306	0.310

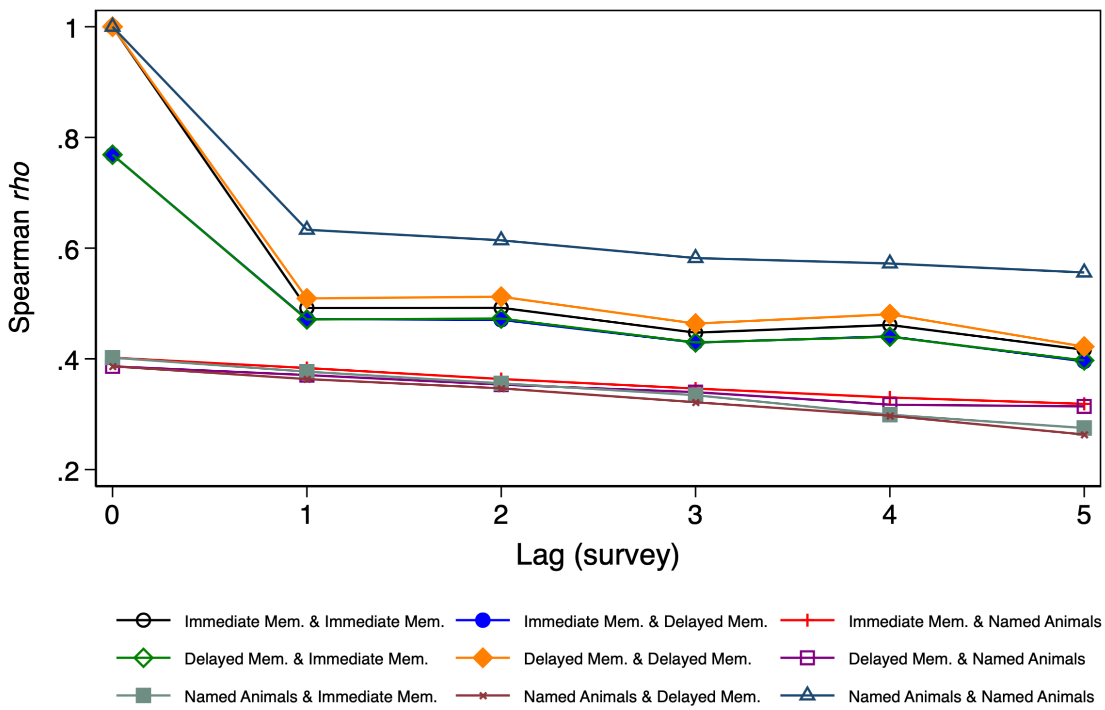
Notes: Variance decomposition of the Memory Index (Recall Fluency), the Math Index (IQ), and all individual cognitive sub-components across HRS waves 2002–2020. $\mathrm{Var}(y_{it})$ is total variance. $\mathrm{Var}_i(\overline{y}_i)$ is the cross-sectional (between-person) component, computed as the variance of individual-level means weighted by the number of non-missing observations per respondent. $\mathrm{E}_i[\mathrm{Var}_t(y_{it})]$ is the mean within-person variance over time. The proportion rows express each component as a share of total variance; by construction the two proportions sum to one. All sub-components are standardized (mean zero, unit variance) before inclusion in the indices.

Table C.3: Within- vs Between-Person Variance Decomposition of Experience Signals and Covariance Terms.

	Stocks		Housing		Health	
	$\overline{s_{it}}$	$\mathrm{cov}(s,(s-s_{it})^2)$	$\overline{s_{it}}$	$\mathrm{cov}(s,(s-s_{it})^2)$	$\overline{s_{it}}$	$\mathrm{cov}(s,(s-s_{it})^2)$
Total variance	1.000	1.000	1.000	1.000	1.000	1.000
$\mathrm{Var}_i(\overline{y}_i)$ (cross-section)	0.798	0.153	0.909	0.112	0.774	0.612
$\mathrm{E}_i[\mathrm{Var}_t(y_{it})]$ (over time)	0.202	0.847	0.091	0.888	0.226	0.388
$\mathrm{Corr}(\overline{s_{it}},\ \mathrm{cov}(s,(s-s_{it})^2))$	0.214		0.094		0.296	

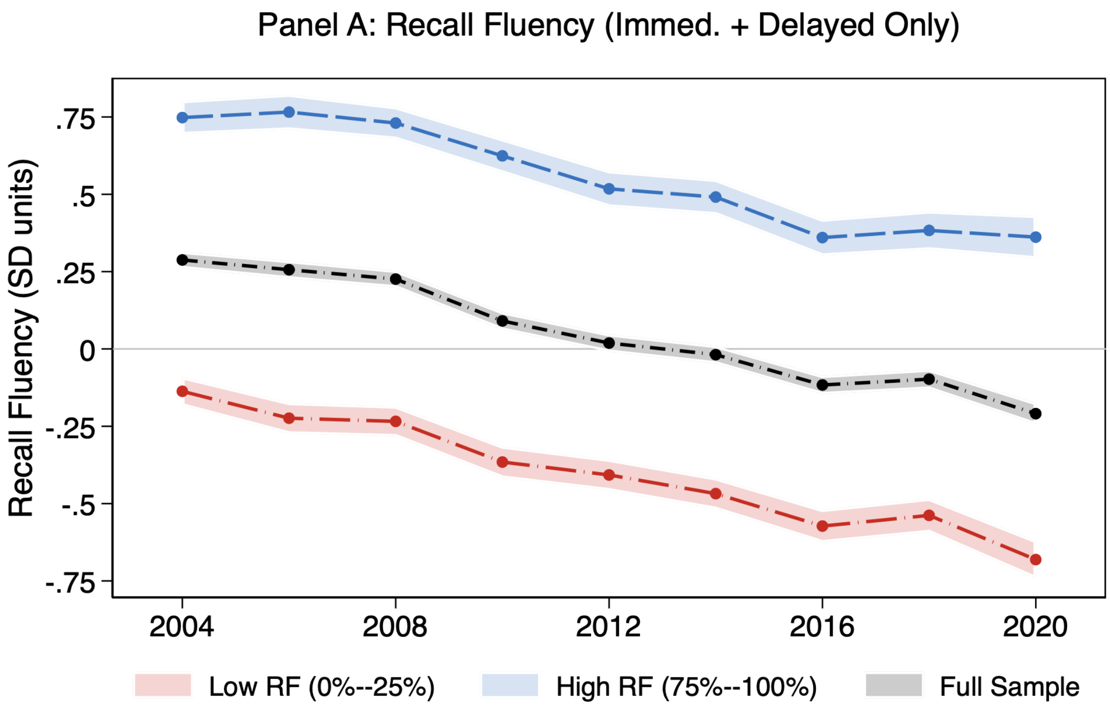
Notes: Proportional variance decomposition of the domain-specific experience signal $\overline{s_{it}}$ and the covariance term $\mathrm{cov}(s,(s-s_{it})^2)$ for stocks, housing, and health. Parallels Table C.2 for the Memory and Math indices; raw variances suppressed here because the source series are on different native scales (monthly returns, quarterly HPI growth, health-PCA units). Rows report shares of total variance: $\mathrm{Var}_i(\overline{y}_i)$ is the cross-sectional (between-person) component, computed as the variance of individual-level means weighted by the number of non-missing observations per respondent; $\mathrm{E}_i[\mathrm{Var}_t(y_{it})]$ is the mean within-person variance over time. The between and within shares sum to one by construction. The bottom row reports the pooled person-wave correlation between $\overline{s_{it}}$ and the covariance term for each category. Stocks series are monthly returns (database 0 / age-25 baseline); housing series are quarterly HPI returns; health series are built from HRS-wave health indices.

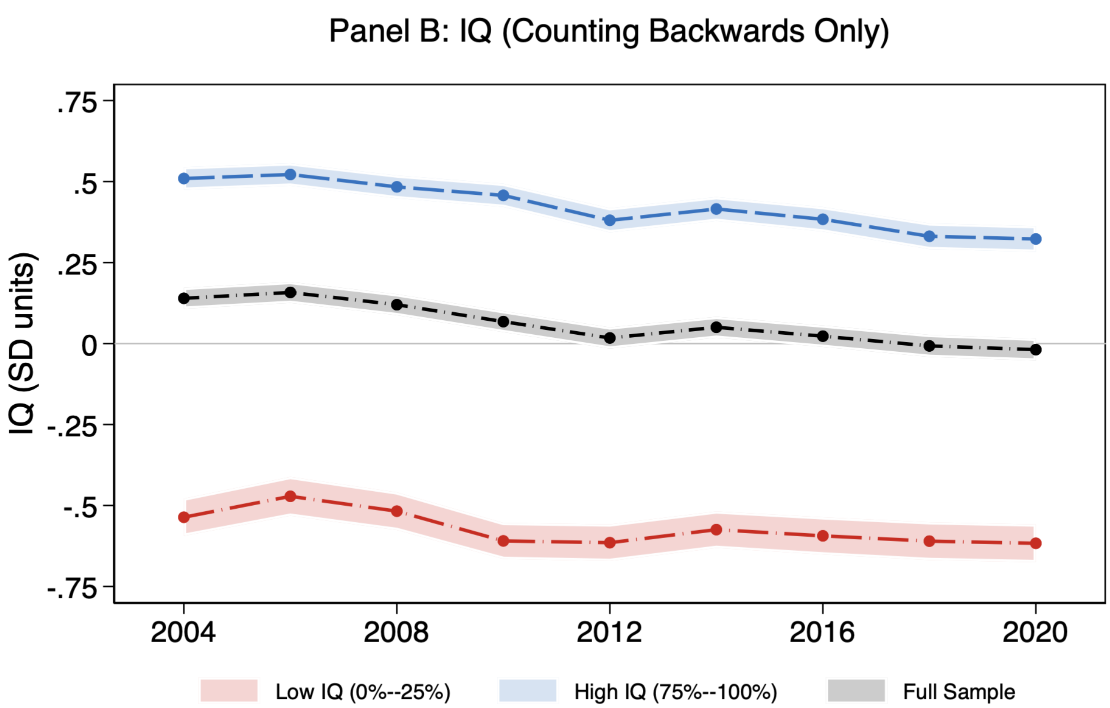
Figure C.1: Memory Trait Stability: Spearman Correlation By Lag.



Notes: Each series plots the Spearman rank correlation ρ (y-axis) between a pair of memory sub-components at increasing survey lags (x-axis; lag 0 = same survey wave, lags 1–5 = up to ten years apart). The nine series correspond to all pairwise combinations of the three sub-components (Immediate Recall, Delayed Recall, Named Animals). High and persistent correlations across lags indicate that individual rankings in memory performance are stable over time. Sample: HRS respondents present in at least two waves, 2002–2020.

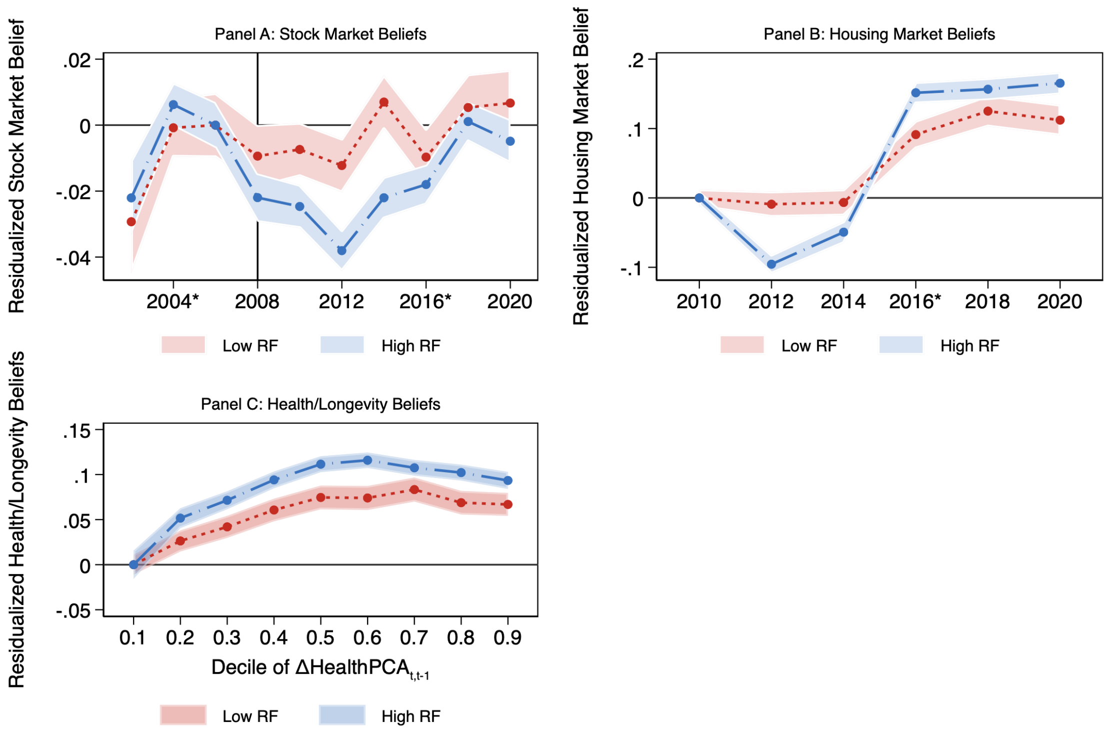
Figure C.2: Recall Fluency and IQ Over Time, By Quartile — Always-Available Sub-Indices





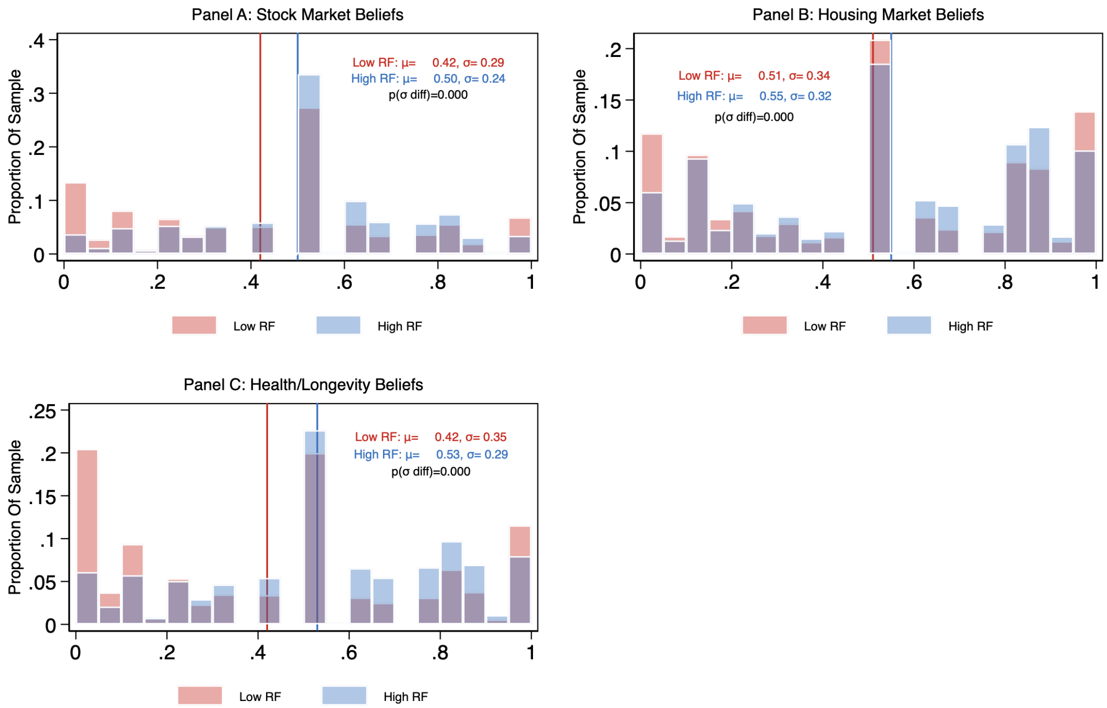
Notes: Robustness version of Figure 2 restricted to cognitive sub-indices available in all survey waves (2002–2020). Panel A uses a restricted Recall Fluency index comprising only immediate and delayed word recall (dropping named-animals retrieval fluency, which is only available from 2010). Panel B uses counting backwards only (dropping number series and numeracy, which start in 2010 and are not administered in re-interviews respectively). Blue lines show respondents in the top 25% of the respective restricted index in 2002; red lines show the bottom 25%; grey lines show the full sample. Shaded bands are 95% confidence intervals. The sample is restricted to COHORT 3 respondents present in all ten survey waves (2002–2020). 2002 is omitted as it is the base year for group assignment. Both indices are expressed in standard deviation units.

Figure C.3: Residualized Beliefs Over Time, By Recall Fluency.



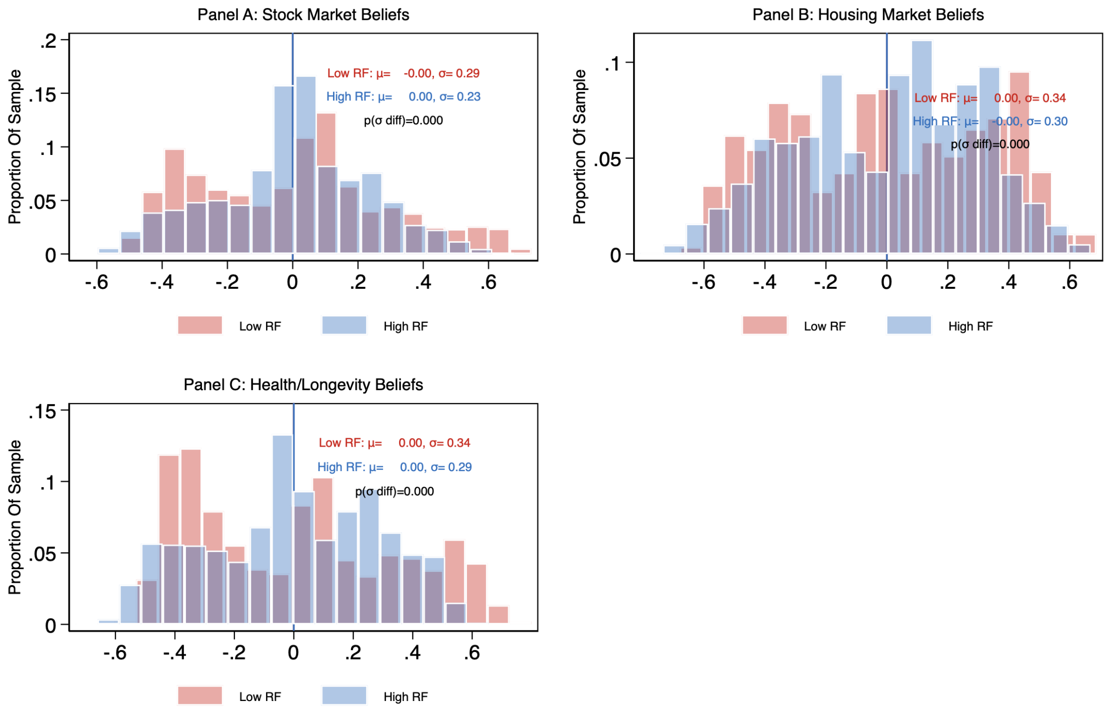
Notes: Each panel shows average residualized beliefs by Recall Fluency (RF) group. Panel A plots stock market beliefs () by survey year (2002–2020). Panel B plots housing market beliefs () by survey year (2010–2020). Panel C plots longevity beliefs () by decile of health change between survey waves. In all panels, beliefs are residualized on: age, gender, marital status, retirement status, income, education, and the Math Index (IQ). The stock market series additionally controls for whether the respondent actively follows the stock market. Each RF group is normalized to zero at its baseline: 2006 for stocks (just prior to the financial crisis), 2010 for housing (first year the question was asked), and the lowest health-change decile for longevity. Low RF (High RF) is respondents in the bottom (top) quartile of the RF distribution. Shaded areas are 90% confidence intervals. Sample: HRS respondents, waves 2002–2020. * denotes years in which the HRS surveys only new entrants.

Figure C.4: Raw Belief Distributions, By Recall Fluency.



Notes: Each panel shows the distribution of raw (un-residualized) beliefs by Recall Fluency (RF) group. Panel A: stock market beliefs (). Panel B: housing market beliefs (; negative-framed responses reversed). Panel C: longevity beliefs (). Low RF (High RF) is respondents in the bottom (top) quartile of the raw RF distribution. Vertical lines show group means. Text on each panel reports group means (μ), standard deviations (σ), and the p-value for the Brown-Forsythe test of equality of variances. Sample: HRS respondents, waves 2002–2020.

Figure C.5: Demographic- and Time-Residualized Belief Distributions, By Recall Fluency.



Notes: Each panel shows the distribution of beliefs residualized on both demographics and time. Panel A: stock market beliefs (). Panel B: housing market beliefs (; negative-framed responses reversed). Panel C: longevity beliefs (). Beliefs are first residualized on: age, gender, marital status, retirement status, income, education, and the Math Index (IQ); the stock market series additionally controls for stock market participation. The memory index is also residualized on the same demographics and used to form RF quartile groups. Beliefs are then further residualized on survey-year dummies separately within each RF group, removing common time variation. Vertical lines show group means. Text on each panel reports group means (μ), standard deviations (σ), and the p-value for the Brown-Forsythe test of equality of variances. Sample: HRS respondents, waves 2002–2020.

Table C.4: DS Regressions with IQ $\times$ Experience Interactions.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance House ↑**		**Chance Live10+**	
Cognition						
RF	.	0.008	.	-0.010	.	0.013***
	.	[0.006]	.	[0.011]	.	[0.003]
IQ	0.017**	0.013	-0.015	-0.009	-0.036***	-0.043***
	[0.008]	[0.008]	[0.014]	[0.015]	[0.004]	[0.004]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	-0.002	0.000	0.015***	0.012*	0.013***	0.017***
	[0.002]	[0.003]	[0.006]	[0.007]	[0.002]	[0.002]
RF Interactions						
$RF * \overline{s_{it}}$ (γ_2)	0.005***	0.003**	0.003***	0.006**	0.006***	0.002**
	[0.000]	[0.001]	[0.001]	[0.003]	[0.000]	[0.001]
$RF * cov(s,(s-r_t)^2)$ (γ_3)	-0.001	-0.001	-0.014***	-0.014***	-0.000***	-0.000***
	[0.001]	[0.001]	[0.002]	[0.002]	[0.000]	[0.000]
IQ Interactions						
$IQ * \overline{s_{it}}$ (κ_1)	0.001	0.002	0.009**	0.008**	0.007***	0.010***
	[0.002]	[0.002]	[0.004]	[0.004]	[0.001]	[0.001]
$IQ * cov(s,(s-r_t)^2)$ (κ_2)	-0.004***	-0.005***	-0.023***	-0.023***	-0.000**	-0.000***
	[0.001]	[0.001]	[0.002]	[0.002]	[0.000]	[0.000]
N	139345	139345	49972	49972	129518	129518
AR2	0.04	0.04	0.05	0.05	0.10	0.10
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

Notes: Each column replicates the corresponding column of the main Table 1 with the addition of IQ $\times$ experience interactions ($\times\overline{s_{it}}$ and $\times cov(s_k,(s_k-r_t)^2)$). This specification tests whether the RF–experience interactions in Table 1 are confounded by respondents having different information sets (proxied by IQ). Dependent variables: cols 1–2 chance stocks rise; cols 3–4 chance house prices rise; cols 5–6 chance of living 10+ years. Columns 2, 4, 6 additionally include RF as a main effect. All regressors standardized by sample IQR prior to estimation. Demographic controls include age, gender, marital status, retirement status, income, and education; housing regressions also include home ownership. Respondent fixed effects absorbed (cols 1–4); age $\times$ cohort fixed effects (cols 5–6). Standard errors clustered by respondent in brackets. * p<0.12, ** p<0.05, *** p<0.01.

Table C.5: DS Regressions with Survey-Year Fixed Effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance House ↑**		**Chance Live10+**	
Cognition						
RF	.	0.003	.	-0.042***	.	0.006**
	.	[0.005]	.	[0.010]	.	[0.003]
IQ	0.024***	0.024***	0.001	0.002	-0.020***	-0.020***
	[0.002]	[0.002]	[0.003]	[0.003]	[0.002]	[0.002]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	-0.004***	-0.003	0.005**	-0.015***	0.021***	0.024***
	[0.001]	[0.003]	[0.002]	[0.005]	[0.001]	[0.002]
RF Interactions						
$RF * \overline{s_{it}}$ (γ_2)	0.005***	0.004***	0.002***	0.013***	0.006***	0.005***
	[0.000]	[0.001]	[0.001]	[0.003]	[0.000]	[0.001]
$RF * cov(s,(s-r_t)^2)$ (γ_3)	-0.003***	-0.003***	-0.005***	-0.005***	-0.000***	-0.000***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.000]	[0.000]
N	139345	139345	49972	49972	129518	129518
AR2	0.04	0.04	0.10	0.10	0.10	0.10
Demographic Controls	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

Notes: Each column replicates the corresponding column of the main Table 1 with the addition of survey-year fixed effects throughout. Year fixed effects absorb aggregate time-series shocks—such as macroeconomic news or market-wide returns—that affect all respondents in the same survey wave simultaneously. Persistence of the RF × experience interactions (γ_2, γ_3) after absorbing year effects indicates that results reflect individual memory reactions to personal past experiences rather than common aggregate information. All other specification choices are identical to Table 1: dependent variables, regression specification, controls, and (non-year) fixed effects structure. Dependent variables: cols 1–2 chance stocks rise; cols 3–4 chance house prices rise; cols 5–6 chance of living 10+ years. Columns 2, 4, 6 additionally include RF as a main effect. Demographic controls include age, gender, marital status, retirement status, income, and education; housing regressions also include home ownership. Standard errors clustered by respondent in brackets. * p<0.12, ** p<0.05, *** p<0.01.

Table C.6: NDS Database Regressions with IQ $\times$ Experience Interactions.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance House ↑**		**Chance Live 10+**	
IQ	0.017**	0.012	-0.017	-0.016	-0.027***	-0.043***
	[0.008]	[0.008]	[0.014]	[0.014]	[0.007]	[0.011]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	0.003*	0.002	0.014**	0.015***	0.017***	0.019***
	[0.002]	[0.002]	[0.006]	[0.006]	[0.002]	[0.003]
RF Interactions						
$RF \times \overline{s_{it}}$ (γ_2)	0.005***	0.005***	0.003***	0.003***	0.005***	0.004***
	[0.000]	[0.000]	[0.001]	[0.001]	[0.001]	[0.001]
$RF \times \text{cov}(s, (s-r_t)^2)$ (γ_3)	-0.002**	-0.002*	-0.014***	-0.014***	-0.000	0.000
	[0.001]	[0.001]	[0.002]	[0.002]	[0.000]	[0.000]
IQ Interactions						
$IQ \times \overline{s_{it}}$ (κ_1)	0.001	0.002	0.010***	0.009**	0.009***	0.009***
	[0.002]	[0.002]	[0.004]	[0.004]	[0.002]	[0.002]
$IQ \times \text{cov}(s, (s-r_t)^2)$ (κ_2)	-0.005***	-0.006***	-0.023***	-0.023***	-0.000	-0.000
	[0.001]	[0.001]	[0.002]	[0.002]	[0.000]	[0.000]
NDS Experiences						
nds (δ_1)	0.000	-0.005***	-0.005	-0.015**	0.036***	-0.006
	[0.001]	[0.002]	[0.003]	[0.006]	[0.005]	[0.004]
$RF \times nds$ (τ)	0.001	0.001	0.001	0.002	0.013***	0.007***
	[0.001]	[0.001]	[0.001]	[0.003]	[0.001]	[0.000]
$IQ \times nds$ (ν)	0.002**	0.014***	0.002	0.011***	0.006*	0.006**
	[0.001]	[0.002]	[0.002]	[0.004]	[0.003]	[0.003]
N	134136	139342	49593	49971	102724	88815
AR2	0.04	0.04	0.06	0.06	0.07	0.07
NDS Var	Chld.Health	Chld.SES	Chld.Health	Chld.SES	$\Delta_L sp500$	$\Delta_L hpi$
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	N	N

Notes: Replicates Table 2 adding IQ $\times$ experience interactions ($IQ \times \overline{s_{it}}$, $IQ \times \text{cov}(s_k, (s_k - r_t)^2)$, $IQ \times nds$), testing whether RF–experience interactions are confounded by information-set differences proxied by IQ. NDS variables: childhood health and SES for stocks and housing; lifetime equity and housing returns for health. All continuous regressors (RF, IQ, $\overline{s_{it}}$, cov, NDS) divided by within-sample IQR; coefficients represent a move from p25 to p75. Demographic controls: age, gender, marital status, retirement status, income, education. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table C.7: NDS Database Regressions with Survey-Year Fixed Effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance House ↑**		**Chance Live 10+**	
Cognition *(η)*						
RF	0.004	0.002	-0.046***	-0.042***	-0.002	-0.003
	[0.006]	[0.006]	[0.010]	[0.010]	[0.004]	[0.007]
IQ	0.024***	0.024***	0.002	0.001	-0.020***	-0.023***
	[0.002]	[0.002]	[0.003]	[0.003]	[0.002]	[0.002]
DS Experiences *(θ)*						
$\overline{s_{it}}$ (γ_1)	0.000	0.000	-0.017***	-0.015***	0.022***	0.025***
	[0.002]	[0.002]	[0.005]	[0.005]	[0.002]	[0.002]
RF Interactions *(β)*						
$RF \times \overline{s_{it}}$ (γ_2)	0.004***	0.004***	0.014***	0.012***	0.008***	0.007***
	[0.001]	[0.001]	[0.003]	[0.003]	[0.001]	[0.001]
$RF \times \text{cov}(s, (s - r_t)^2)$ (γ_3)	-0.003***	-0.002***	-0.005***	-0.005***	-0.000	-0.000
	[0.001]	[0.001]	[0.001]	[0.001]	[0.000]	[0.000]
NDS Experiences *(δ, τ)*						
nds (δ_1)	0.001	0.001	-0.002	-0.004	0.046***	-0.003
	[0.001]	[0.001]	[0.002]	[0.005]	[0.005]	[0.004]
$RF \times nds$ (τ)	0.001**	0.005***	0.002	0.005**	0.014***	0.008***
	[0.001]	[0.001]	[0.001]	[0.003]	[0.002]	[0.002]
N	134136	139342	49593	49971	102724	88815
AR2	0.04	0.04	0.10	0.10	0.07	0.07
NDS Var	Chld.Health	Chld.SES	Chld.Health	Chld.SES	$\Delta_L sp500$	$\Delta_L hpi$
Demographic Controls	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	N	N

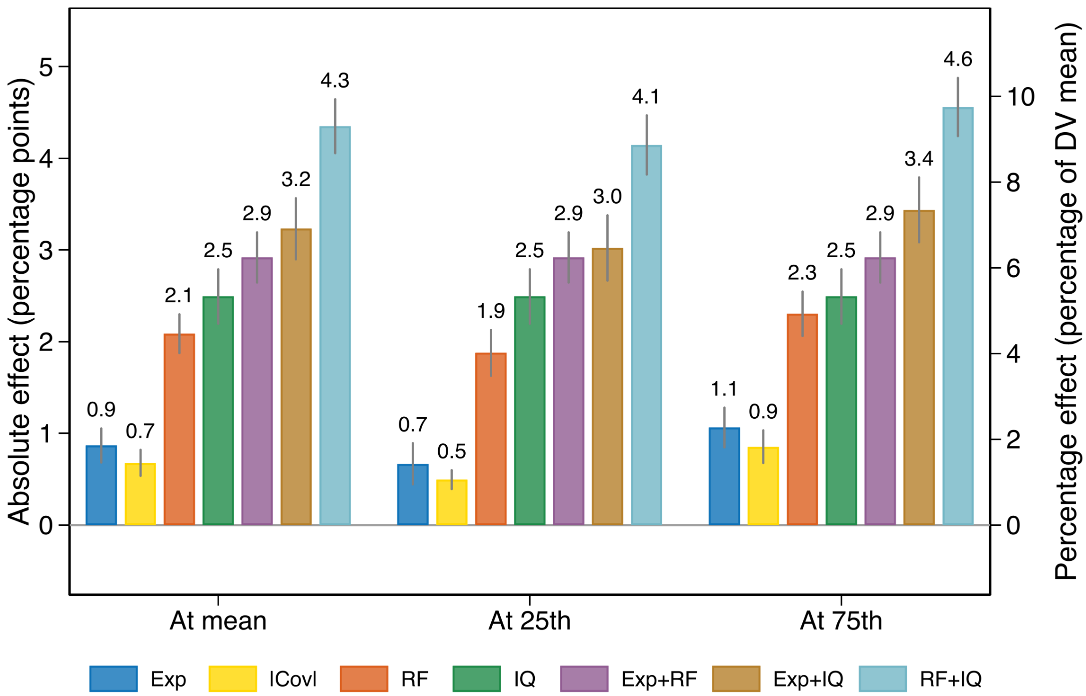
Notes: Replicates Table 2 adding survey-year fixed effects throughout. Year FE absorb common time trends; RF–experience interactions identify within-year, cross-person variation in belief sensitivity. NDS variables: childhood health and SES for stocks and housing; lifetime equity and housing returns for health. All continuous regressors (RF, IQ, $\overline{s_{it}}$, $\text{cov}(s_k, (s_k - r_t)^2)$, NDS) divided by within-sample IQR; coefficients represent a move from p25 to p75. Demographic controls: age, gender, marital status, retirement status, income, education. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table C.8: Health Regressions: Sensitivity to Minimum Personal Survey Count.

	(≥ 2)	(≥ 4)	(≥ 6)	(≥ 8)	(≥ 10)
			Chance Live10+		
Cognition					
RF	.	.	.	.	.
	.	.	.	.	.
IQ	-0.019***	-0.020***	-0.019***	-0.023***	-0.020**
	[0.002]	[0.002]	[0.003]	[0.005]	[0.010]
DS Experiences					
$\overline{s_{it}}$ (γ_1)	0.029***	0.041***	0.053***	0.057***	0.061***
	[0.001]	[0.001]	[0.002]	[0.003]	[0.007]
RF Interactions					
$RF * \overline{s_{it}}$ (γ_2)	0.007***	0.007***	0.006***	0.005***	0.003**
	[0.000]	[0.001]	[0.001]	[0.001]	[0.002]
$RF * cov(s,(s-r_t)^2)$ (γ_3)	-0.001***	-0.001***	-0.001***	-0.002***	-0.002***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.001]
N	116262	80141	44002	18983	3825
AR2	0.11	0.12	0.14	0.14	0.12
SD of $cov(s,(s-r_t)^2)$	1.750	1.420	1.214	1.059	0.886
Demographic Controls	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y
Age FE	Y	Y	Y	Y	Y

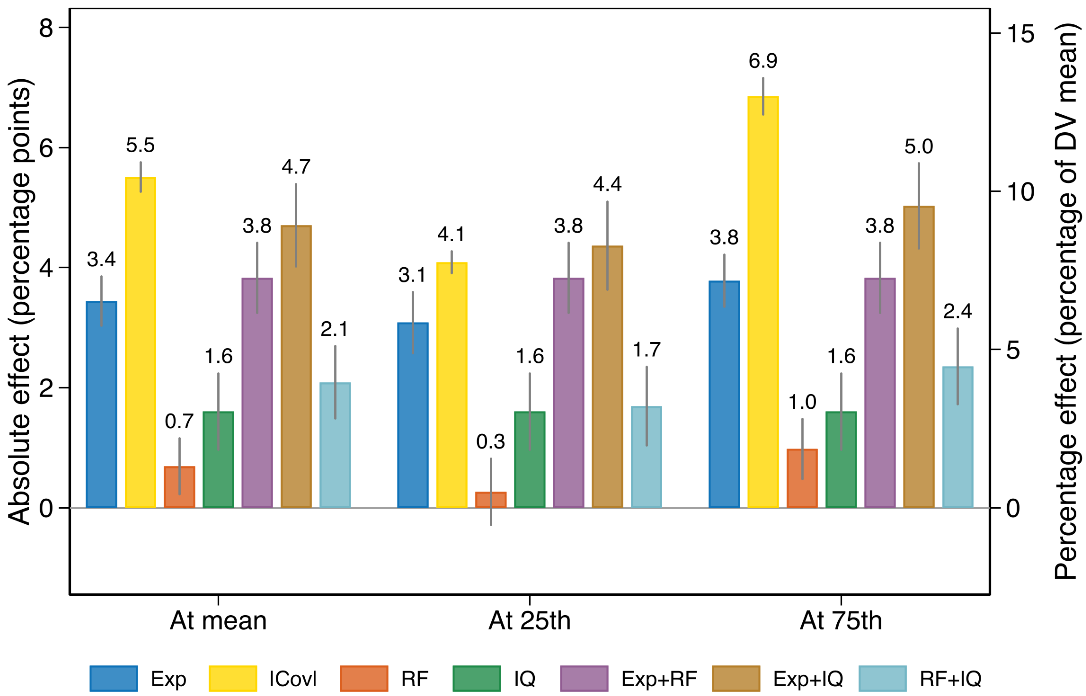
Notes: Each column replicates the health specification of main Table 1 (column 5: Chance Live10+ regressed on RF and IQ cognition indices, average health experience $\overline{s_{it}}$, and their interactions with RF) on progressively restricted samples. Column header ($\geq k$) restricts the sample to respondents who have contributed at least k personal survey observations, so that the health covariance regressor is computed from a longer within-person history. The SD of Cov row reports the standard deviation of the unscaled covariance regressor in each sample, illustrating how the distribution stabilises as the sample is restricted to longer-tenured respondents. Outcome: subjective probability of surviving 10+ years (0–100). Regressors standardised by sample IQR. Demographic controls: gender, marital status, retirement status, income, education. Fixed effects: age and cohort. Standard errors clustered by respondent in brackets. * p<0.12, ** p<0.05, *** p<0.01.

Figure C.6: Impact Coefficient Sizes: Chance Stocks Will Go Up.



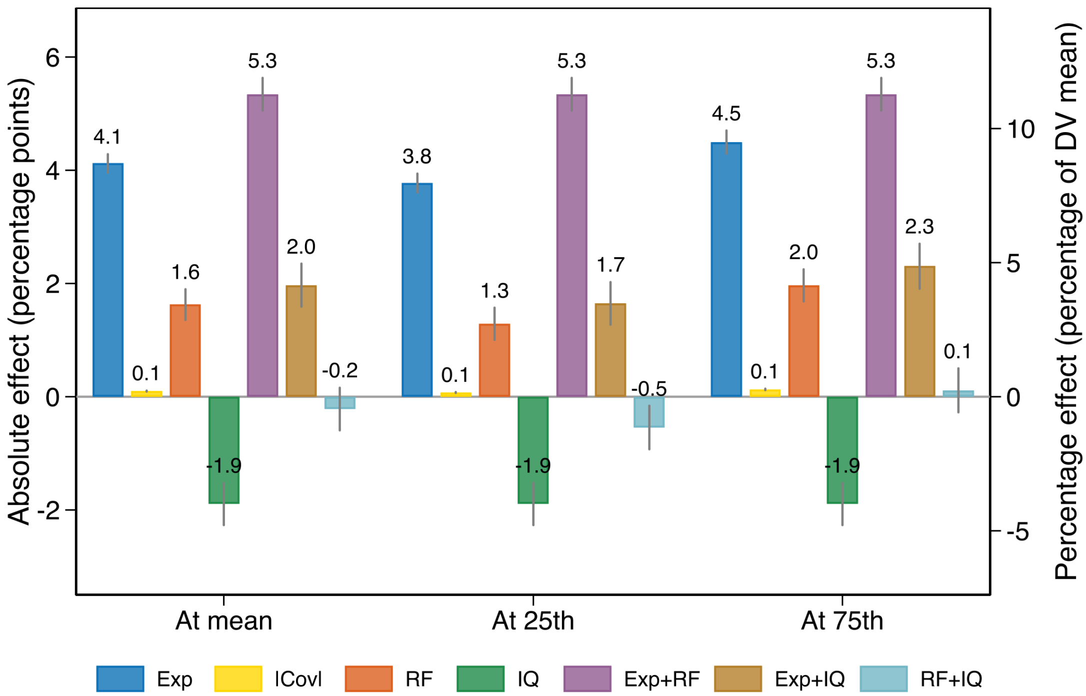
Notes: This figure translates the regression coefficients from column 2 of Table 1 into economically interpretable magnitudes. The dependent variable is the subjective probability that stock prices will go up. Each bar shows a marginal effect evaluated with the non-focal regressors held at the sample mean, 25th percentile, or 75th percentile (indicated by the x-axis grouping). X-axis groupings ("At mean", "At 25th", "At 75th") indicate the percentile at which the *other* regressors are held fixed when computing a given marginal effect. For example, "At 25th" means the non-focal variables are evaluated at their 25th percentiles. Seven bar types within each group: Exp $= \overline{s_{it}}$, the marginal effect of past experiences (holding RF and IQ at the evaluation point); —Cov—, the absolute marginal effect of the perceived-covariance term ($|\partial y/\partial r_{\text{cov}}|$, evaluated at the same memory/math point; in this specification r_{cov} enters only through the $c.mem \times c.r_{\text{cov}}$ interaction, so this equals $|\beta_{mem\times cov} \times mem_{at}|$); RF, the marginal effect of the Recall Fluency index (holding Exp and IQ at the evaluation point); IQ, the marginal effect of the Math index (holding Exp and RF at the evaluation point). The combined bars—Exp+RF, Exp+IQ, RF+IQ—show the total predicted change in the belief when both named variables move simultaneously from the 25th to the 75th percentile of their sample distributions, with the third variable held at the group's evaluation point. These combined effects capture the interaction between RF and experiences: if RF amplifies the response to experience ($\gamma_2 > 0$), then Exp+RF will exceed the sum of the two individual bars. Y-axes: the left axis shows the percentage-point change in the dependent variable; the right axis re-expresses the same bars as a percent of the sample DV mean (a linear rescaling of the left axis). Numbers above each bar give the percentage-point effect size. Error bars are 95% delta-method confidence intervals. All regressors are standardized by the sample inter-quartile range before estimation; units are therefore comparable across the three belief domains. Specification and sample are identical to column 2 of Table 1.

Figure C.7: Impact Coefficient Sizes: Chance House Prices Will Go Up.



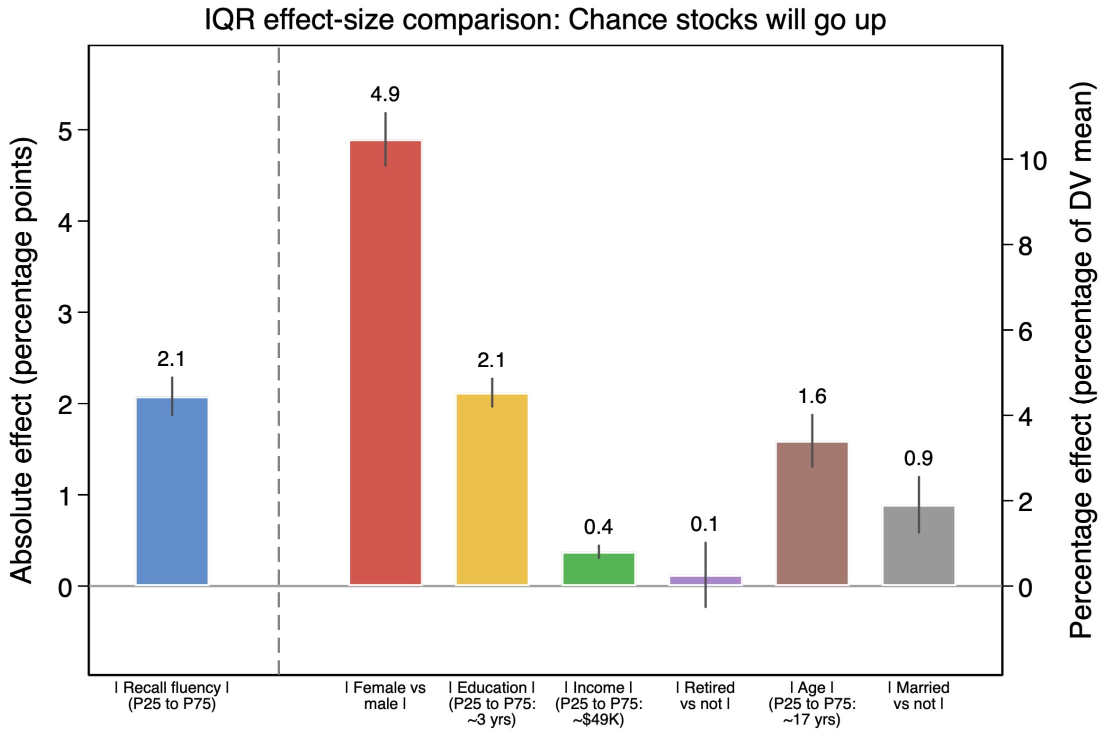
Notes: This figure translates the regression coefficients from column 4 of Table 1 into economically interpretable magnitudes. The dependent variable is the subjective probability that house prices will go up (negatively framed responses are reversed to "up" framing). Each bar shows a marginal effect evaluated with the non-focal regressors held at the sample mean, 25th percentile, or 75th percentile (indicated by the x-axis grouping). The housing sample excludes 2008. X-axis groupings ("At mean", "At 25th", "At 75th") indicate the percentile at which the *other* regressors are held fixed when computing a given marginal effect. Seven bar types within each group: Exp = $\overline{hpi}_t$, the marginal effect of past house-price experiences (holding RF and IQ at the evaluation point); —Cov—, the absolute marginal effect of the perceived-covariance term ($|\partial y/\partial r_{\text{cov}}|$, evaluated at the same memory/math point; in this specification r_{cov} enters only through the $c.mem \times c.r_{\text{cov}}$ interaction, so this equals $|\beta_{mem\times cov} \times mem_{at}|$); RF, the marginal effect of the Recall Fluency index (holding Exp and IQ at the evaluation point); IQ, the marginal effect of the Math index (holding Exp and RF at the evaluation point). The combined bars—Exp+RF, Exp+IQ, RF+IQ—show the total predicted change in the belief when both named variables move simultaneously from the 25th to the 75th percentile, with the third variable held at the group's evaluation point. A large Exp+RF bar relative to Exp alone indicates that high-RF individuals react more strongly to their housing experience history. Y-axes: the left axis shows the percentage-point change in the dependent variable; the right axis re-expresses the same bars as a percent of the sample DV mean (a linear rescaling of the left axis). Numbers above each bar give the percentage-point effect size. Error bars are 95% confidence intervals. All regressors are standardized by the sample inter-quartile range. Specification and sample are identical to column 4 of Table 1.

Figure C.8: Impact Coefficient Sizes: Chance of Living 10+ Years.



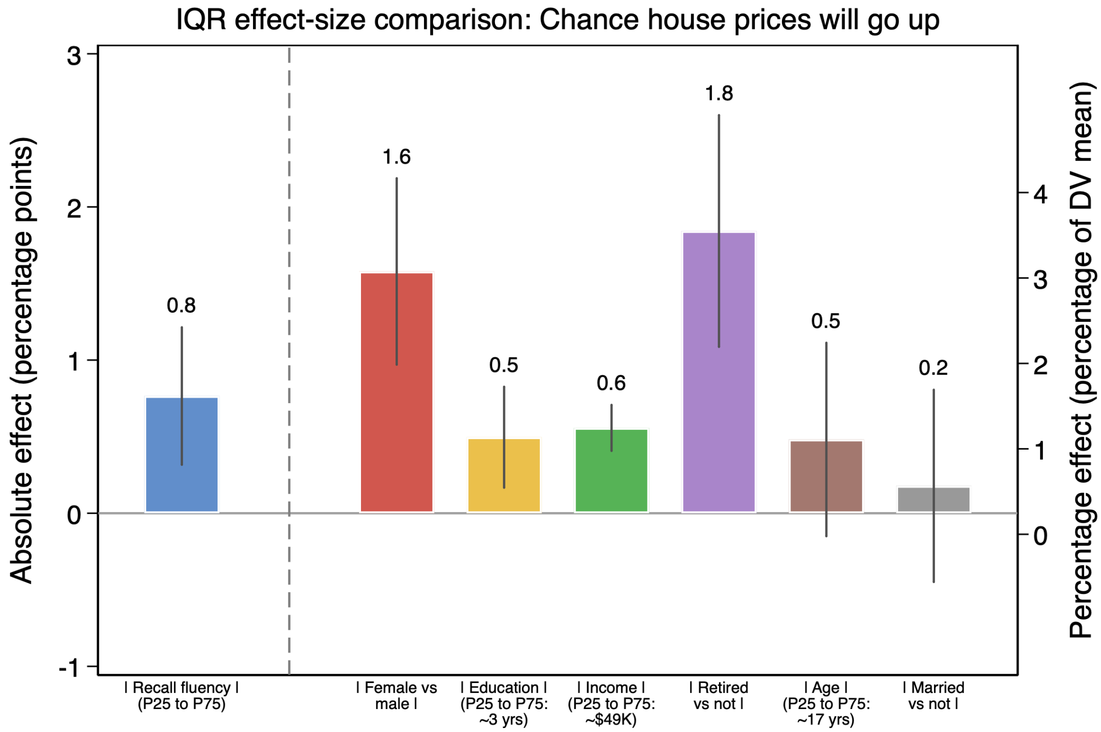
Notes: This figure translates the regression coefficients from column 6 of Table 1 into economically interpretable magnitudes. The dependent variable is the subjective probability of surviving at least 10 more years. Each bar shows a marginal effect evaluated with the non-focal regressors held at the sample mean, 25th percentile, or 75th percentile (indicated by the x-axis grouping). Health experiences are measured as cumulative changes in self-reported health, shifted so the sample mean is positive. X-axis groupings ("At mean", "At 25th", "At 75th") indicate the percentile at which the *other* regressors are held fixed when computing a given marginal effect. Seven bar types within each group: Exp = $\bar{h}_{it}$, the marginal effect of past health experiences (holding RF and IQ at the evaluation point); —Cov—, the absolute marginal effect of the perceived-covariance term ($|\partial y/\partial r_{\text{cov}}|$, evaluated at the same memory/math point; in this specification r_{cov} enters only through the $c.mem \times c.r_{\text{cov}}$ interaction, so this equals $|\beta_{mem\times cov} \times mem_{at}|$); RF, the marginal effect of the Recall Fluency index (holding Exp and IQ at the evaluation point); IQ, the marginal effect of the Math index (holding Exp and RF at the evaluation point). The combined bars—Exp+RF, Exp+IQ, RF+IQ—show the total predicted change in the belief when both named variables move simultaneously from the 25th to the 75th percentile, with the third variable held at the group's evaluation point. The health domain uses age and cohort fixed effects (rather than individual FE) to allow for the strong life-cycle component in longevity beliefs. Y-axes: the left axis shows the percentage-point change in the dependent variable; the right axis re-expresses the same bars as a percent of the sample DV mean (a linear rescaling of the left axis). Numbers above each bar give the percentage-point effect size. Error bars are 95% confidence intervals. All regressors are standardized by the sample inter-quartile range. Specification and sample are identical to column 6 of Table 1.

Figure C.9: Economic magnitude of Recall Fluency versus main demographic controls: Chance stocks will go up.



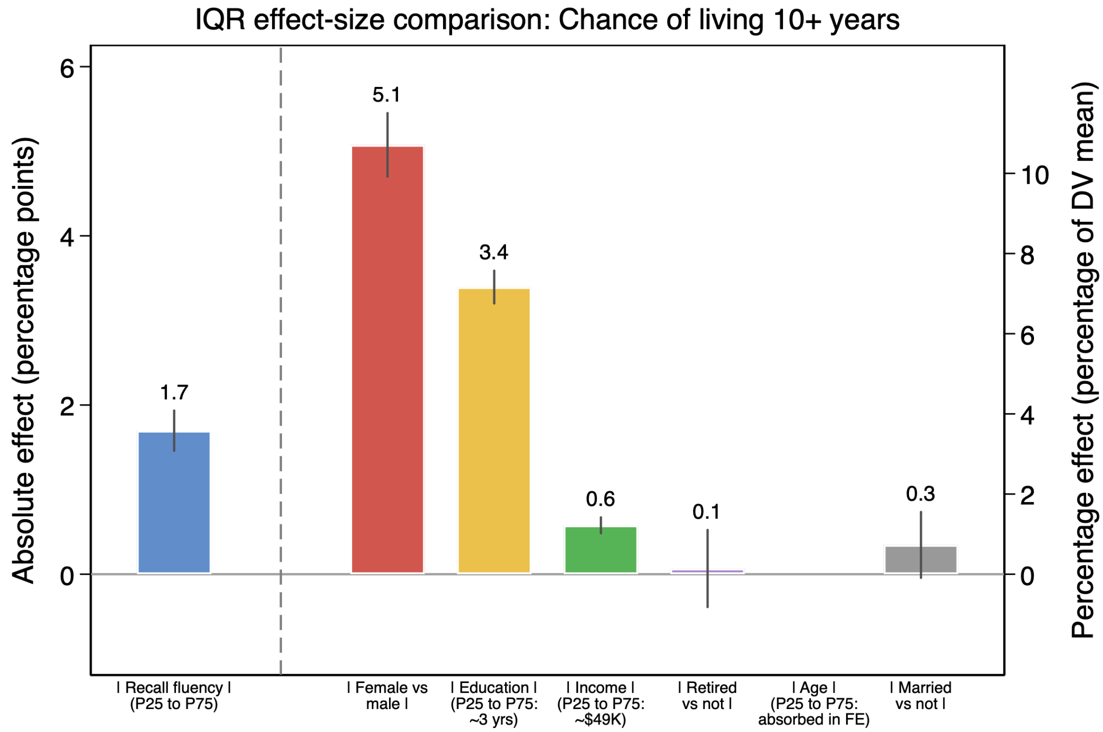
Notes: Bars show the absolute change in the predicted belief (in percentage points) implied by moving each regressor by a pre-specified amount, holding all other regressors fixed, in column 1 of Table 1 (subjective-belief outcome: Chance stocks will go up; N = 139,345). For continuous regressors (recall fluency, education, income, age) the bar is the predicted change from a P25-to-P75 move in the estimation sample; for binary regressors (gender, retired, married) it is a 0-to-1 change. Gender is plotted as female relative to male. The recall-fluency bar is $\hat{\gamma}_2 \cdot \bar{s}_{it}$ evaluated at the sample mean of $\bar{s}_{it}$, following Equation 10; this is the same quantity plotted in Figure C.6. Interquartile ranges (continuous variables only): education ≈ 3 years; household income ≈ \$49K; age ≈ 17 years. Error bars are 95% confidence intervals.

Figure C.10: Economic magnitude of Recall Fluency versus main demographic controls: Chance house prices will go up.



Notes: Bars show the absolute change in the predicted belief (in percentage points) implied by moving each regressor by a pre-specified amount, holding all other regressors fixed, in column 3 of Table 1 (subjective-belief outcome: Chance house prices will go up; $N = 49{,}972$). For continuous regressors (recall fluency, education, income, age) the bar is the predicted change from a P25-to-P75 move in the estimation sample; for binary regressors (gender, retired, married) it is a 0-to-1 change. Gender is plotted as female relative to male. The recall-fluency bar is $\hat{\gamma}_2 \cdot \bar{s}_{it}$ evaluated at the sample mean of $\bar{s}_{it}$, following Equation 10; this is the same quantity plotted in Figure C.7. Interquartile ranges (continuous variables only): education $\approx$ 3 years; household income $\approx$ \$49K; age $\approx$ 17 years. Error bars are 95% confidence intervals.

Figure C.11: Economic magnitude of Recall Fluency versus main demographic controls: Chance of living 10+ years.



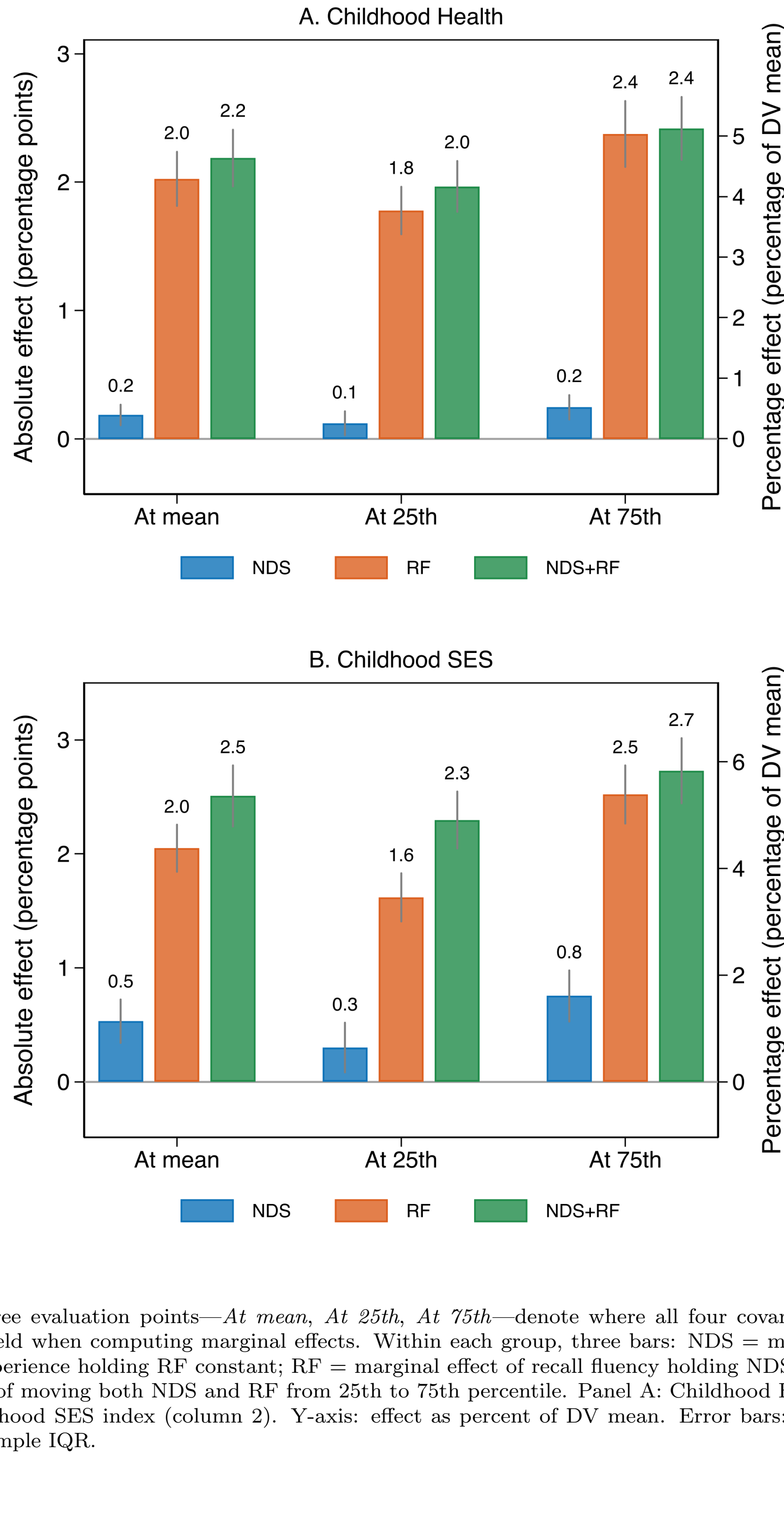
Notes: Bars show the absolute change in the predicted belief (in percentage points) implied by moving each regressor by a pre-specified amount, holding all other regressors fixed, in column 5 of Table 1 (subjective-belief outcome: Chance of living 10+ years; N = 116,262). For continuous regressors (recall fluency, education, income, age) the bar is the predicted change from a P25-to-P75 move in the estimation sample; for binary regressors (gender, retired, married) it is a 0-to-1 change. Gender is plotted as female relative to male. The recall-fluency bar is $\hat{\gamma}_2 \cdot \bar{s}_{it}$ evaluated at the sample mean of $\bar{s}_{it}$, following Equation 10; this is the same quantity plotted in Figure C.8. Interquartile ranges (continuous variables only): education $\approx$ 3 years; household income $\approx$ \$49K; age is absorbed in the column fixed effects. Error bars are 95% confidence intervals.

Table C.9: Robustness of Housing Belief Results: Progressive Controls and Fixed Effects.

	Dependent Variable: Chance House Price ↑						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Baseline (Table 1 Column 3 specification)							
$\bar{s}_{it}$ ($\hat{\gamma}_1$)	0.027***	0.023***	0.023***	0.092***	0.005**	0.006	0.006
	[0.002]	[0.002]	[0.002]	[0.005]	[0.002]	[0.006]	[0.006]
RF × $\bar{s}_{it}$ ($\hat{\gamma}_2$)	0.004***	0.004***	0.003***	0.004***	0.002***	0.002***	0.002***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
N	49,972	49,072	49,072	49,908	49,972	49,908	49,009
Adj. R^2	0.053	0.054	0.068	0.058	0.101	0.102	0.107
DV Mean	0.53	0.53	0.53	0.53	0.53	0.53	0.53
Panel B: With RF Base Effect (Table 1 Column 4 specification)							
$\bar{s}_{it}$ ($\hat{\gamma}_1$)	0.021***	0.017***	0.014**	0.093***	-0.015***	-0.020**	-0.020**
	[0.006]	[0.006]	[0.006]	[0.007]	[0.005]	[0.008]	[0.008]
RF × $\bar{s}_{it}$ ($\hat{\gamma}_2$)	0.008***	0.007***	0.008***	0.004	0.013***	0.014***	0.013***
	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]
N	49,972	49,072	49,072	49,908	49,972	49,908	49,009
Adj. R^2	0.053	0.054	0.068	0.058	0.101	0.102	0.107
DV Mean	0.53	0.53	0.53	0.53	0.53	0.53	0.53
Demographic Controls	Y	Y	Y	Y	Y	Y	Y
Home Value & Wealth	N	Y	Y	N	N	N	Y
Childhood Rural	N	Y	Y	N	N	N	Y
Housing Chars.	N	N	Y	N	N	N	Y
Region FE	N	N	N	Y	N	Y	Y
Year FE	N	N	N	N	Y	Y	Y

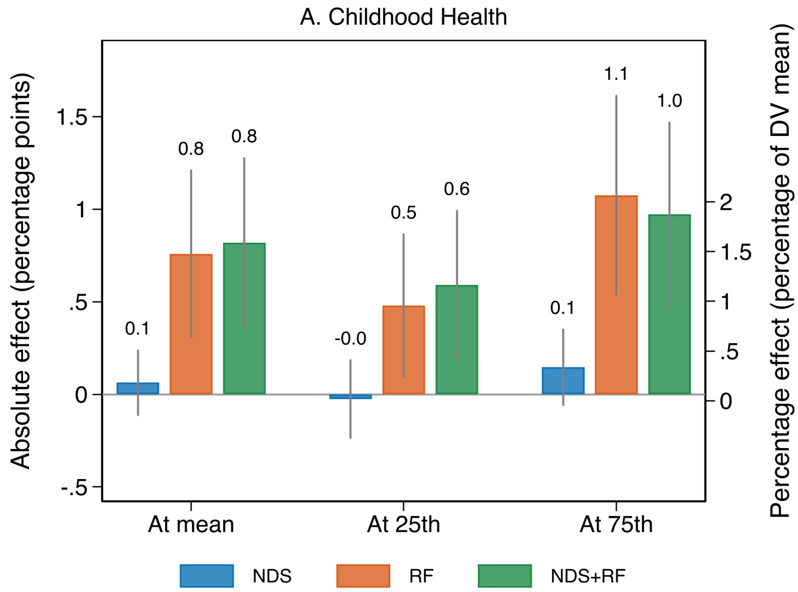
Notes: Each column reports a cross-sectional regression of the subjective probability that the respondent's home price will rise (0–1) on lifetime experienced housing price growth ($\bar{s}_{it}$, measured as the average quarterly HPI return in the respondent's childhood census division from birth to survey date), its interaction with Recall Fluency (RF), and controls. The sample excludes 2008. Panel A replicates the Table 1 Column 3 specification (interactions only); Panel B replicates the Table 1 Column 4 specification, which additionally includes the RF and IQ base effects. RF, IQ, and $\bar{s}_{it}$-related regressors are IQR-standardized. Column 1 replicates the baseline. Column 2 adds home value, real estate, total wealth, and a childhood-rural indicator. Column 3 adds housing characteristics (dwelling type, neighborhood safety, number of rooms, tenure, mortgage status, self-rated home quality, farm residence, and second-home ownership; missing-value indicators included). Columns 4–6 add, respectively, census-division fixed effects (current and childhood), year fixed effects, and both. Column 7 combines all controls with region and year fixed effects. Standard errors in brackets; * $p<0.12$, ** $p<0.05$, *** $p<0.01$.

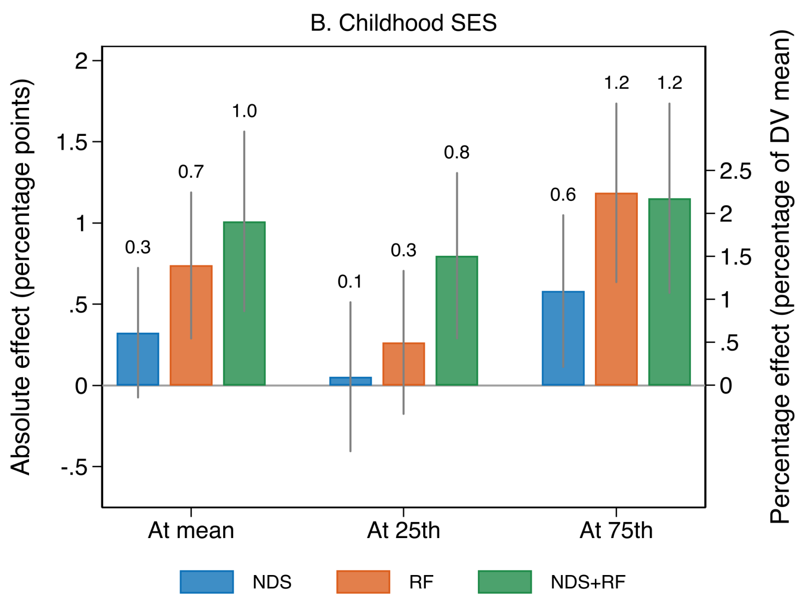
Figure C.12: NDS impact coefficient sizes (% of DV mean): Chance stocks will go up.



Notes: X-axis: three evaluation points—*At mean*, *At 25th*, *At 75th*—denote where all four covariates (RF, NDS, IQ, Experiences) are held when computing marginal effects. Within each group, three bars: NDS = marginal effect of non-domain-specific experience holding RF constant; RF = marginal effect of recall fluency holding NDS constant; NDS+RF = combined effect of moving both NDS and RF from 25th to 75th percentile. Panel A: Childhood Health index (column 1). Panel B: Childhood SES index (column 2). Y-axis: effect as percent of DV mean. Error bars: 95% CIs. Variables standardized by sample IQR.

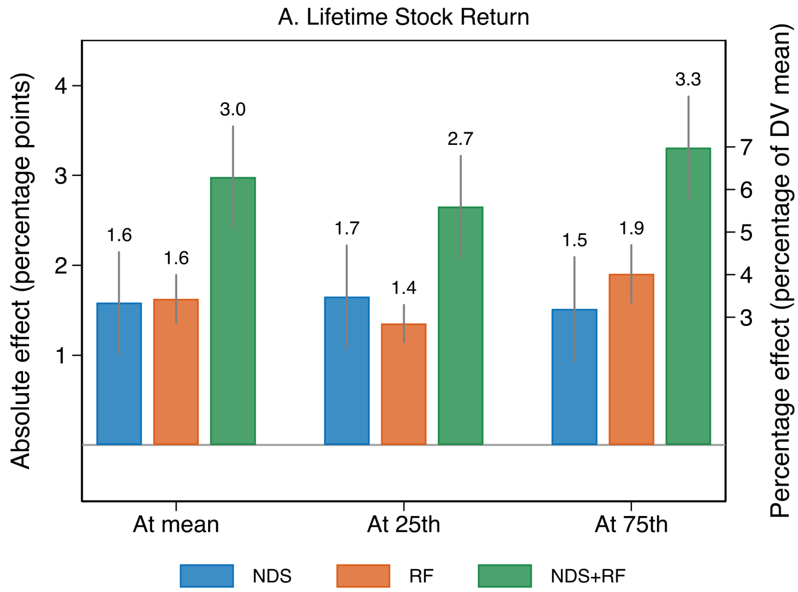
Figure C.13: NDS impact coefficient sizes (% of DV mean): Chance house will go up.

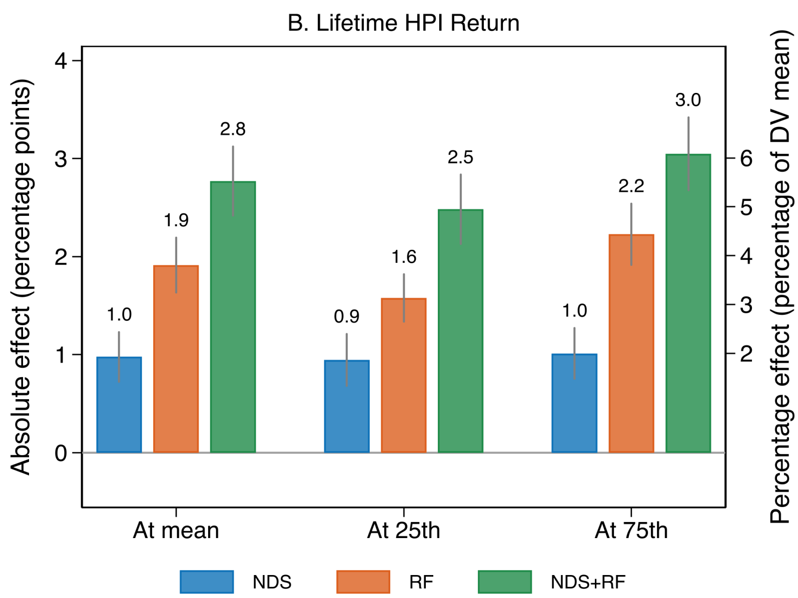




Notes: X-axis: three evaluation points—*At mean*, *At 25th*, *At 75th*—denote where all four covariates (RF, NDS, IQ, Experiences) are held when computing marginal effects. Within each group, three bars: NDS = marginal effect of non-domain-specific experience holding RF constant; RF = marginal effect of recall fluency holding NDS constant; NDS+RF = combined effect of moving both NDS and RF from 25th to 75th percentile. Panel A: Childhood Health index (column 3). Panel B: Childhood SES index (column 4). Y-axis: effect as percent of DV mean. Error bars: 95% CIs. Variables standardized by sample IQR.

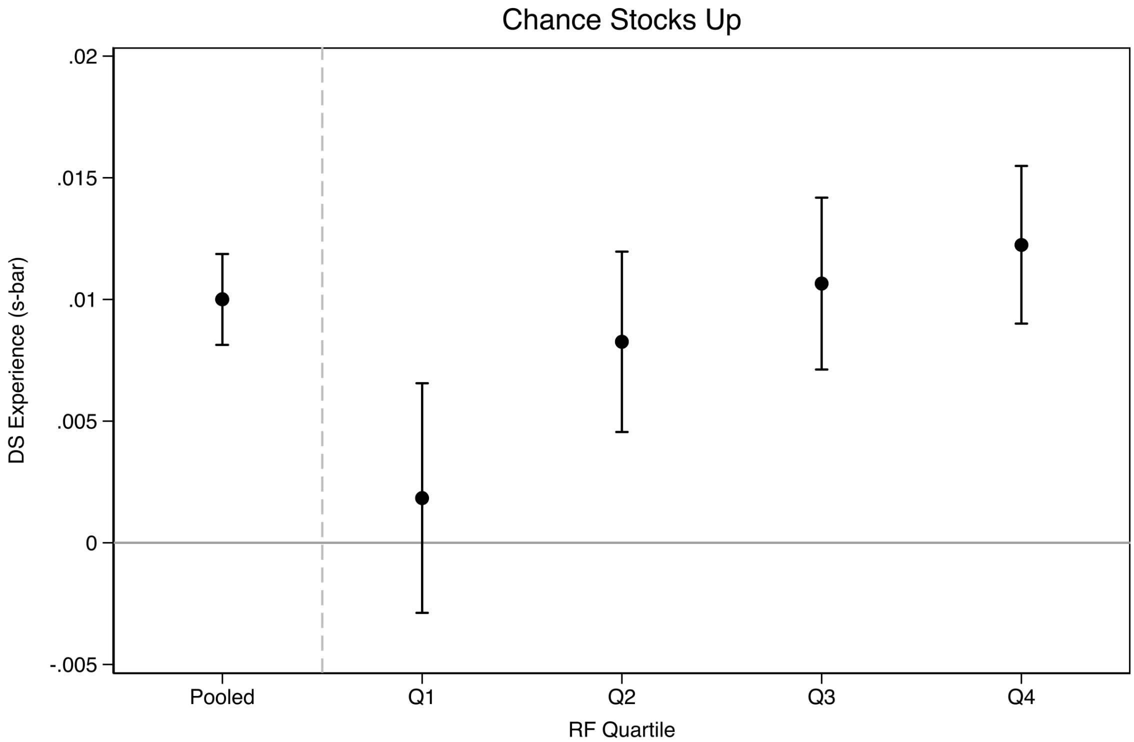
Figure C.14: NDS impact coefficient sizes (% of DV mean): Chance live 10+ years.





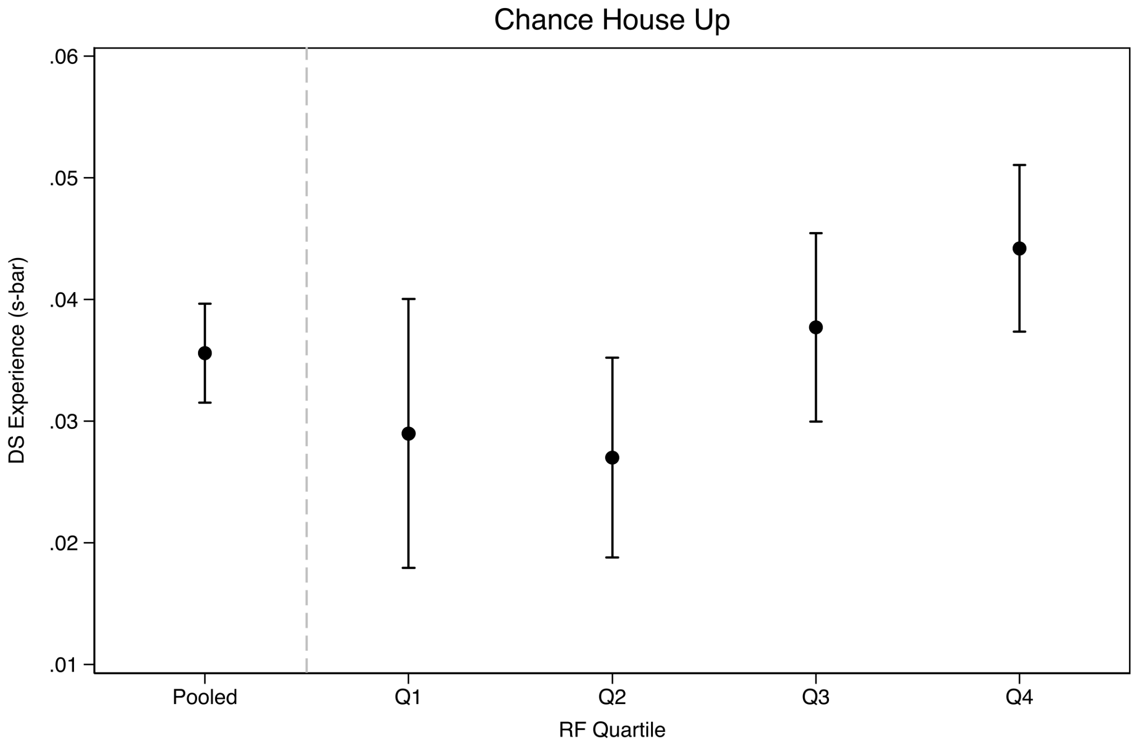
Notes: X-axis: three evaluation points—*At mean*, *At 25th*, *At 75th*—denote where all four covariates (RF, NDS, IQ, Experiences) are held when computing marginal effects. Within each group, three bars: NDS = marginal effect of non-domain-specific experience holding RF constant; RF = marginal effect of recall fluency holding NDS constant; NDS+RF = combined effect of moving both NDS and RF from 25th to 75th percentile. Panel A: Lifetime S&P 500 return Δ_L (column 5). Panel B: Lifetime HPI return Δ_L (column 6). Y-axis: effect as percent of DV mean. Error bars: 95% CIs. Variables standardized by sample IQR.

Figure C.15: Non-Parametric RF Quartile Coefficients: Stock Market Beliefs.



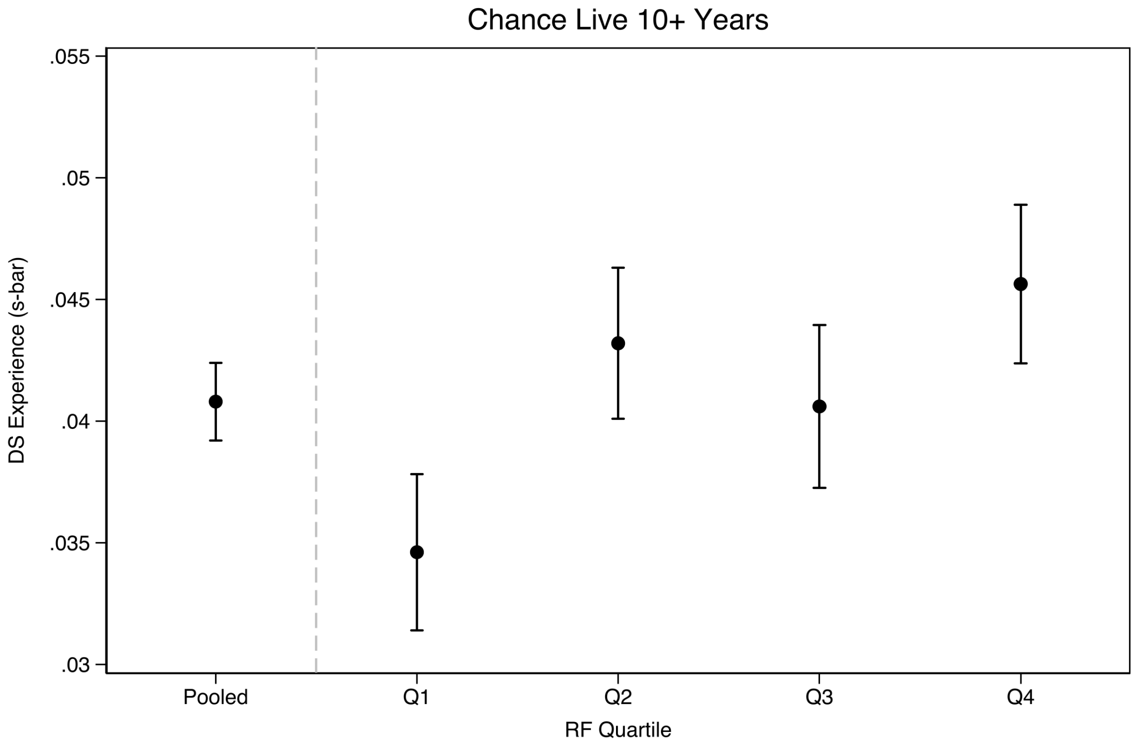
Notes: Each group reports the DS experience coefficient ($\hat{\gamma}_1$ on $\bar{s}_{it}$) and 95% confidence interval from the column 1/3/5 specification of Table 1, estimated separately within each recall-fluency (RF) quartile; the RF $\times$ Experience interaction is omitted so the figure isolates the pure experience effect within each quartile. The Pooled group gives the full-sample estimate. All continuous regressors are IQR-standardized within each subsample. Demographic controls included throughout.

Figure C.16: Non-Parametric RF Quartile Coefficients: Housing Beliefs.



Notes: Each group reports the DS experience coefficient ($\hat{\gamma}_1$ on $\bar{s}_{it}$) and 95% confidence interval from the column 1/3/5 specification of Table 1, estimated separately within each recall-fluency (RF) quartile; the RF × Experience interaction is omitted so the figure isolates the pure experience effect within each quartile. The Pooled group gives the full-sample estimate. All continuous regressors are IQR-standardized within each subsample. Demographic controls included throughout.

Figure C.17: Non-Parametric RF Quartile Coefficients: Health/Longevity Beliefs.



Notes: Each group reports the DS experience coefficient ($\hat{\gamma}_1$ on $\bar{s}_{it}$) and 95% confidence interval from the column 1/3/5 specification of Table 1, estimated separately within each recall-fluency (RF) quartile; the RF × Experience interaction is omitted so the figure isolates the pure experience effect within each quartile. The Pooled group gives the full-sample estimate. All continuous regressors are IQR-standardized within each subsample. Demographic controls included throughout.

Table C.10: Robustness: Choice Regressions Including RF Directly in Second Stage.

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Stock/Housing Market Choices					
	New Stock Market Inv.		**% IRA in Stocks**		**Bought a New Home**	
Cognition						
RF (1 sd) (κ^M)	0.006***	0.009***	0.015***	0.024***	0.003**	0.001
	[0.002]	[0.002]	[0.005]	[0.005]	[0.001]	[0.002]
IQ (1 sd)	0.011***	-0.003*	0.037***	0.007*	0.005***	-0.002
	[0.001]	[0.001]	[0.004]	[0.004]	[0.001]	[0.002]
Beliefs (ϕ)						
$\widehat{B}_{it}$ (θ)	0.040***	.	0.037***	.	0.008***	.
	[0.004]	.	[0.011]	.	[0.001]	.
$\widehat{B(E)}$ (γ)	.	-0.011***	.	-0.018***	.	0.002**
	.	[0.001]	.	[0.003]	.	[0.001]
$B(\widehat{RF \times E})$ (μ)	.	0.012***	.	0.002	.	0.004**
	.	[0.002]	.	[0.004]	.	[0.002]
$B(\widehat{RF \times \text{Cov}})$ (ν)	.	-0.003***	.	-0.004*	.	0.006***
	.	[0.001]	.	[0.002]	.	[0.001]
Risk Taking						
Risk Taking (1 sd)	.	0.004***	.	0.030***	.	0.003***
	.	[0.001]	.	[0.003]	.	[0.001]
Avg DV	0.10	0.10	0.56	0.56	0.04	0.04
N	129529	120357	36078	34263	45249	43865
Demographic Controls	Y	Y	Y	Y	Y	Y
	Panel B: Health Choices					
	Chance Alive Past 70		**Retired**		**% of Income to Pension**	
Cognition						
RF (1 sd) (κ^M)	0.006**	0.004	-0.004***	-0.024***	-0.001	-0.006***
	[0.003]	[0.004]	[0.002]	[0.002]	[0.002]	[0.002]
IQ (1 sd)	0.016***	0.014***	-0.005***	0.001	0.002	0.003*
	[0.003]	[0.003]	[0.001]	[0.001]	[0.002]	[0.002]
Beliefs (ϕ)						
$\widehat{B}_{it}$ (θ)	0.032***	.	-0.009***	.	0.004	.
	[0.004]	.	[0.002]	.	[0.002]	.
$\widehat{B(E)}$ (γ)	.	0.015***	.	-0.033***	.	-0.006**
	.	[0.004]	.	[0.002]	.	[0.003]
$B(\widehat{RF \times E})$ (μ)	.	0.014***	.	0.027***	.	0.009***
	.	[0.004]	.	[0.003]	.	[0.002]
$B(\widehat{RF \times \text{Cov}})$ (ν)	.	-0.001	.	0.003**	.	-0.001
	.	[0.004]	.	[0.001]	.	[0.004]
Risk Taking						
Risk Taking (1 sd)	.	0.029***	.	0.002	.	0.001
	.	[0.003]	.	[0.001]	.	[0.002]
Avg DV	0.26	0.26	0.64	0.64	0.08	0.08
N	20280	20129	100979	91946	4352	4334
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort/Age FE	Y	Y	Y	Y	Y	Y

Notes: Robustness version of Table 3 (Table 3). The two-stage structure is identical: Stage 1 predicts beliefs from RF, IQ, experience interactions, and the RF × experience-covariance interaction (same specification as Table 2). Stage 2 decomposes fitted beliefs into four additive components: experience ($\widehat{B(E)}$, γ); memory amplification of past experiences ($B(\widehat{RF \times E})$, μ); the interaction of recall fluency with the domain-specific experience covariance ($B(\widehat{RF \times \text{Cov}})$, ν); and a controls component ($\widehat{B(Dems)}$). The only difference from Table 3 is that RF (Recall Fluency) is also included as a direct second-stage regressor (κ^M), allowing it to affect choices through channels beyond its belief-shaping effect. Odd columns (1, 3, 5): total predicted belief $\widehat{B}_{it}$ (θ). Even columns (2, 4, 6): full decomposition with risk taking. Panel A outcomes: *New Stock Market Inv.*, an indicator equal to one if the respondent made a new stock market investment since the prior survey wave; *% IRA in Stocks*, the share of IRA assets held in stocks (0–1); *Bought a New Home*, an indicator for home purchase since the prior wave; person fixed effects. Panel B outcomes: *Chance Alive Past 70*, subjective probability of surviving past age 70 (0–100); *Retired*, an indicator for retirement status; *% of Income to Pension*, the share of income contributed to a pension plan (0–1); age and cohort fixed effects. Demographic controls in all columns.

Table C.11: Choice Regressions with Recall Fluency × Belief Interaction.

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Stock/Housing Market Choices					
	New Stock Market Inv.		**% IRA in Stocks**		**Bought a New Home**	
Cognition						
RF (1 sd) (κ^M)		-0.059***		0.125***		0.004
		[0.007]		[0.022]		[0.008]
IQ (1 sd)	0.013***	0.014***	0.035***	0.032***	0.005***	0.005***
	[0.002]	[0.002]	[0.004]	[0.004]	[0.001]	[0.001]
Beliefs						
$\widehat{B_{it}}$ (θ)	0.029***	0.013**	0.044***	0.078***	0.006***	0.008**
	[0.005]	[0.005]	[0.012]	[0.014]	[0.001]	[0.003]
$\widehat{B_{it}} \times$ RF	0.001***	0.008***	0.001**	-0.012***	0.000**	-0.000
	[0.000]	[0.001]	[0.000]	[0.002]	[0.000]	[0.001]
Avg DV	0.10	0.10	0.56	0.56	0.04	0.04
N	129529	129529	36078	36078	45249	45249
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes
	Panel B: Health Choices					
	Chance Past 70		**Retired**		**% of Income to Pension**	
Cognition						
RF (1 sd) (κ^M)		0.006		-0.049***		-0.002
		[0.018]		[0.008]		[0.011]
IQ (1 sd)	0.016***	0.016***	-0.006***	-0.005***	0.002	0.002
	[0.003]	[0.003]	[0.001]	[0.001]	[0.002]	[0.002]
Beliefs						
$\widehat{B_{it}}$ (θ)	0.030***	0.032***	-0.009***	-0.024***	0.004	0.004
	[0.004]	[0.008]	[0.002]	[0.003]	[0.003]	[0.005]
$\widehat{B_{it}} \times$ RF	0.001**	-0.000	-0.000*	0.006***	-0.000	0.000
	[0.000]	[0.002]	[0.000]	[0.001]	[0.000]	[0.001]
Avg DV	0.26	0.26	0.64	0.64	0.08	0.08
N	20280	20280	100979	100979	4352	4352
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cohort/Age FE	Y	Y	Y	Y	Y	Y

Notes: Two-stage robustness analysis complementing Table 3. Stage 1 predicts beliefs using the same specification as Table 3 (recall fluency, IQ, domain-specific experiences and their interactions with RF, and the RF × experience-covariance reminder term). Stage 2 regresses each choice on the total predicted belief $\widehat{B}_{it}$, the interaction of $\widehat{B}_{it}$ with recall fluency, and IQ. Within each dependent variable, the first column reports this specification; the second additionally includes recall fluency as a direct second-stage regressor (shown above the IQ row). Coefficients on $\widehat{B}_{it} \times$RF test whether the mapping from beliefs into choices is itself modulated by recall fluency. Panel A outcomes: new stock-market investment since the prior survey wave; share of IRA assets held in stocks (0–1); purchase of a new home since the prior wave. Panel B outcomes: subjective probability of working past age 70 (0–100); an indicator for retirement status; share of income contributed to a pension plan (0–1); age and cohort fixed effects in Panel B only. Demographic controls (age, gender, marital status, income, education) included in all columns. Coefficients scaled to 1 SD of each regressor. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table C.12: Database Regressions for Depression and Inflation Beliefs.

	(1)	(2)	(3)	(4)	(5)	(6)
		Chance Depression			**Chance Inflation > 5%**	
Cognition						
RF	.	.	-0.101***	.	.	-0.035
	.	.	[0.022]	.	.	[0.099]
IQ	-0.030***	-0.028***	-0.028***	0.003	-0.000	-0.000
	[0.003]	[0.003]	[0.003]	[0.004]	[0.004]	[0.004]
DS experiences						
$\overline{s_{it}}$ (γ_1)	-0.005***	0.004**	-0.005**	-0.001***	-0.001**	-0.002
	[0.002]	[0.002]	[0.002]	[0.000]	[0.000]	[0.001]
RF Interactions						
$RF * \overline{s_{it}}$ (γ_2)	.	0.001***	0.006***	.	0.000***	0.000
	.	[0.000]	[0.001]	.	[0.000]	[0.001]
$RF * cov(s,(s-r_t)^2)$ (γ_3)	.	-0.058***	-0.058***	.	-0.013***	-0.013***
	.	[0.001]	[0.001]	.	[0.001]	[0.001]
N	46880	46576	46576	29869	29682	29682
AR2	0.02	0.05	0.05	0.01	0.01	0.01
Min year	2002	2002	2002	2006	2006	2006
Max year	2008	2008	2008	2008	2008	2008
Demographic Controls	Y	Y	Y	Y	Y	Y

Notes: Each column reports a panel regression of subjective beliefs on domain-specific (DS) experience variables ($\overline{s_{it}}$, $cov(s_k,(s_k-r_t)^2)$), optionally Recall Fluency (RF), IQ, and RF interactions, mirroring the specification in Table 1. Within each panel, columns 1 and 4 report the unconditional experience effect (experience variable with IQ and demographic controls; no RF or RF interactions); columns 2 and 5 add cognition and RF interactions without the RF base; columns 3 and 6 additionally include the RF base effect. **Chance Depression** is the respondent's subjective probability that the U.S. economy will experience a major depression sometime during the next 10 years, paired with lifetime NBER recession experience. **Chance Inflation > 5%** is the subjective probability that the cost of living will rise by more than 5% on average over the next 10 years, paired with lifetime inflation experience. Min/Max year report the earliest and latest HRS wave year in each column's estimation sample. Standard errors in brackets; * p<0.12, ** p<0.05, *** p<0.01.

Table C.13: Database Regressions for Chance of Major Depression: Specification Sweep.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No Memory				**With Memory Interactions**			
IQ	-0.025***	-0.025***	-0.025***	-0.023***	-0.024***	-0.026***	-0.024***	-0.024***
	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]
DS experiences								
$\overline{s_{it}}$ (γ_1)	-0.007***	0.009***	-0.018***	-0.010***	0.006**	0.009***	-0.014***	-0.010***
	[0.003]	[0.003]	[0.002]	[0.002]	[0.002]	[0.003]	[0.002]	[0.002]
RF Interactions								
$RF * \overline{s_{it}}$ (γ_2)	.	.	.	.	0.001***	-0.001	0.001***	0.000
	.	.	.	.	[0.000]	[0.000]	[0.000]	[0.000]
$RF * cov(s, (s - r_t)^2)$ (γ_3)	.	.	.	.	-0.002***	0.001*	-0.002***	0.000
	.	.	.	.	[0.000]	[0.000]	[0.000]	[0.000]
N	46880	32126	46880	46880	46576	31912	46576	46576
AR2	0.02	0.02	0.02	0.06	0.05	0.02	0.05	0.06
Min year	2002	2002	2002	2002	2002	2002	2002	2002
Max year	2008	2006	2008	2008	2008	2006	2008	2008
Demographic Controls	Y	Y	Y	Y	Y	Y	Y	Y
Age Control	Y	Y	N	N	Y	Y	N	N
Year FE	N	N	N	Y	N	N	N	Y
Excl. Financial Crisis	N	Y	N	N	N	Y	N	N

Notes: Each column reports a panel regression of the subjective probability of a major U.S. depression within the next 10 years on lifetime NBER recession experience (the experience mean $\overline{s_{it}}$ and the experience covariance term $cov(s_k, (s_k - r_t)^2)$), along with IQ and demographic controls. Columns 1–4 omit recall-fluency measures entirely. Columns 5–8 add interactions of Recall Fluency (RF) with the experience mean and with the covariance term, while omitting the RF base effect. Within each block of four columns, the first column uses the full estimation sample and retains the baseline demographic control set; the second column excludes the financial-crisis wave from the sample; the third column drops the age control from the demographic set; and the fourth column additionally adds year fixed effects. Min/Max year report the earliest and latest HRS wave year in each column's estimation sample. Standard errors in brackets; * p<0.12, ** p<0.05, *** p<0.01.

Table C.14: Database Regressions for Chance Inflation > 5%: Specification Sweep.

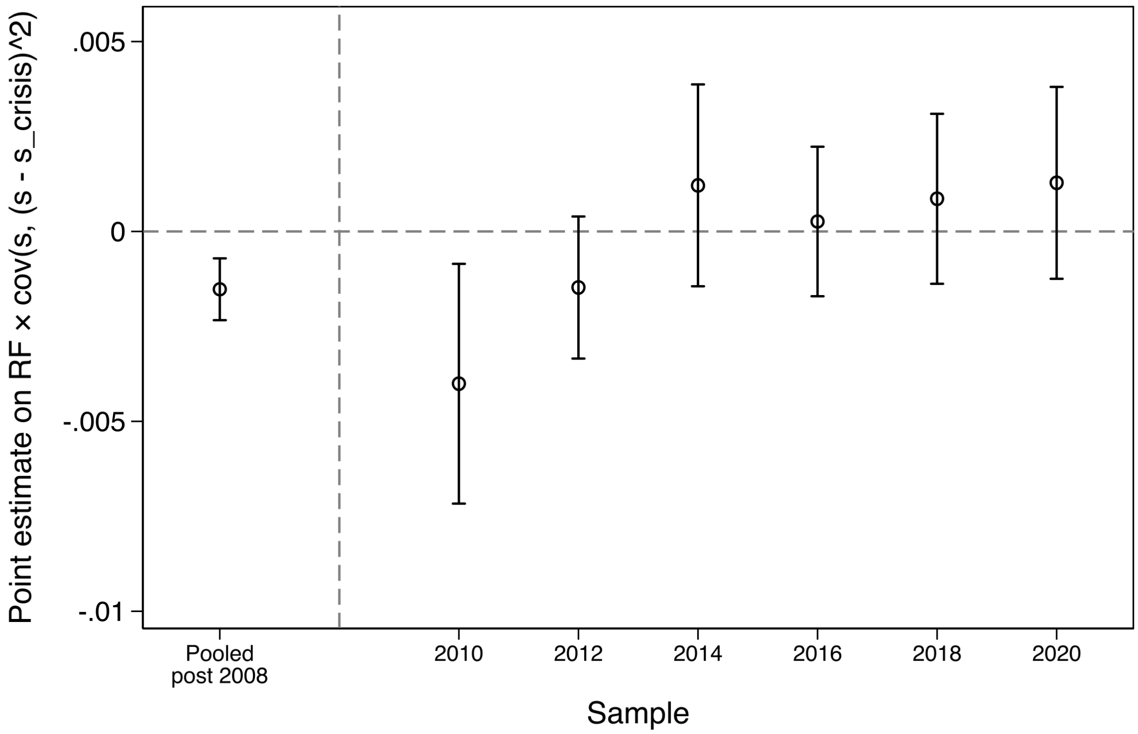
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No Memory				**With Memory Interactions**			
IQ	0.003	0.005	0.003	0.005*	-0.000	0.002	-0.001	0.001
	[0.003]	[0.005]	[0.003]	[0.003]	[0.003]	[0.006]	[0.003]	[0.003]
DS experiences								
$\overline{s_{it}}$ (γ_1)	-0.012***	0.001	-0.002	0.008**	-0.009**	0.000	0.001	0.006
	[0.004]	[0.006]	[0.004]	[0.004]	[0.004]	[0.006]	[0.004]	[0.004]
RF Interactions								
$RF * \overline{s_{it}}$ (γ_2)	.	.	.	.	0.001***	0.000	0.001***	0.001***
	.	.	.	.	[0.000]	[0.000]	[0.000]	[0.000]
$RF * cov(s,(s-r_t)^2)$ (γ_3)	.	.	.	.	-0.007***	0.002	-0.007***	0.000
	.	.	.	.	[0.001]	[0.002]	[0.001]	[0.001]
N	29869	15206	29869	29869	29682	15107	29682	29682
AR2	0.01	0.01	0.01	0.03	0.01	0.01	0.01	0.03
Min year	2006	2006	2006	2006	2006	2006	2006	2006
Max year	2008	2006	2008	2008	2008	2006	2008	2008
Demographic Controls	Y	Y	Y	Y	Y	Y	Y	Y
Age Control	Y	Y	N	N	Y	Y	N	N
Year FE	N	N	N	Y	N	N	N	Y
Excl. Financial Crisis	N	Y	N	N	N	Y	N	N

Notes: Each column reports a panel regression of the subjective probability that the cost of living rises by more than 5% on average over the next 10 years on lifetime inflation experience (the experience mean $\overline{s_{it}}$ and the experience covariance term $cov(s_k,(s_k-r_t)^2)$), along with IQ and demographic controls. Columns 1–4 omit recall-fluency measures entirely. Columns 5–8 add interactions of Recall Fluency (RF) with the experience mean and with the covariance term, while omitting the RF base effect. Within each block of four columns, the first column uses the full estimation sample and retains the baseline demographic control set; the second column excludes the 2008 financial-crisis wave (leaving only the 2006 wave, the other year this outcome is elicited); the third column drops the age control from the demographic set; and the fourth column additionally adds year fixed effects. Min/Max year report the earliest and latest HRS wave year in each column's estimation sample. Standard errors in brackets; * p<0.12, ** p<0.05, *** p<0.01.

Figure C.18: Coefficient on RF × crisis covariance (full spec), pooled and by survey year (post-2008)

Regression specification:

$$Y_{it} = \alpha + \eta^I \mathrm{IQ} + \theta \bar{s}_{it} + \beta_1(\mathrm{RF} \times \bar{s}_{it}) + \beta_2(\mathrm{RF} \times \mathrm{cov}(s, (s - r_t)^2)) + \beta_3(\mathrm{RF} \times \mathrm{cov}(s, (s - s_{\mathrm{crisis}})^2)) + \mathbf{X}_{it}\gamma$$



Notes: Each point is the OLS coefficient on the interaction RF × crisis covariance, $\mathrm{cov}(s, (s - s_{\mathrm{crisis}})^2)$. The left-most point is the *pooled* post-2008 estimate (all survey waves with year > 2008); the remaining points restrict the regression to the single survey year indicated on the horizontal axis. The vertical dashed line separates the pooled estimate from the year-by-year estimates. Vertical bars: 95% confidence intervals. The specification is analogous to that of Table 1, which estimates Equation 10, except that we additionally include the covariance term with the financial crisis and condition the sample on post-2008 observations. Y = chance stocks will go up; RF = recall-fluency (memory) index; covariates $\mathbf{X}$ = age, gender, married, retired, income, education. All regressors IQR-standardized within sample.

Table C.15: Stock-Market Belief Regressions, by Whether Respondent Follows the Stock Market.

	(1)	(2)	(3)	(4)
	Follow Stocks		**Don't Follow Stocks**	
Cognition				
RF	.	0.005	.	-0.005
	.	[0.008]	.	[0.008]
IQ	0.018***	0.018***	0.017***	0.017***
	[0.002]	[0.002]	[0.002]	[0.002]
DS Experiences				
$\overline{s_{it}}$ (γ_1)	0.006***	0.008**	0.000	-0.002
	[0.002]	[0.004]	[0.002]	[0.003]
RF Interactions				
$RF * \overline{s_{it}}$ (γ_2)	0.004***	0.003	0.004***	0.005***
	[0.000]	[0.002]	[0.000]	[0.002]
$RF * cov(s, (s - r_t)^2)$ (γ_3)	-0.004***	-0.004***	-0.003***	-0.003***
	[0.000]	[0.001]	[0.001]	[0.001]
N	71080	71080	67730	67730
AR2	0.03	0.03	0.02	0.02
Demographic Controls	Y	Y	Y	Y

Notes: Each column reports a panel regression of the subjective probability that the stock market will rise on Recall Fluency (RF), IQ, stock-return experience variables ($\overline{s_{it}}$, $cov(s_k, (s_k - r_t)^2)$), and their interactions with RF. The specification mirrors columns 1–2 of Table 1: within each sample split, column (1)/(3) omits the RF main effect, and column (2)/(4) adds it back. Columns 1–2 restrict the sample to respondents who report following the stock market; columns 3–4 restrict to respondents who do not. Demographic controls include age, gender, marital status, retirement status, income, and education. All regressors IQR-standardized within the estimation sample. Standard errors in brackets; * p<0.12, ** p<0.05, *** p<0.01.